

Incompressible Knowledge Probes: Estimating Black-Box LLM Parameter Counts via Factual Capacity

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Abstract

Closed-source frontier labs do not disclose parameter counts, and the standard alternative—*inference economics*—carries $2\times+$ uncertainty from hardware, batching, and serving-stack assumptions external to the model. We exploit a tighter intrinsic bound: storing F facts requires at least $F/(\text{bits per parameter})$ weights, so measuring how much a model *knows* lower-bounds how many parameters it *has*. We introduce **Incompressible Knowledge Probes (IKPs)**, a benchmark of 1,400 factual questions spanning 7 tiers of obscurity, designed to isolate knowledge that cannot be derived by reasoning or compressed by architectural improvements.

We score with no hallucination penalty ($\lambda = 0$: IKP accuracy is simply the fraction of probed facts answered correctly), which removes both the penalty hyperparameter and the per-tier flooring choice; a full $\lambda\times$ flooring ablation (Appendix B) shows the calibration is robust across scoring choices while individual estimates are not, motivating the no-penalty default. We calibrate a log-linear mapping from IKP accuracy to parameter count on 93 open-weight models (135M–1,600B) spanning 19 vendors, achieving $R^2 = 0.910$; leave-one-out cross-validation confirms generalization (median fold error $1.48\times$, 72% within $2\times$ and 86% within $3\times$). For Mixture-of-Experts models, total parameters predict knowledge ($R^2 = 0.67$) better than active parameters ($R^2 = 0.41$). We evaluate 201 models from 27 vendors on a curated probe set (1,311 of 1,400 probes surviving name-collision and label-ambiguity filters) and report effective knowledge capacity for all major proprietary frontier models as prediction bands rather than point estimates; for heavily safety-tuned models these are lower bounds, since refusal policy can hide tens of percentage points of “refused but known” capacity. A black-box hallucination-similarity test—rare-fact agreement on *wrong* answers—separates weight-sharing siblings, post-training lineages, and full retrains without model weights.

The widely-reported saturation of reasoning benchmarks does not imply the end of scaling. Procedural capability compresses under the “Densing Law,” but across 96 dated open-weight models the IKP time coefficient is $-0.0010/\text{month}$ (95% CI $[-0.0031, +0.0008]$)—indistinguishable from zero, and rejecting the Densing prediction of $+0.0117/\text{month}$ at $p < 10^{-15}$. Factual capacity continues to scale log-linearly with parameters across generations and across vendors.

1 Introduction

For the past three years, the author and a few peers have stress-tested newly released models against a standing probe—“What do you know about USTC Hackergame?”—an annual Capture-the-Flag contest, run by the author’s university Linux User Group, with idiosyncratic Chinese-language challenge titles. The same prompt, asked at each model, traced a single fact arriving in parameters over time. In May 2024, GPT-4o knew the contest existed but invented fake challenge names when asked. Just nine months later, in February 2025, Claude 3.7 Sonnet listed nineteen verified Hackergame 2023 titles, verifiable against the official writeup repository. By April 2026, Kimi

K2.6, Claude Opus 4.7 and Gemini 3.1 Pro list specific challenges from multiple consecutive years (Appendix I.1).

This is not a quirk of one event. Transformer feed-forward layers function as key-value memories for factual associations [Geva et al., 2021, Meng et al., 2022, Dai et al., 2022], with an empirical storage capacity of $\sim 2\text{--}4$ bits of factual knowledge per parameter [Allen-Zhu and Li, 2025, Morris et al., 2025].

Curiosity drove me to perform a systematic probe across fellow researchers, and the pattern that emerged was unexpected: *citation count and h-index alone do not determine whether a frontier model recognizes a researcher*. Two researchers with similar citation counts and h-indices can elicit completely different model responses—those whose work has broader impact appear in model outputs reliably, while those whose citations spread across many incremental papers often do not.

A second interesting pattern appeared across vendors. As of evaluation in April 2026, Gemini 3.1 Pro recognizes researchers and named systems that GPT-5.5 hesitates on; GPT-5.5 in turn recognizes more than Claude Opus 4.7. The ranking is also not stable across release cycles, which means a single instrument that can be re-run at each release is itself a useful artifact.

The generalization is that frontier foundation models have become compressed images of the expert discourse of their era. Human experts work in the open—papers, code, talks, documentation—and each successive training run distills more of that output into parameters. Over time, the useful knowledge of entire research communities (which ideas matter, who produced them, what artifacts they built) flows from the long tail of the open web into a handful of proprietary models. How much of that knowledge a particular model has absorbed is bounded below by the Shannon entropy [Shannon, 1948] of the stored facts and bounded above by the model’s parameter budget; and these bounds are tight in practice—accuracy on rare factual questions scales log-linearly with model size across three orders of magnitude [Petroni et al., 2019, Roberts et al., 2020, Kandpal et al., 2023, Lu et al., 2024]. Consequently, the fraction of an expert’s contributions—or of the facts about any entity—that a given model has internalized is an *empirically measurable* quantity. The appropriate instrument is a set of probes spanning (i) rare-but-real researchers and their subfields and (ii) specific factual attributes of specific entities, scored through the public API. This is what we build.

The same instrument, inverted, addresses a problem practitioners care about for independent reasons. Closed-source frontier labs do not disclose parameter counts, and the standard alternative—inference economics [Epoch AI, 2024, Cai et al., 2025]—estimates size from API throughput, pricing, and hardware cost models, carrying acknowledged $2\times+$ uncertainty from factors (hardware generation, batching, quantization, serving stack) that are external to the model itself. A knowledge-based estimator is *intrinsic*: because storing F facts requires at least $F/(\text{bits per parameter})$ weights, the furthest tier of the frequency long tail a model has internalized directly constrains its parameter count from below. Measuring how much a model *knows* therefore estimates how many parameters it *has*—an inverse problem not previously exploited for parameter estimation, and one that uses only black-box API access.

This reframing matters because of a now-widespread misreading of the “Densing Law” [Huang et al., 2025]: capability per parameter doubles every ~ 3.5 months, a 2026 7B model matches a 2023 70B on MMLU [Hendrycks et al., 2021] or HELM [Liang et al., 2023], therefore scaling has stopped mattering. We argue this conflates two distinct resources stored in the same parameter budget. *Procedural* capability—reasoning, parsing, instruction following—is a compressible function over its inputs; better architectures and training recipes genuinely do pack more of it into fewer parameters, and the Densing Law is real for it. *Factual* knowledge is not. “USTC Hackergame started in 2014” cannot be derived, computed, or inferred from general knowledge; it must be stored explicitly, and Shannon entropy provides a hard lower bound on the bits required. IKPs are designed

to isolate this incompressible component, and empirically they do: across 96 dated open-weight models the time coefficient on IKP accuracy is indistinguishable from zero ($-0.0010/\text{month}$, 95% CI $[-0.0031, +0.0008]$), rejecting the Densing prediction of $+0.0117/\text{month}$ at $p < 10^{-15}$. Benchmark saturation is therefore not evidence that scaling has ended—it is evidence that benchmarks have stopped measuring the part of scaling that cannot compress. Factual capacity continues to grow log-linearly with parameters as a function of total parameter count.

Figure 1 summarises the picture: IKP accuracy grows log-linearly with parameter count across three orders of magnitude of open-weight models, and projecting proprietary frontier scores onto the same line places every major closed model at a specific effective size, including several above the largest open-weight release.

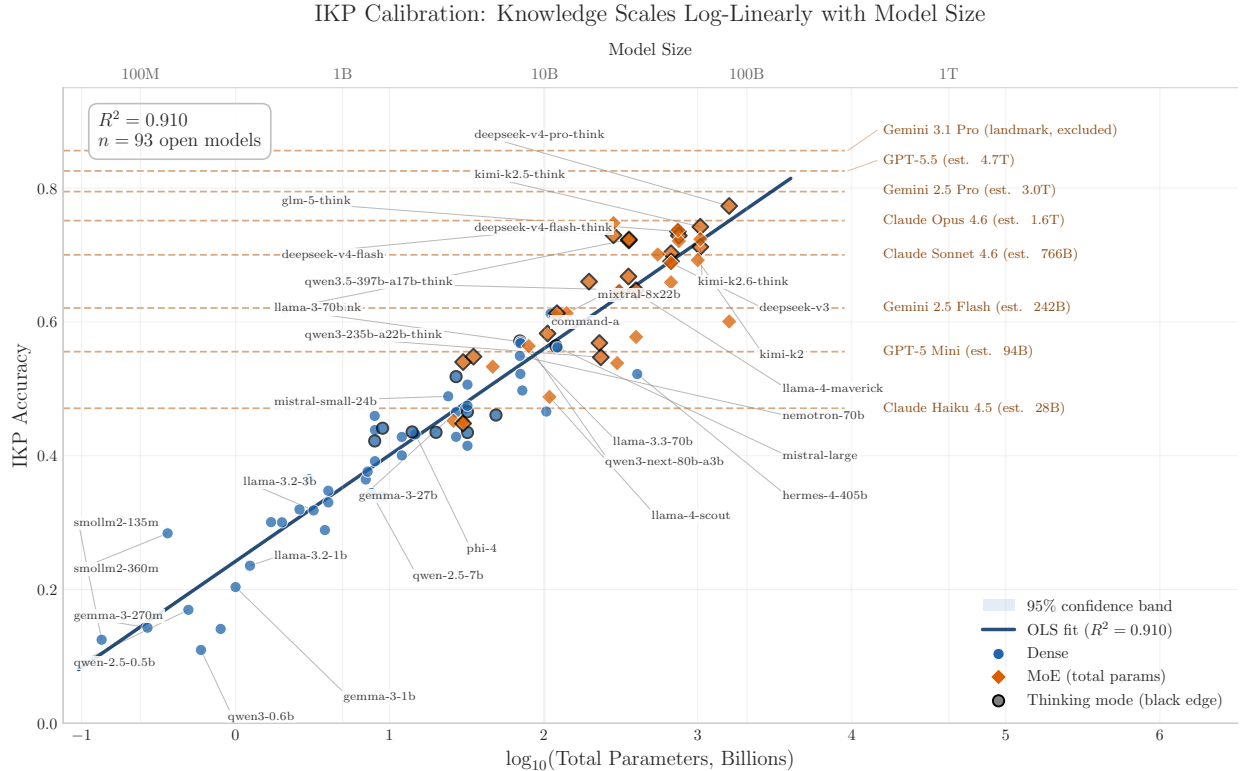


Figure 1: IKP calibration curve. Each point is an open-weight model with known parameter count. Blue circles: dense models; orange diamonds: MoE models (plotted at total parameter count). The regression line ($R^2 = 0.910$, 93 models from 19 vendors, no-penalty scoring) enables parameter estimation for proprietary models (shown as horizontal dashed lines on the right). Gemini 3.1 Pro is excluded from estimation (landmark, T6 score inflated by construction).

Contributions.

- 1. Incompressibility framework.** We distinguish *compressible* procedural capability (subject to the Densing Law) from *incompressible* factual capacity (bounded by Shannon entropy), and introduce Incompressible Knowledge Probes (IKPs), a tiered benchmark designed to measure only the latter from black-box API access.
- 2. Calibration and validation.** We calibrate IKP on 93 open-weight models across 19 vendors ($R^2 = 0.910$, no-penalty scoring), with leave-one-out cross-validation placing 72% of models

within $2\times$ and 86% within $3\times$ of their known size.

3. **Densing Law falsification.** Across 96 dated open-weight models, the time coefficient on IKP is $-0.0010/\text{month}$ (95% CI $[-0.0031, +0.0008]$)—indistinguishable from zero and rejecting the Densing prediction of $+0.0117/\text{month}$ at $p < 10^{-15}$. Factual capacity does not compress.
4. **Frontier estimates and the MoE total-params rule.** We estimate parameter counts for 97 proprietary frontier models and show that total (not active) parameters predict MoE knowledge capacity ($R^2 = 0.67$ vs. 0.41).
5. **Knowledge fingerprinting.** Combining rare-fact Jaccard overlap with *hallucination similarity* (the rate at which two models produce the same wrong answer on rare facts) yields a training-free test that distinguishes weight-sharing siblings, post-training lineages, and full retrains—including across closed-vendor version families—without requiring model weights.
6. **Open-source release.** We release the IKP evaluation toolkit, the full probe set, and the results so that any researcher can estimate a model’s effective knowledge capacity from OpenRouter’s standard API endpoint. Code: <https://github.com/19PINE-AI/ikp>. Companion website: <https://01.me/research/ikp>.

2 Background and Related Work

2.1 Knowledge Capacity of Language Models

[Allen-Zhu and Li \[2025\]](#) establish that transformer models store approximately 2 bits of knowledge per parameter, derived from synthetic factual tuples. [Morris et al. \[2025\]](#) refine this estimate to ~ 3.6 bits per parameter for GPT-style models, distinguishing unintended memorization from generalization and showing that models “memorize until their capacity fills, at which point grokking begins.” Together, these provide a theoretical link between parameter count and factual storage capacity with a 2–4 bits/param range. [Lu et al. \[2024\]](#) show that fact capacity scales linearly with model size and exponentially with training epochs, estimating that memorizing all Wikidata facts requires $\sim 1,000\text{B}$ parameters for 100 epochs—providing direct empirical support for the calibration curve underlying IKP. [Pan et al. \[2025\]](#) formalize knowledge acquisition through the lens of Kolmogorov complexity, proposing a Syntax-Knowledge model grounded in Heap’s and Zipf’s laws that explains scaling laws and hallucination patterns from first principles. [Chang et al. \[2024\]](#) show that 7B models have significantly greater factual knowledge acquisition effectivity than 1B models, demonstrating that model size qualitatively changes knowledge acquisition. [Petroni et al. \[2019\]](#) first demonstrated that pretrained language models serve as implicit knowledge bases, and [Roberts et al. \[2020\]](#) showed that model size directly determines how much knowledge can be packed into parameters. At the mechanistic level, transformer feed-forward layers function as key-value memories [[Geva et al., 2021](#)], with specific “knowledge neurons” responsible for individual facts [[Dai et al., 2022](#)] that can be causally localized and edited [[Meng et al., 2022](#)].

2.2 Long-Tail Knowledge and Scaling

Neural scaling laws [[Kaplan et al., 2020](#), [Hoffmann et al., 2022](#)] establish power-law relationships between model size, compute, data, and loss. [Kandpal et al. \[2023\]](#) demonstrate that accuracy on rare facts scales log-linearly with model size ($R^2 = 0.98$ within the BLOOM family). Larger models extend further down the power-law frequency tail of web knowledge [[Tirumala et al., 2022](#)], consistent with the Zipfian distribution of entity popularity [[Zipf, 1949](#), [Piantadosi, 2014](#)]. [Badhe](#)

et al. [2026] provide a structured taxonomy of long-tail knowledge loss mechanisms—gradient dilution, representational interference, tokenization effects, and post-training compression—formalizing the challenges that IKP tiers exploit for discrimination. Mallen et al. [2023] confirm that LLMs are unreliable on less popular facts, with parametric memory failing where retrieval succeeds. The “Law of Knowledge Overshadowing” [Zhang et al., 2025] shows that popular knowledge suppresses less popular knowledge, with hallucination rate increasing log-linearly with knowledge popularity—implying that obscure probes are more discriminating. Carlini et al. [2021, 2023] demonstrate that memorization scales with model size, and Chen et al. [2025] show that factoid retention degrades severely across fine-tuning stages, explaining why post-training can shift IKP scores between model generations. These findings provide further empirical support for the link between parameters and stored knowledge.

2.3 Existing Model Size Estimation

Epoch AI’s inference economics approach [Epoch AI, 2024] estimates parameter counts from token throughput and API pricing, with acknowledged $2\times+$ uncertainty. Gao et al. [2025] formalize API model verification as a two-sample testing problem, finding that 11 of 31 Llama API endpoints serve distributions different from Meta’s reference weights with just ~ 10 samples per prompt—directly motivating the need for knowledge-based verification. LLMmap [Pasquini et al., 2025] fingerprints identify *which* model is behind an API ($> 95\%$ accuracy across 42 versions) but do not estimate size of unknown models. More recent fingerprinting work includes RoFL [Tsai et al., 2025], which identifies models from statistical output patterns without model modification, and Nasery et al. [2025], who embed 24,576 fingerprints into a single model that persist after fine-tuning. Both are complementary to IKP: they identify *which* model is served but do not estimate size of unknown models. Cai et al. [2025] show that text-output statistical tests fail at $\sim 50\%$ for detecting quantization, motivating knowledge-based approaches. The Densing Law [Huang et al., 2025] demonstrates why standard benchmarks [Hendrycks et al., 2021, Joshi et al., 2017, Kwiatkowski et al., 2019, Liang et al., 2023] fail as size proxies: capability density doubles every ~ 3.5 months, making a 7B model from 2026 match a 70B model from 2023 on reasoning tasks.

2.4 Parameter Specialization

Hong et al. [2025] show that stronger models develop more specialized parameter vectors for factual knowledge storage, with the ratio of parameters dedicated to factual knowledge trending predictably with model quality. This partially addresses the concern about what fraction of total parameters serves factual storage. For MoE models, joint scaling laws [Ludziejewski et al., 2025] across 280+ experiments demonstrate that MoE models can be more memory-efficient than dense models, while Zhao et al. [2025] identify five key factors (data size, total model size, activated model size, active expert count, and shared expert ratio) governing MoE performance across 446 controlled experiments—informing our analysis of effective parameters in MoE architectures (Section 6).

2.5 Distillation Detection

Knowledge distillation detection is an emerging problem. Shi et al. [2025] formalize the task for open-weight models, using data-free input synthesis and statistical scoring when only the student’s weights and the teacher’s API are available. Li et al. [2025] detect distillation via MoE expert routing patterns that persist through distillation, achieving $> 94\%$ detection accuracy with a Shadow-MoE approach for black-box settings. IKP offers a complementary signal: knowledge fingerprints

(Section 6) measure rare-fact overlap and shared-error rates between model pairs through black-box API access alone, without requiring model weights or MoE structure.

3 Theoretical Framework

3.1 Factual Knowledge as Incompressible Information

We define *incompressible factual knowledge* as a factual association (e, a, v) —entity e , attribute a , value v —where v cannot be derived, computed, or inferred from other known facts or from structural regularities in language.

We acknowledge that factual knowledge is not *perfectly* incompressible: prior distributions over values provide some compression. For example, a founding year can be narrowed to a plausible era (say, 1500–2025), reducing the search space from 2^{32} possibilities to ~ 500 , or ~ 9 bits. But the exact year within that range still requires ~ 9 bits per fact, which cannot be compressed further. Similarly, while statistical regularities exist across entities (e.g., European capitals tend to be large cities), IKP probes are specifically designed to minimize such compressibility: researcher probes exclude ML/AI researchers (whose field is trivially predictable from venue co-occurrence), founding-year probes target the specific year (not the century), and name-collision-prone entities are filtered out (Section 4). The residual compressibility of each probe is small relative to the total information content of the 1,400-probe set, preserving the log-linear relationship between stored facts and parameter count.

3.2 The Capacity Bound

From Allen-Zhu and Li [2025], a model with N parameters stores at most $\sim 2N$ bits of factual knowledge under ideal training (with empirical estimates ranging up to ~ 3.6 bits/param [Morris et al., 2025]). From Kandpal et al. [2023], observed accuracy scales as $\text{acc} \approx a \cdot \log(N) + b$, confirmed by Lu et al. [2024] who show linear scaling of fact capacity with model size. Combining these: for a probe set P with known information content, observed accuracy constrains N from below.

3.3 Why Procedural Improvements Do Not Help

We decompose model parameters into three functional roles: $N = N_{\text{fact}} + N_{\text{proc}} + N_{\text{ling}}$, where N_{fact} is the share devoted to storing specific factual associations (entity attributes, dates, names), N_{proc} is the share devoted to procedural skills (reasoning, parsing, instruction following, tool use), and N_{ling} is the share devoted to linguistic competence (syntax, morphology, vocabulary, register). The Densing Law improves the efficiency of N_{proc} and N_{ling} , freeing capacity for N_{fact} or allowing smaller total N . But N_{fact} is bounded below by the information content of stored facts. IKPs exclusively probe N_{fact} , which provides a lower bound on N .

3.4 The Frequency-Capacity Relationship

Web knowledge follows a power-law frequency distribution (Zipf’s law [Zipf, 1949]). Models with more capacity memorize facts further down the long tail. This creates a natural frequency cutoff per model: facts above the cutoff are memorized, those below are not. Since the number of facts above a power-law cutoff grows logarithmically with the cutoff position, aggregate accuracy scales as $\log(N)$ —yielding the log-linear relationship we observe empirically.

Per-tier accuracy follows a logistic sigmoid: $T_i(N) = L_i / (1 + \exp(-k_i \cdot (\log N - m_i)))$, where m_i is the midpoint parameter count for tier i . The aggregate (mean of 7 shifted sigmoids) approximates a straight line over a wide range of $\log N$.

4 Methodology

4.1 Probe Generation

IKP probes are generated through a two-phase pipeline that targets seven *difficulty tiers* (T1–T7) defined by entity-popularity proxies (Common Crawl document frequency) and validated empirically against the landmark ladder (Section 4.2). The full pipeline produces 1,400 probes (200 per tier); the design and per-tier composition are summarized here, with full reproducibility details in Appendix A.

Phase A: LLM-generated candidates (T1–T2 primary, T3–T4 supplementary). A capable LLM (GPT-5) generates candidate questions with answers, prompted with tier-specific frequency targets and rotating region/domain emphasis to enforce coverage across people, places, organizations, events, publications, and measurements. Of the 1,400 final probes, 401 originate from this phase, dominantly at T1 ($n = 166$) and T2 ($n = 152$), with smaller supplementary contributions at T3 ($n = 51$) and T4 ($n = 32$). Empirically, $\sim 82\%$ of LLM-generated candidates land in T1–T2 regardless of difficulty prompting, confirming that an LLM cannot reliably generate factual probes beyond its own knowledge horizon.

Phase B: Corpus-grounded probes (T3–T7). For harder tiers, LLM recall is circular by construction (the generator can only produce facts it knows). We instead sample entities from external corpora and verify answers against the source. Two corpora supply the bulk of T3–T7:

- **Wikidata** [Vrandečić and Krötzsch, 2014] via SPARQL: 557 probes targeting founding years and capital-style attributes for institutions (universities, journals, museums), civic landmarks (bridges, sports clubs), and geographic places. Wikidata supplies ground truth via published property values; we sample entities by Wikipedia view-count quartiles to populate T3 ($n = 94$), T4 ($n = 111$), T5 ($n = 100$), T6 ($n = 141$), and T7 ($n = 100$).
- **DBLP / arXiv researcher records:** 345 probes asking the model to identify a CS researcher’s primary subfield *and* name one verifiable artifact—a paper title, named system, institutional affiliation, or co-author—associated with that researcher’s work. Researchers are sampled from DBLP and OpenAlex by citation-count buckets and assigned a primary subfield via venue tags (e.g., SOSP/OSDI $\rightarrow$ “operating systems”). The evidence requirement separates models that genuinely know the researcher from those that emit a plausible but unattested CS subfield label; the judge accepts a response only if the subfield matches *and* the response cites a verifiable artifact from a per-researcher OpenAlex-derived evidence bundle (Appendix A). This source dominates T5–T7 ($n = 100, 59, 100$) and supplies the steepest discrimination at the frontier: even a 1.6T model cannot reliably name the subfield and a real artifact for a researcher with 10–50 citations.

The remaining 97 probes are manual or supplementary additions used to balance T1–T4 coverage.

Probe quality filters. We exclude: (1) *computable* probes whose answers can be derived by rule (e.g., IUPAC element naming, arithmetic on years, alphabetical ordering), since these test reasoning rather than memorization; (2) *ambiguous* probes that larger models legitimately disagree on, identified by the monotonicity filter (Appendix A): if a more-capable landmark gets a probe wrong while a less-capable one gets it right, the probe is dropped; (3) *collision-prone* researcher probes with common names (two-character Chinese names, single-initial given names) where multiple dis-

tinct researchers share an identifier; (4) *contamination-prone* probes drawn from machine-learning or AI subfields, where researchers’ own work tends to dominate frontier-model training corpora and inflates accuracy beyond what generic factual recall would predict; (5) *grounding embedded in the question template* for Wikidata-sourced probes, following the researcher-probe pattern: every entity-naming question carries a discriminating attribute (year, country, admin region, genre, or publisher) extracted from a Wikidata field other than the answer field. This prevents title-collision failures (“Madonna and Child”—which one?, “Putnam”—in which state?) without leaking the answer. We arrived at this template after a 10-round audit (Section 7.7) revealed that $\sim 12\%$ of T5–T7 Wikidata probes carried a bare-name ambiguity defect and $\sim 33\text{--}46\%$ of diverse-fact-type Wikidata candidates failed independent web cross-check at the obscure tier.

Automated collision/ambiguity audit and robustness. Beyond the template filters, we run an automated post-hoc audit of the released set. For every researcher probe we query OpenAlex for same-name authors and flag a probe when a distinct researcher (different institution, ≥ 100 citations) shares the exact name, so the model cannot know which person is meant; for every Wikidata probe we fetch the entity’s label and flag it when ≥ 2 items of the same type share that label and the question carries no disambiguating qualifier. This flags 89/1400 probes (6.4%: 59 researcher name-collisions, 29 Wikidata label-ambiguities, one known-wrong manual answer), concentrated in the hard tiers (T3–T7) as expected. Re-scoring every model on the cleaned 1,311-probe subset *improves* the calibration— R^2 rises from 0.900 to 0.910 and the 90% prediction-interval factor tightens from $3.45\times$ to $3.20\times$ —and individual frontier estimates move by $<5\%$. The ambiguous probes were adding noise, not signal, and the parameter estimates are robust to their removal. All flags are reproducible via `scripts/clean_flag_researchers.py` and `scripts/clean_flag_wikidata.py`; results in the paper are on the cleaned set.

The complete generation pipeline, per-tier composition table, sample probes, verification procedure for each source type, and the audit log are detailed in Appendix A.

4.2 Landmark-Based Tier Assignment

Tiers are assigned empirically, not by proxy metrics. Six **landmark models** spanning from 0.5B to frontier define the tier boundaries:

Tier landmark	Model	Params	Tier boundary
T1	Qwen 2.5 0.5B	0.5B	T1/T2
T2	Qwen 2.5 7B	7.6B	T2/T3
T3	Qwen 3 32B	32B	T3/T4
T4	Qwen 3 235B	235B	T4/T5
T5	Kimi K2.5	1T	T5/T6
T6	Gemini 3.1 Pro	Frontier	T6/T7

A probe is assigned to tier k if the T_k landmark answers correctly but the T_{k-1} landmark does not. Probes with non-monotonic correctness across the ladder are dropped ($\sim 15\%$), as non-monotonicity indicates ambiguous answers that larger models legitimately contest.

4.3 Scoring

Each model receives the question with a system prompt instructing direct, concise answers. All queries use temperature = 0.

Judge. Gemini 3 Flash Preview classifies each response. Non-researcher probes (Wikidata, LLM-generated, manual) use a 3-way judge: CORRECT, REFUSAL (“I don’t know”), or WRONG (confident incorrect answer). Researcher subfield probes use a 4-way *evidence-aware* judge that consumes the per-researcher evidence bundle (Section 5, Appendix A): CORRECT_STRONG (right subfield *and* cites a verifiable evidence item—paper title fragment, named system, venue, affiliation, or co-author), CORRECT_WEAK (right subfield only, no specific evidence; or correct evidence with adjacent-but-not-listed subfield), WRONG (subfield outside {primary, secondary} or fabricated specifics), or REFUSAL.

Per-probe score. Each probe scores in $\{+1.0, +0.5, 0, \lambda\}$:

Verdict	Non-researcher	Researcher
CORRECT (= STRONG)	+1.0	+1.0
CORRECT_WEAK	—	+0.5
REFUSAL	0	0
WRONG	λ	λ

We use $\lambda = 0$ (no penalty) throughout: IKP accuracy is then simply the fraction of probed facts a model answers correctly, with *no* scoring hyperparameter to tune and *no* per-tier flooring decision to make (per-tier scores are correct/total ≥ 0 by construction, so flooring at zero is a no-op). This choice is deliberate. A nonzero penalty must be justified, and its exact value materially moves individual estimates; moreover, under a negative penalty the per-tier score can go negative and must either be floored—which flattens the calibration slope and inflates frontier estimates—or left unfloored, which destabilizes the fit at hard tiers. Setting $\lambda = 0$ removes both degrees of freedom at once. We adopt it by a *parsimony principle* (Appendix B, Figure 11): take the parameter-free operating point unless a nonzero penalty *materially* improves the calibration—which, across $\lambda \in [0, -2]$, it does not (fit quality varies by ≤ 0.03 in R^2 while individual estimates and their vendor ordering swing by 2–3 $\times$). Appendix B reports the full $\lambda \times$ flooring ablation: $\lambda = 0$ sits within ~ 0.02 – 0.03 R^2 of the shallow mild-penalty optimum near $\lambda = -0.5$ and is the only point at which the flooring choice does not arise. For readers who prefer a hallucination-aware score, the calibration and per-model estimates under any λ are read directly from Table 10. At $\lambda = 0$ the researcher judge’s four verdicts collapse to the ordering STRONG (1.0) > WEAK (0.5) > REFUSAL = WRONG (0); the evidence-aware STRONG/WEAK distinction is retained, and a wrong subfield guess is scored the same as an honest “I don’t know.”

Aggregate accuracy. Per-tier score is the mean per-probe score in that tier. Overall accuracy is the unweighted mean of the seven per-tier scores. Per-tier scores are not floored at zero in the released results—a strongly-bluffing model can score negative on T6/T7—to preserve the bluff signal in the calibration; this is a change from earlier versions of IKP that floored at zero. The mean-of-tier-means construction prevents hard-tier behavior from erasing easy-tier knowledge.

4.4 Calibration Curve

We fit $A = \alpha \cdot \log_{10}(N) + \beta$ where A is aggregate accuracy (no-penalty, $\lambda = 0$) and N is parameter count in billions, using ordinary least squares on open-weight models with known parameter counts. To estimate the parameter count of a target model, we invert the regression: $\hat{N} = 10^{(A-\beta)/\alpha}$. We report the forward-direction R^2 (predicting accuracy from size), which equals the squared Pearson correlation and is invariant to inversion.

5 Experimental Setup

Probe set. 1,400 probes (200 per tier, T1–T7) across 16 domains: researcher subfields (345 probes), founding years (532), geography, history, science, culture, and others.

Models. 201 models from 27 vendors, accessed via OpenRouter API and local Ollama server. Open-weight models with known parameter counts (93 models after excluding pathological-refuser and post-training-degraded outliers, 135M to 1,600B including DeepSeek V4 Pro) serve as calibration data, including models from the Llama [Touvron et al., 2023a,b], Qwen [Yang et al., 2024], Gemma [Gemma Team et al., 2024], DeepSeek [DeepSeek-AI, 2024, 2025], and OLMo [Groeneveld et al., 2024] families. Proprietary models (97 models), including GPT-4 [OpenAI, 2023], are estimation targets.

Judge. Gemini 3 Flash Preview with temperature = 0 and low reasoning effort. Estimated judge error rate: 0.1–0.2% (verified by manual audit).

Researcher probes. All 345 researcher probes use a two-part format: “In computer science, what is the research subfield of [Name], and name one paper, system, institution, or co-author associated with their work? If you don’t know who this person is, say so.” The CS scoping reduces cross-field name collisions; the artifact requirement forces models to produce verifiable evidence rather than emit a plausible-sounding but unattested subfield label. Each researcher carries a manually- or OpenAlex-derived evidence bundle (subfield, top venues, named systems, affiliations, top co-authors), and the four-way judge classifies a response as CORRECT-STRONG only when the response names the right subfield *and* cites at least one matching evidence item; subfield-only responses without specific evidence get CORRECT-WEAK; subfield mismatch with fabricated specifics gets WRONG; “I don’t know” is REFUSAL. See Appendix A for the full evidence schema, sample probes, and judge rubric.

6 Results

6.1 Calibration Curve Quality

Table 1: Scaling law fits: $\text{acc} = \text{slope} \cdot \log_{10}(N_B) + \text{intercept}$

Subset	n	Slope	R^2 (tier-mean)	R^2 (overall)
All open models	93	0.159	0.910	0.910
Dense only	52	0.151	0.875	0.855
Dense non-thinking	42	0.150	0.877	0.866
MoE (total params)	41	0.145	0.667	0.712
MoE (active params)	41	0.144	0.412	0.483

The all-open fit achieves $R^2 = 0.910$ under no-penalty scoring ($\lambda = 0$), with each $10\times$ increase in parameters adding ~ 15.9 percentage points (Figure 1). The tier-mean and overall-accuracy fits agree ($R^2 = 0.910$ both). For Mixture-of-Experts models [Shazeer et al., 2017, Fedus et al., 2022], total parameters ($R^2 = 0.67$) clearly outperform active parameters ($R^2 = 0.41$), confirming that factual knowledge is stored across all expert weights, not just those activated per token (Figure 2). Dense and MoE share nearly the same slope; we calibrate on the combined set because fitting MoE anchors alone does not improve recovery of known MoE sizes in leave-one-out (Appendix C), though it would raise the highest frontier estimates by $\sim 18\%$. This is consistent with unified MoE scaling laws [Clark et al., 2022], joint MoE scaling laws showing that total capacity matters for

knowledge storage [Ludziejewski et al., 2025, Zhao et al., 2025], and practical MoE architectures such as Mixtral [Jiang et al., 2024].

MoE Models: Total vs Active Parameters

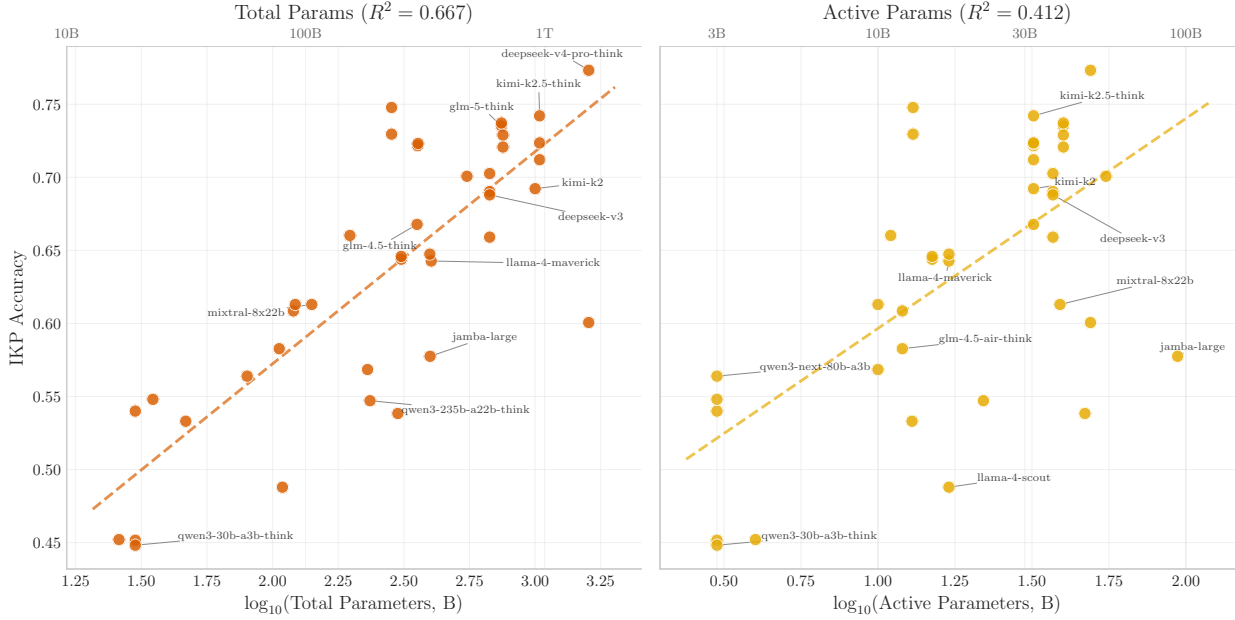


Figure 2: MoE models: accuracy vs total parameters (left, $R^2 = 0.67$) and vs active parameters (right, $R^2 = 0.41$). Total parameters are clearly the better predictor of factual knowledge capacity.

6.2 Leave-One-Out Cross-Validation

To validate that the calibration generalizes rather than overfitting to the 93-model training set, we perform leave-one-out cross-validation (LOO-CV): for each open model, we refit the regression on the remaining 92 models and predict the held-out model’s parameter count from its accuracy.

Figure 3 shows predicted versus actual parameter counts. The LOO median multiplicative fold error is $1.48\times$: 72% of models are predicted within $2\times$ of their true size and 86% within $3\times$, with a 90% prediction interval factor of $3.20\times$. The worst outliers in either direction are Nemotron-70B (predicted $\sim 490\text{B}$, ECR $7.0\times$, NVIDIA’s heavy RLHF post-training pass), Llama 3.1-70B and 3.3-70B (ECRs $\sim 3.5\text{--}4.2\times$), Llama 4 Scout (109B predicted as $\sim 18\text{B}$, ECR $0.17\times$, suggesting heavy refusal calibration), and a cluster of small Qwen variants under-predicted around the $0.3\text{--}0.4\times$ band.

Leave-One-Out Cross-Validation: Predicted vs Actual Parameter Count

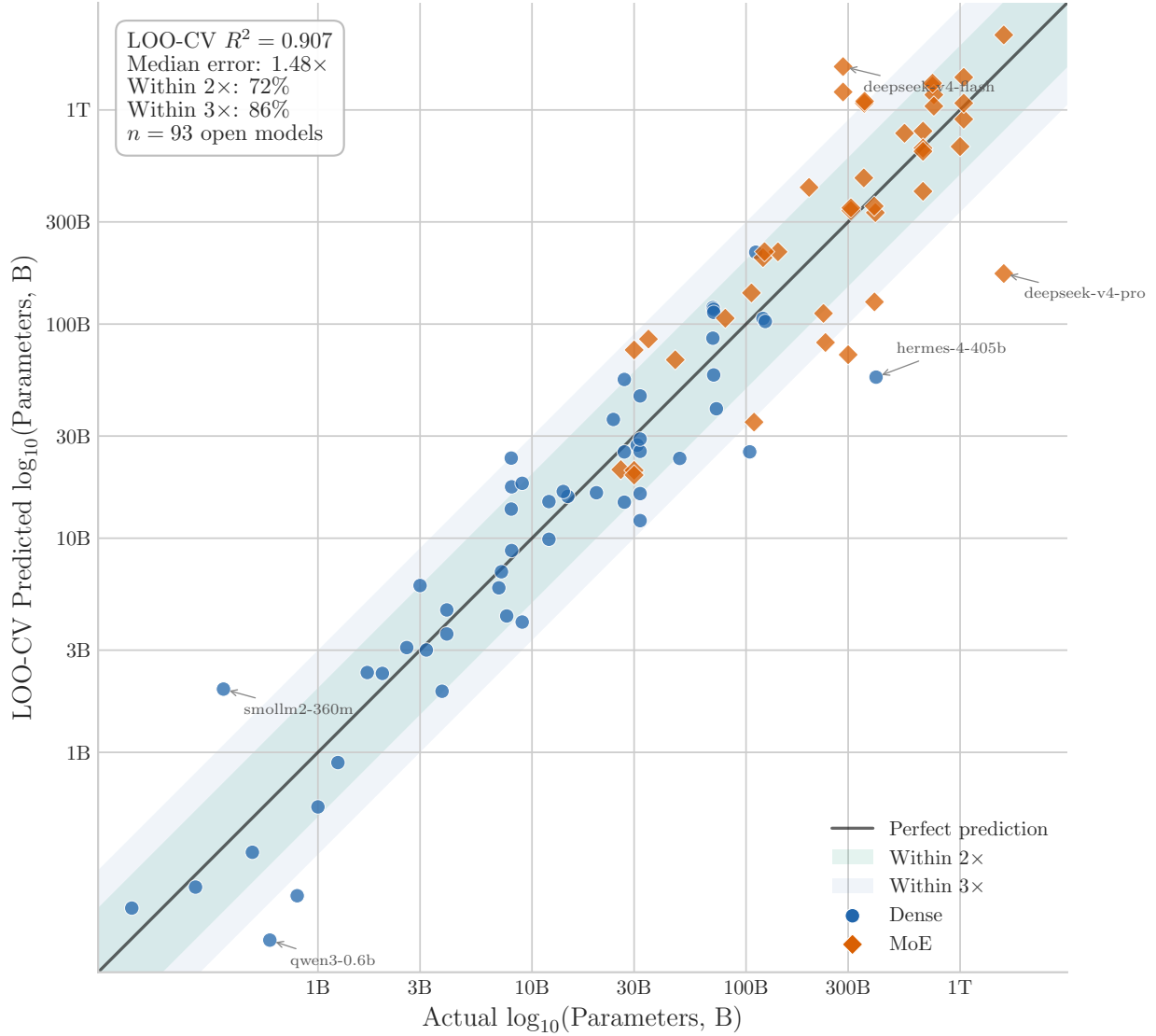


Figure 3: Leave-one-out cross-validation: predicted vs actual parameter count for 93 open models. Green band: within $2\times$; blue band: within $3\times$. Median fold error is $1.48\times$; 72% within $2\times$, 86% within $3\times$. The 90% prediction interval factor is $3.20\times$.

6.3 Frontier Model Estimates

Table 2 shows parameter estimates for the major proprietary models, all derived by inverting the calibration curve fit on 93 open-weight models. None of the entries below were used to fit the curve; the full ranking is therefore an out-of-sample readout of where each closed-source frontier model sits along the same log-linear scaling axis that produced the open-weight regression.

Table 2: Effective knowledge capacity of proprietary frontier models, in open-model-equivalent parameters, as 90% prediction bands (calibration: 93 open models, $R^2 = 0.910$, $\lambda = 0$; PI factor $\sim 3.2\times$). Higher-scoring variant shown per model. The Gemini 3.x family is excluded (Gemini 3.1 Pro is the T6 landmark; the Flash line inherits its inflation). Generated by `scripts/frontier_table.py`.

Model	Vendor	Accuracy	Est. Size	90% PI
GPT-5.5 Pro (think)	OpenAI	83.4%	$\sim 5.3\text{T}$	[1.7T–17.1T]
GPT-5.5	OpenAI	82.6%	$\sim 4.7\text{T}$	[1.5T–15.1T]
Claude Fable 5	Anthropic	80.5%	$\sim 3.5\text{T}$	[1.1T–11.2T]
Gemini 2.5 Pro	Google	79.5%	$\sim 3.0\text{T}$	[943B–9.6T]
GPT-5.4 Pro	OpenAI	77.8%	$\sim 2.4\text{T}$	[744B–7.6T]
GPT-4.1	OpenAI	77.2%	$\sim 2.2\text{T}$	[677B–6.9T]
GPT-5 Pro	OpenAI	77.0%	$\sim 2.1\text{T}$	[657B–6.7T]
o3	OpenAI	76.9%	$\sim 2.1\text{T}$	[650B–6.7T]
Grok-3	xAI	76.8%	$\sim 2.1\text{T}$	[642B–6.6T]
Grok-4	xAI	76.6%	$\sim 2.0\text{T}$	[619B–6.3T]
GPT-5 (think)	OpenAI	76.5%	$\sim 2.0\text{T}$	[614B–6.3T]
Claude Opus 4.6 (think)	Anthropic	75.1%	$\sim 1.6\text{T}$	[503B–5.1T]
MiMo v2.5 Pro	Xiaomi	74.4%	$\sim 1.4\text{T}$	[451B–4.6T]
Hy3 Preview	Tencent	74.1%	$\sim 1.4\text{T}$	[433B–4.4T]
GPT-5.4	OpenAI	72.3%	$\sim 1.1\text{T}$	[334B–3.4T]
GPT-5.3	OpenAI	72.2%	$\sim 1.1\text{T}$	[330B–3.4T]
GLM-5.2	Z.ai	72.2%	$\sim 1.0\text{T}$	[326B–3.3T]
GLM-5 Turbo	Z.ai	71.1%	$\sim 896\text{B}$	[280B–2.9T]
Claude Opus 4.5	Anthropic	71.1%	$\sim 892\text{B}$	[279B–2.9T]
Qwen3.7 Max	Alibaba	71.0%	$\sim 887\text{B}$	[277B–2.8T]
Claude Opus 4.1 (think)	Anthropic	70.7%	$\sim 847\text{B}$	[265B–2.7T]
Claude Sonnet 4.6 (think)	Anthropic	70.0%	$\sim 766\text{B}$	[240B–2.5T]
Claude Opus 4.7 (think)	Anthropic	69.8%	$\sim 738\text{B}$	[231B–2.4T]
MiMo v2.5	Xiaomi	69.5%	$\sim 705\text{B}$	[220B–2.3T]
Grok-4.20	xAI	69.3%	$\sim 689\text{B}$	[215B–2.2T]
GPT-5.2 Pro	OpenAI	68.8%	$\sim 639\text{B}$	[200B–2.0T]
GPT-5.1	OpenAI	68.7%	$\sim 634\text{B}$	[198B–2.0T]
Qwen3 Max	Alibaba	68.4%	$\sim 606\text{B}$	[189B–1.9T]
Claude Sonnet 5	Anthropic	68.2%	$\sim 584\text{B}$	[183B–1.9T]
Qwen3.7 Plus	Alibaba	68.1%	$\sim 579\text{B}$	[181B–1.9T]
Step-3.7 Flash	StepFun	66.7%	$\sim 471\text{B}$	[147B–1.5T]
MiniMax M3	MiniMax	63.9%	$\sim 313\text{B}$	[98B–1.0T]
Claude Opus 4.8	Anthropic	61.9%	$\sim 236\text{B}$	[74B–753B]

The shape of the frontier. Estimates are reported as 90% prediction intervals, not point values. At the top, GPT-5.5 Pro and GPT-5.5 lead at $\sim 5.3\text{T}$ and $\sim 4.7\text{T}$, followed by Claude Fable 5 ($\sim 3.5\text{T}$) and Gemini 2.5 Pro ($\sim 3.0\text{T}$). A broad $\sim 2\text{T}$ cluster gathers GPT-4.1, GPT-5.4 Pro, GPT-5 Pro, o3, Grok-3, and Grok-4 (all within $[0.6, 2.2]\text{T}$), indicating that competitive frontier development has converged to within $\sim 1.5\times$ effective capacity. The ratio between the top of the proprietary fleet and the smallest estimated model ($\sim 9\text{B}$) is $\sim 580\times$, spanning the same dynamic range as the open-weight calibration set. Because we score with no hallucination penalty, refusal-heavy models are systematically under-placed (Section 7.1); their bands should be read as lower bounds.

Pro tiers add little factual capacity. OpenAI’s “Pro” variants sit only $+1.2$ (GPT-5), $+0.8$

(GPT-5.2), and +0.9 (GPT-5.5) pp above their non-Pro siblings—effective-capacity premiums of 1.1–1.2 $\times$ —consistent with the Pro tier targeting reasoning, agentic, and long-context capability through stronger post-training and longer inference-time budgets rather than added *stored facts*. The lone exception is GPT-5.4 Pro (+5.5 pp, 2.2 $\times$), which we do not over-interpret given the $\sim 3.2\times$ prediction-interval width. IKP measures factual capacity specifically, so a near-flat Pro-vs-base gap is the expected outcome and does not contradict the Pro tier’s purpose (Section 3).

Point releases reshuffle; major versions scale up. Within OpenAI’s GPT-5.x series the non-thinking scores span 68–83%: GPT-5 at 75.8%, the 5.1–5.4 point releases clustered at 68–72%, and GPT-5.5 stepping clear to 82.6%. Read literally, the point releases share a similar parameter budget while GPT-5.5 is a genuine scale-up; the fingerprint analysis (Section 6.9) agrees, placing every consecutive GPT-5 $\rightarrow$ 5.x transition in the *retrained* regime ($HSS < 0.10$) rather than the lineage regime. The Claude Opus non-thinking line, by contrast, is *non-monotonic* under no-penalty scoring (4.5 at $\sim 892B$, 4.6 at $\sim 1.3T$, 4.7 at $\sim 538B$, 4.8 at $\sim 236B$). This is not evidence of shrinking models: the later Opus releases refuse far more often (Section 7.1), which depresses raw accuracy and hence the no-penalty estimate. Under a mild penalty (Table 10) the ordering largely reverts, and the Opus estimates should be read as lower bounds—a caution that applies to every heavily refusal-trained model under $\lambda = 0$.

6.4 Per-Tier Discrimination

Each obscurity tier acts as a different rung of the parameter ladder. Fitting a log-linear regression of per-tier accuracy against $\log_{10}(N)$ on the 93-model calibration set reveals a striking stratification (no-penalty scoring): T1’s slope is only 0.072 per decade; T2 0.144; T3 0.275; T4 has the steepest slope (0.297 per decade) and does not saturate within the calibration range; T5 0.224; T6 drops to 0.077 (nearly flat in log-parameters within the open-weight set); and T7 ≈ 0.022 . Each tier is a step function whose midpoint sits at a different parameter scale, and the seven midpoints together span the ladder. Figure 4 shows the resulting step-function pattern.

T3 is the most informative single tier. It combines the steepest slope (0.324 per decade), the highest Spearman correlation with overall accuracy ($\rho = 0.983$), and the widest within-set spread (from near-zero on sub-1B models up to 97% on the largest open-weights). A model’s T3 score alone is therefore a strong proxy for its overall ranking—useful when limited compute precludes a full 1,400-probe evaluation. T2 is nearly as informative ($\rho = 0.874$) but saturates earlier; T1 is approximately a binary check (is this even a real LM?) and contributes essentially nothing to the discrimination of frontier models. Practically, a 200-probe sub-evaluation drawing primarily from T3 (with a thin T1/T2 sanity check and a thin T5/T6 ceiling check) recovers most of the signal of the full benchmark.

T6 discriminates the frontier. Under no-penalty scoring the open-weight T6 slope is a modest 0.077/decade, and large open models do register some T6 accuracy (DeepSeek V4 Pro at 16.5% non-thinking, 46.4% thinking), so T6 is a softer separator than under penalized scoring. Even so, the strongest proprietary models lead clearly: GPT-5.5 at 67.4%, GPT-5.5-Pro at 72.2%, GPT-5.5-Think at 69.5%, GPT-5 at 43.9%, Claude Opus 4.6-Think at 38.0%, o1 at 32.1%, and Opus 4.7 at 20.3%. The Gemini 3.x family posts the highest T6 scores (82–94%), inflated by construction because Gemini 3.1 Pro is the T6 calibration landmark—hence their exclusion from the frontier table. The T6 gap between the leading proprietary models and the open-weight cluster remains the clearest signal of an effective-capacity step beyond $\sim 2T$.

T7 is a hard ceiling. Only two of 188 evaluated models score above 1% on T7 (Jamba-large at 2.8%, Grok-4 at 1.0%); every other model lands at 0% to within sampling noise. Critically, this includes models we estimate at multi-trillion effective capacity. The implication is structural

rather than parametric: T7 probes target facts whose effective mention frequency on the open web is below what current pretraining corpora preserve at any model size we have observed, so even a model trained on a comparable web snapshot cannot memorize them. T7 functions as an *anchor*—it certifies that the obscurity ladder has not run out of room, that the long tail still extends beyond the largest model in our roster, and that any future “saturated on IKP” claim would have to be tested by extending the obscurity scale rather than the parameter axis.

Saturation thresholds give a back-of-envelope size estimator. Reading the table of slopes and intercepts in reverse, the parameter scale at which each tier crosses a given accuracy threshold is informative. For instance, 90% T2 accuracy implies $\geq 117\text{B}$; 90% T3 implies $\geq 600\text{B}$; 50% T4 implies $\geq 100\text{B}$; 50% T5 implies $\sim 300\text{B}$; 20% T6 implies effective capacity above the calibration range. A practitioner who knows nothing about a black-box API but observes its per-tier accuracies can therefore read the bands directly without inverting the joint regression.

6.5 Thinking Mode Analysis

Across 30 base/think pairs, thinking mode [Wei et al., 2022] improves accuracy in 18 cases (mean +2.3 pp, range -11.0 to $+17.3$ pp; Figure 5). The largest gains are DeepSeek V4 Pro (+17.3 pp) and Claude Opus 4.1 (+14.6 pp); the largest regression is Claude 3.7 Sonnet (-11.0 pp), with smaller regressions on Grok-4.20 and MiMo variants plausibly reflecting increased refusal conservatism in thinking mode. The benefit peaks at T3–T4 (medium-hard tiers) and vanishes at T7, supporting the interpretation that chain-of-thought helps with knowledge *retrieval* but does not create new *stored* knowledge.

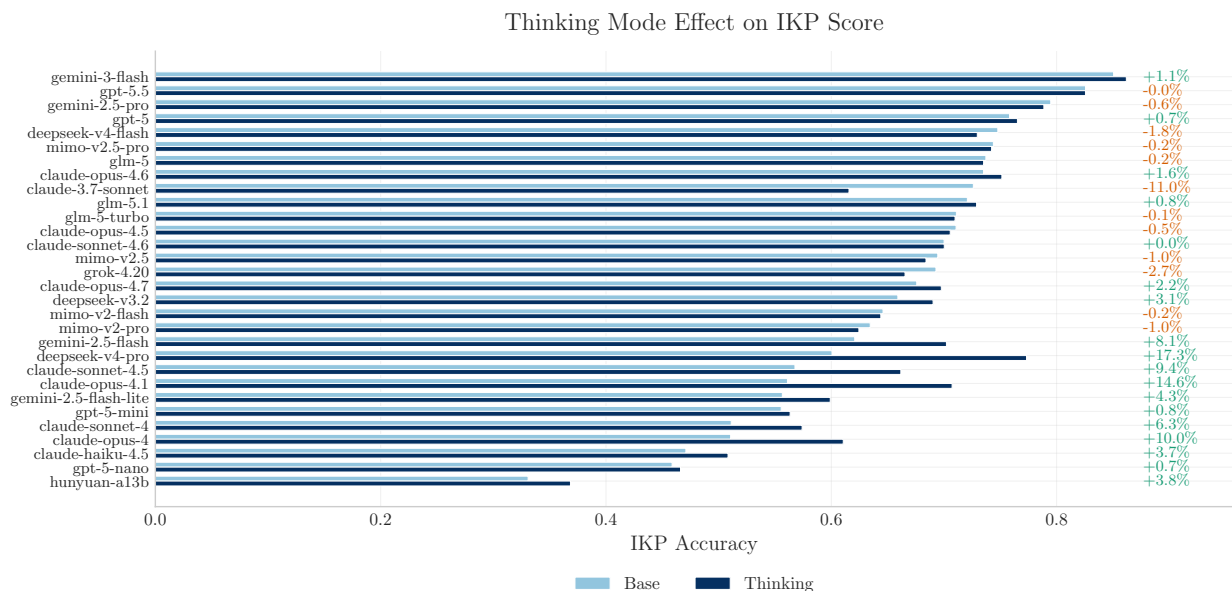


Figure 5: Thinking mode effect across 27 base/think model pairs. Light bars: base model accuracy; dark bars: thinking variant. Thinking improves accuracy in 18 of 30 cases (mean +2.3 pp). Regressions on the remaining pairs (up to -11 pp for Claude 3.7 Sonnet) are largely due to increased refusal conservatism in thinking mode.

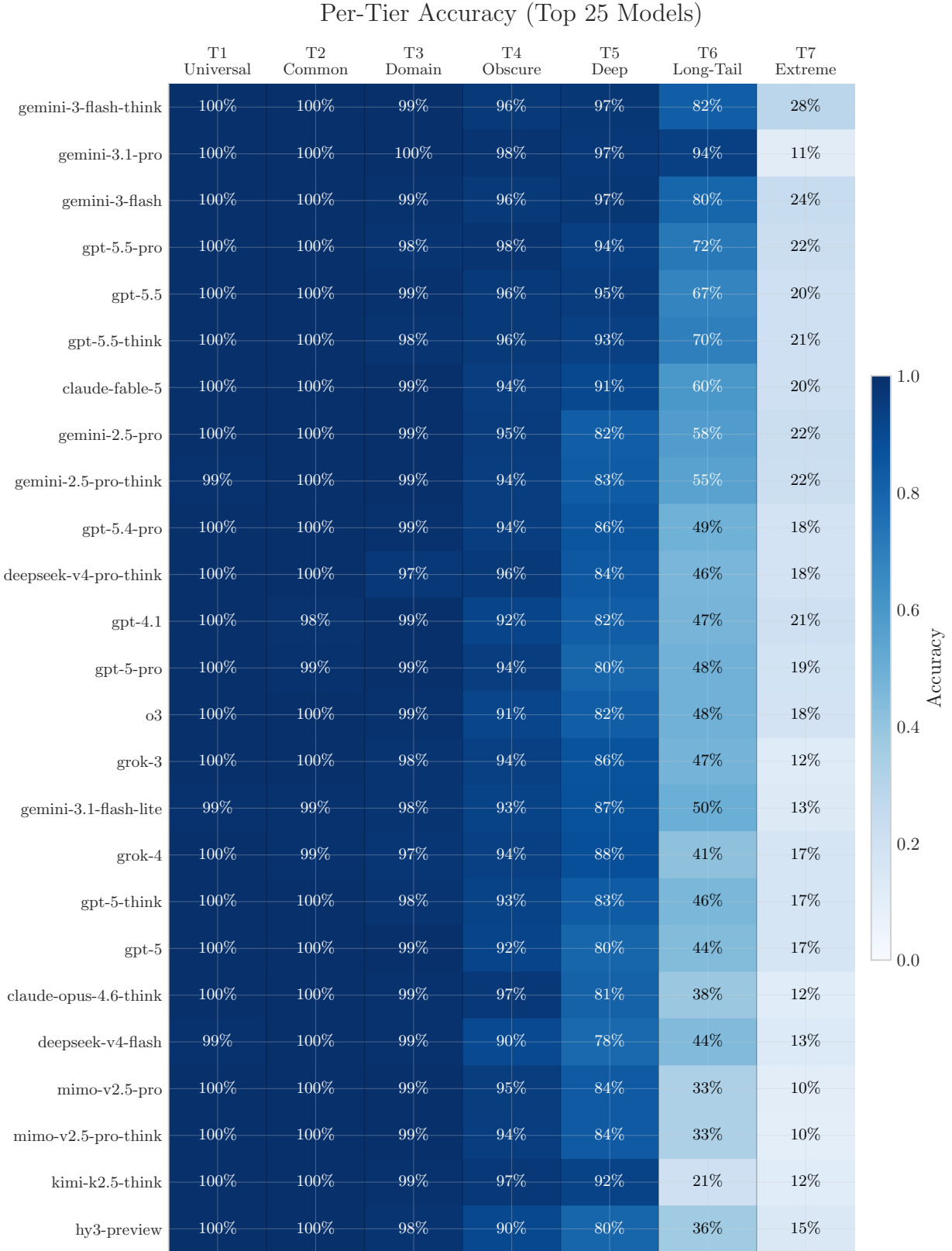


Figure 4: Per-tier accuracy for the top 25 models. Each row is a model (sorted by overall accuracy); each column is a tier (T1–T7). The step-function pattern is clear: T1–T2 are saturated (dark blue), T3–T5 provide the main discrimination, and T6–T7 separate only the strongest frontier models.

6.6 Cross-Generation Improvement

Within model families, newer generations generally know more: Claude Opus base 4 $\rightarrow$ 4.1 $\rightarrow$ 4.5 $\rightarrow$ 4.6 $\rightarrow$ 4.7 gains +13.3 pp end-to-end (52.4 $\rightarrow$ 57.9 $\rightarrow$ 61.1 $\rightarrow$ 63.1 $\rightarrow$ 65.7%); Qwen 2.5 $\rightarrow$ 3 $\rightarrow$ 3.5 at $\sim$ 8B gains a few percentage points per generation; GLM 4 $\rightarrow$ 5.1 gains steadily across each version bump. However, generation transitions can also cause regression: GPT-4 $\rightarrow$ 4o is essentially flat at 54.8% $\rightarrow$ 55.3% but with a different per-tier shape (4o is more conservative on hard tiers), and Claude 3.5 Haiku $\rightarrow$ Haiku 4.5 *loses* -5.7 pp (45.6% $\rightarrow$ 39.9%), driven by tighter refusal calibration in the newer release rather than knowledge loss (Section 7.1). Within the Claude Opus line, Opus 4.7 (65.7%) sits a hair above Opus 4.6 (63.1%) on the non-thinking axis but *below* it under thinking (66.4% vs 68.0%), confirming that within-family ordering near the frontier is sensitive to refusal calibration in each release rather than a single underlying capacity ranking.

6.7 Densing Law Falsification

A central empirical implication of our framework (Section 3) is that the Densing Law [Huang et al., 2025] should *not* hold on IKP. If factual capacity is incompressible, then at fixed $\log_{10}(N)$ there should be no per-month improvement in IKP accuracy—in contrast with reasoning benchmarks, on which capability-per-parameter doubles every $\sim$ 3.5 months.

Design. We assembled release dates (YYYY-MM-DD, verified against vendor announcements and HuggingFace model cards) for all 201 models in our evaluation. For the 96 open-weight models with published parameter counts and IKP results, spanning 2023-09-27 to 2026-04-24, we fit

$$\text{pen_acc} = \beta_0 + \beta_1 \log_{10}(N_B) + \beta_2 \cdot \text{months},$$

where `months` is release date measured relative to 2024-01-01. The baseline $\beta_1 = 0.136$ ($R^2 = 0.761$) implies, under the Densing Law, a monthly accuracy gain of $\beta_2^{\text{Densing}} = \beta_1 \log_{10}(2)/3.5 \approx +0.0117/\text{month}$ ($\approx +14.1$ pp/year).

Result. The fitted coefficient is $\hat{\beta}_2 = -0.0010/\text{month}$ [95% bootstrap CI: -0.0031, +0.0008]. The point estimate is indistinguishable from zero ($p = 0.34$) and rejects the Densing prediction at $p < 10^{-15}$. Adding release date to the scaling regression increases R^2 by only +0.0024. Controls for thinking mode and MoE architecture do not change this conclusion (Appendix F, Table 17): across specifications the time coefficient stays within $\pm 0.002/\text{month}$, while the Densing prediction is never within the confidence interval. Figure 6 partials out $\log_{10}(N)$ and contrasts the flat observed residual-vs-date relationship with the steep Densing-Law prediction line.

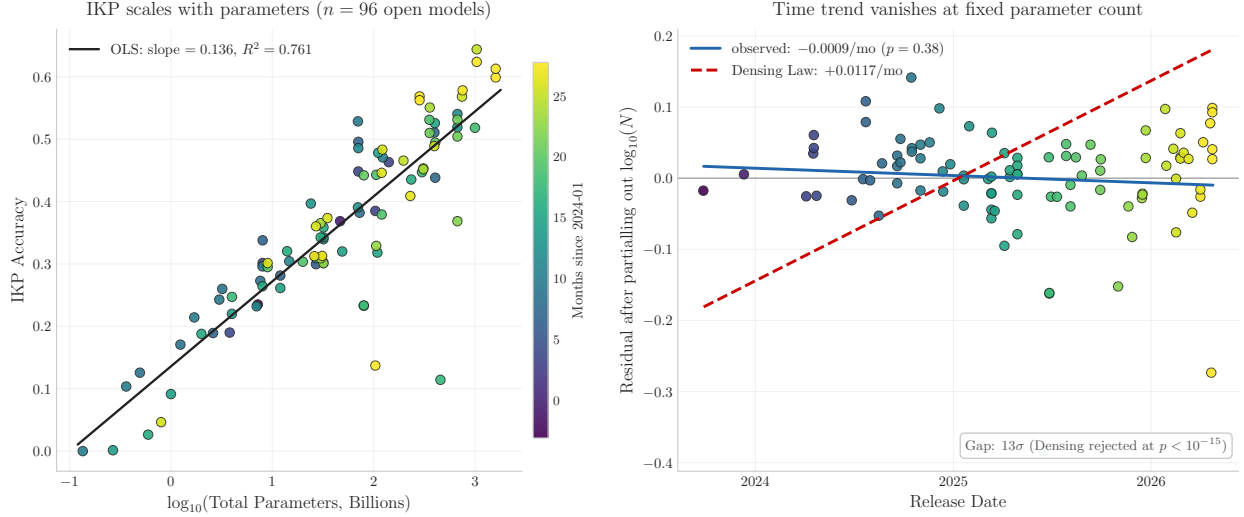


Figure 6: Densing Law falsification on IKP across 96 open-weight models (2023-09 to 2026-04). **Left:** IKP accuracy vs $\log_{10}(\text{params})$; points colored by release month. **Right:** residuals after partialling out $\log_{10}(\text{params})$, vs release date. The observed trend (solid blue, $-0.0010/\text{month}$) is indistinguishable from zero, while the Densing-Law prediction (dashed red, $+0.0117/\text{month}$) would imply ~ 14 pp/year of per-parameter improvement. The gap exceeds $11\times$ the bootstrap confidence half-width.

Interpretation. Capability density has indeed increased on reasoning benchmarks over 2024–2026, but the improvement has come from more efficient use of procedural and linguistic parameters ($N_{\text{proc}}, N_{\text{ling}}$), not from more efficient storage of facts. Smaller recent models know no more than older models of the same size—exactly the prediction of incompressibility (Section 3). The positive marginal R^2 of time alone (Table 17) is fully accounted for by the selection effect that newer releases skew larger.

6.8 Why not just use MMLU? Standard knowledge benchmarks as size proxies

A natural question is whether existing knowledge benchmarks could substitute for IKP. We collected vendor-published official scores from primary sources (model cards, system cards, technical reports, vendor blog posts) for four widely-reported benchmarks: MMLU [Hendrycks et al., 2021], MMLU-Pro [Wang et al., 2024], GPQA Diamond [Rein et al., 2024], and SimpleQA [Wei et al., 2024]. We then fit each benchmark as a parameter-count proxy (score $\sim \log_{10} N$) and as a Densing-style joint fit (score $\sim \log_{10} N + \text{months}$), restricting to the same calibration set used for IKP. Crucially, we re-fit IKP on each matched subset so the comparison is apples-to-apples.

Metric	N	R^2 vs $\log_{10} N$	IKP R^2 (same subset)	Time slope (pp/month)
IKP (full set)	93	0.910	—	−0.06
SimpleQA	10	0.904	0.991	+0.03
MMLU	30	0.705	0.886	+0.58
MMLU-Pro	25	0.689	0.900	+0.82
GPQA Diamond	30	0.520	0.903	+1.99

Table 3: Standard knowledge benchmarks vs IKP as parameter-count proxies. On every matched subset of dated open-weight models, IKP explains substantially more variance in $\log_{10}(N)$ than the standard benchmark does. The right-most column reports the joint fit’s time slope (percentage points per month) at fixed $\log_{10}(N)$: reasoning-heavy benchmarks (GPQA Diamond, MMLU-Pro) drift fastest, exactly as the Densing Law predicts; the pure-factual benchmark (SimpleQA) and IKP drift near zero. Sample sizes differ because vendors do not report all benchmarks for all models, and we exclude scores from with-search or tools-augmented configurations.

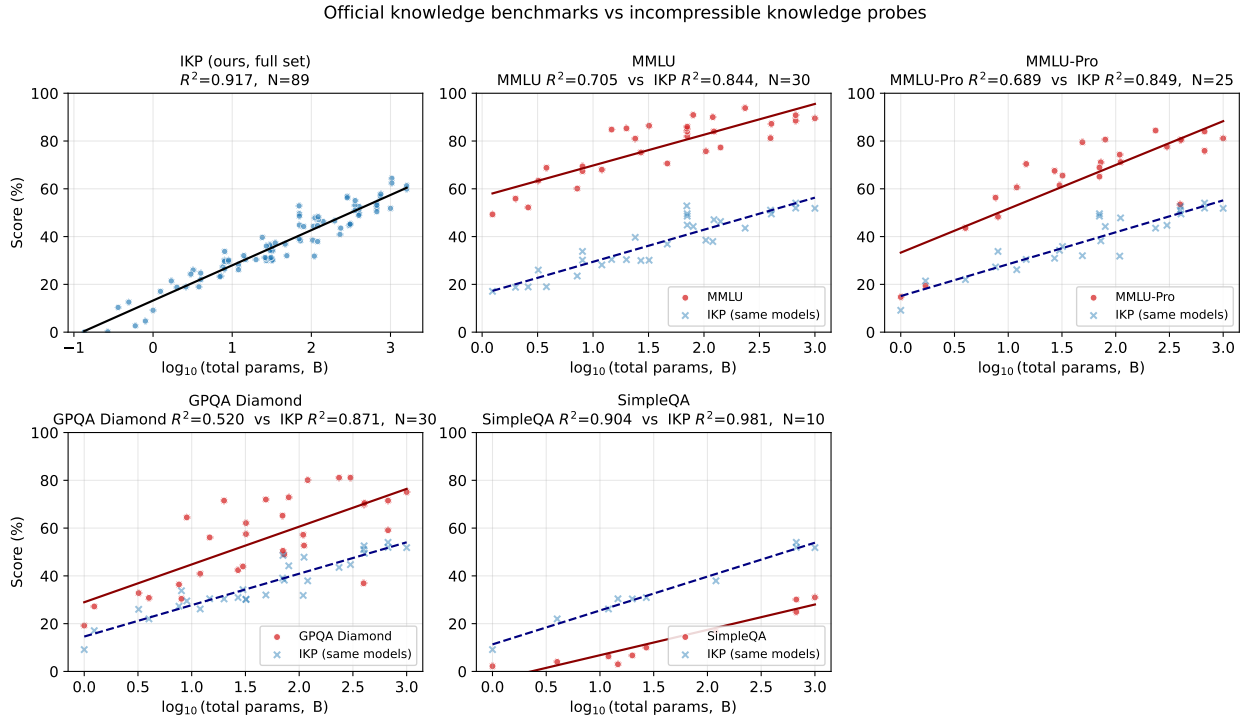


Figure 7: Vendor-reported benchmark scores vs $\log_{10}(\text{total params, } B)$. Top-left: IKP across all 93 calibration models ($R^2 = 0.910$). Other panels: each standard benchmark (red) plotted against the same models with IKP (blue $\times$, dashed line) overlaid for direct comparison. On every matched subset, IKP yields a tighter fit; the gap is largest for GPQA Diamond (reasoning-heavy) and smallest for SimpleQA (factual).

Three findings. (i) On every matched subset IKP wins on R^2 , with the gap largest for the most reasoning-heavy benchmark (GPQA Diamond: 0.52 vs 0.90) and smallest for the only purely-factual benchmark (SimpleQA: 0.90 vs 0.99). (ii) Reasoning benchmarks drift sharply over time at fixed $\log_{10}(N)$: GPQA Diamond gains ≈ 2 pp/month, meaning a 33B model improves by ≈ 24 points across one year of releases without growing. This is exactly the Densing prediction, and it directly

invalidates these benchmarks as parameter proxies. (iii) Factual benchmarks behave like IKP: SimpleQA’s time slope is +0.03 pp/month, statistically indistinguishable from zero, supporting the broader claim that the incompressibility property holds for the factual subspace specifically rather than for “benchmarks” as a category.

Coverage matters. Vendors publish MMLU-Pro, GPQA, and SimpleQA scores predominantly for upper-mid and frontier models. Tiny pretrained models (smollm2-135m, gemma-3-270m) and many older mid-range models do not appear on these benchmark tables at all because they score near random. IKP, evaluated identically on all 89 models from 135M to 1.6T parameters (> 4 orders of magnitude), avoids this sample-selection asymmetry. Even setting aside the Densing-Law time confound, this coverage difference is itself a methodological argument: a parameter-estimation instrument needs to work across the full size range, including the small models that anchor the bottom of the calibration curve.

6.9 Knowledge Fingerprinting: Lineage vs. Retraining

The specific set of rare facts (T5–T6) a model knows constitutes a *knowledge fingerprint* (Figure 8), complementing output-distribution fingerprints [Tsai et al., 2025] and embedded watermarks [Nasery et al., 2025] with a training-free, knowledge-based signal. For each pair of models we compute three metrics on the 400 T5–T6 probes:

- **Jaccard similarity** J on correct-answer sets. Inflated by probes that almost every frontier model knows.
- **Lift** = observed intersection / expected-under-independence. Controls for common knowledge.
- **Hallucination similarity** HSS: among probes where both models are wrong (non-refusal), the fraction on which they produce the *same* normalized wrong answer. HSS is the most diagnostic of the three because independently trained models essentially never converge on an identical wrong rare fact, whereas weight-sharing siblings do so on 30–55% of shared-wrong probes.

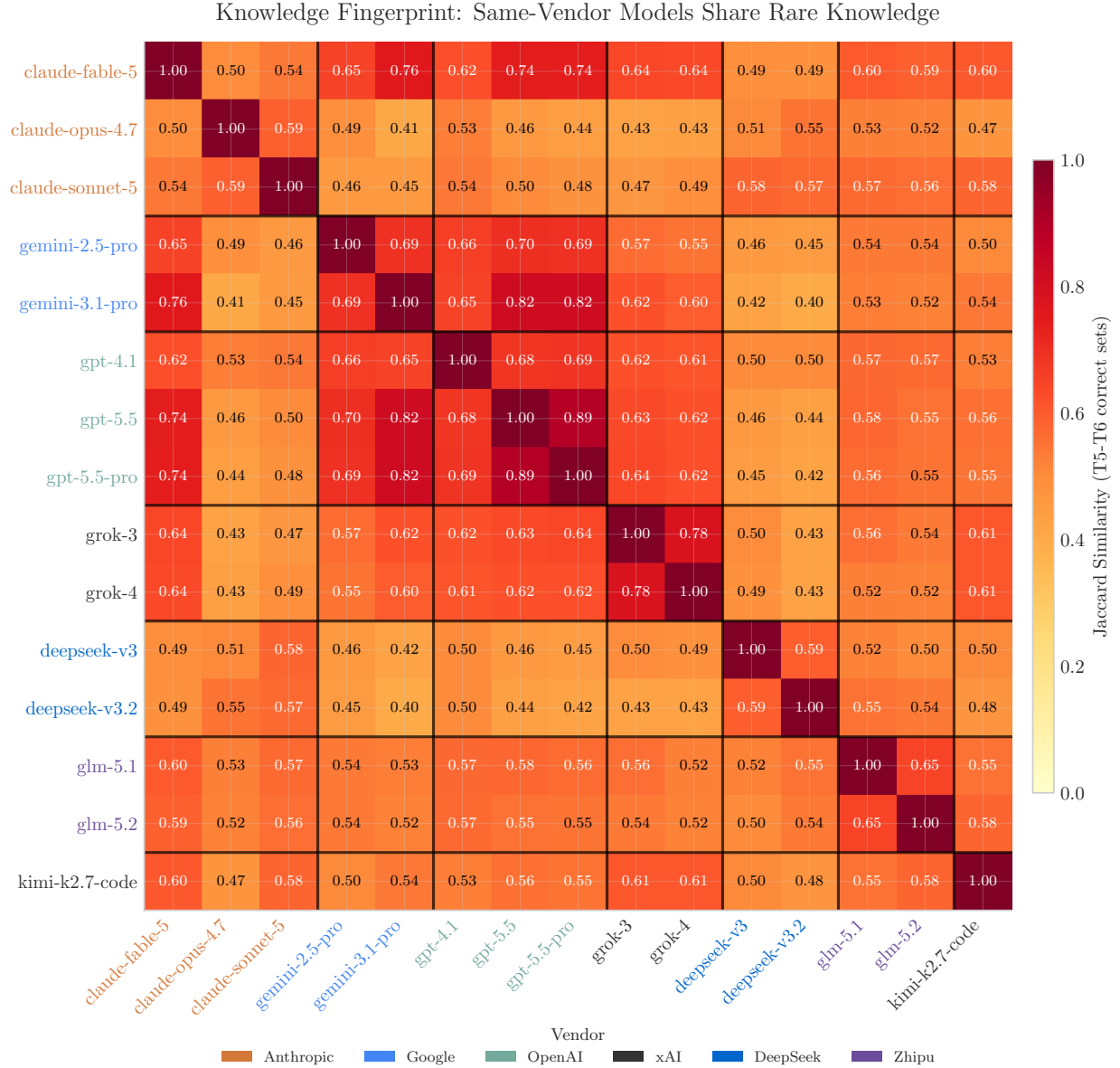


Figure 8: Jaccard similarity on T5–T6 correct-answer sets for 15 frontier models, clustered by vendor. Within-vendor similarity (diagonal blocks) is consistently higher than cross-vendor (off-diagonal). Jaccard alone does not separate shared-base from independent-but-competitive models; see Figure 9 for the sharper HSS view.

Three regimes. Applied pairwise across 201 models, the three metrics cleanly separate into three regimes (Figure 9a).¹ *Shared base* pairs ($\text{HSS} \geq 0.30$, $J \geq 0.60$) represent the same weights served with different inference or light alignment; *lineage* pairs ($0.10 \leq \text{HSS} < 0.30$) are consistent with post-training, continued pretraining, or distillation on top of a shared ancestor; *retrained* pairs ($\text{HSS} < 0.10$ on ≥ 15 joint-wrong probes) fall in the same statistical regime as cross-vendor

¹The regime boundaries are empirical: weight-sharing siblings (e.g., GPT-5 / GPT-5-pro, GPT-5.2 / GPT-5.2-pro, Claude Opus 4.5 / 4.5-think) concentrate at $\text{HSS} \geq 0.30$; cross-vendor pairs with ≥ 10 joint-wrong probes have median $\text{HSS} = 0.05$.

independent pairs even when the models share a version-family label.

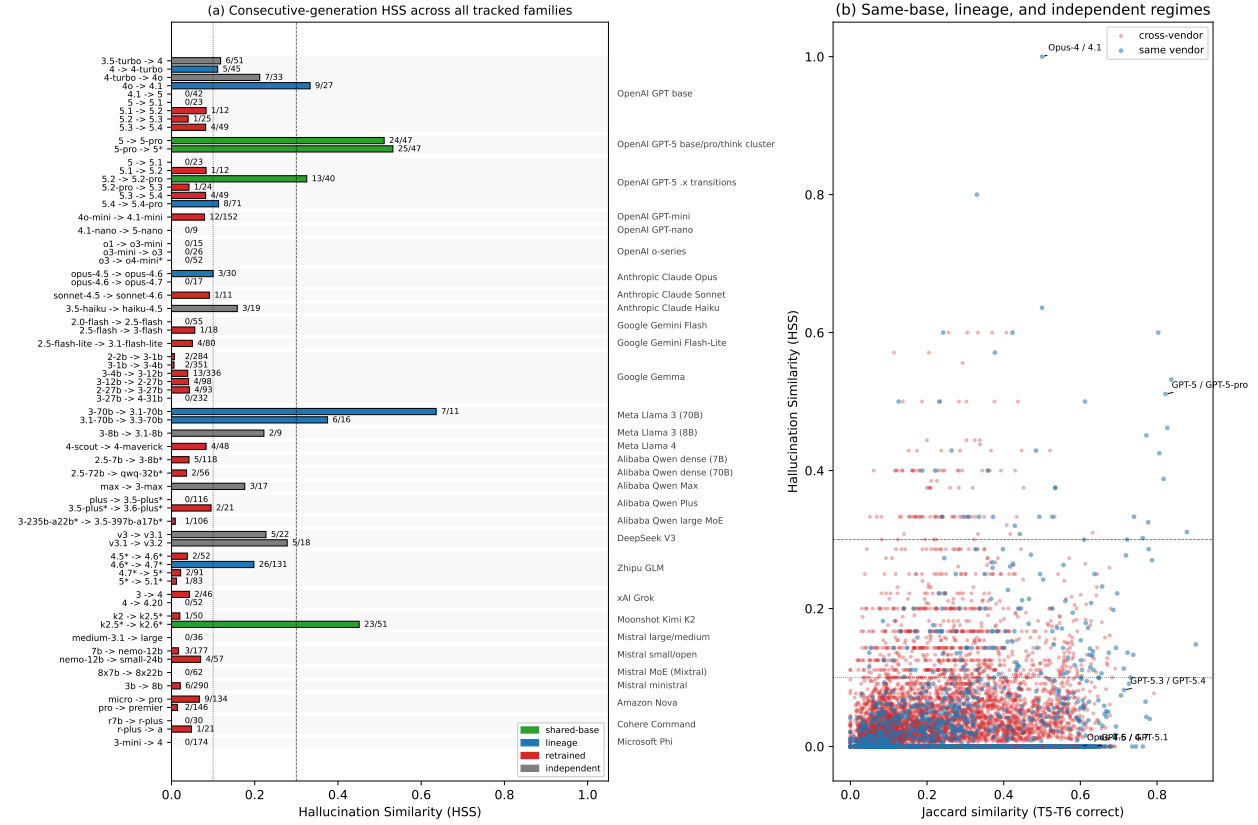


Figure 9: (a) Hallucination-similarity (HSS) for consecutive-generation pairs within each family. Annotation is same_wrong/both_wrong. Dashed line: shared-base threshold (0.30); dotted line: lineage threshold (0.10). GPT-5 through 5.4 transitions all fall in the retrained regime. (b) All pairs with ≥ 5 joint-wrong probes in the (J, HSS) plane. Same-vendor (blue) and cross-vendor (red) pairs largely overlap on J but separate on HSS.

Within-family lineage (selected). Applied to consecutive-generation pairs within each vendor’s family, this test gives a surprisingly varied picture of release practices:

- **OpenAI GPT-5 family.** GPT-5, GPT-5-pro and GPT-5-think form a tight shared-base cluster (HSS = 0.51, 0.53). But every “.x” transition—GPT-5 $\rightarrow$ 5.1, 5.1 $\rightarrow$ 5.2, 5.2-pro $\rightarrow$ 5.3, and 5.3 $\rightarrow$ 5.4—has $\text{HSS} \leq 0.08$ on 12–49 joint-wrong probes. This is the retrained regime. The data supports the folklore that GPT-5.3 and GPT-5.4 are not post-trains of GPT-5 but independent training runs, most likely from scratch or from a significantly changed base. GPT-5.4 vs. GPT-5.4-pro (HSS = 0.11, 71 joint-wrong) is instead a lineage pair, consistent with a shared base differentiated only at the alignment stage.
- **Anthropic Claude Opus.** Opus 4 $\rightarrow$ 4.1 is clear lineage (HSS = 1.00, 7 joint-wrong; also supported by Anthropic’s own description of 4.1 as a post-training update). Opus 4.5 $\rightarrow$ 4.6 is weak lineage (HSS = 0.10, 30 joint-wrong) and Opus 4.6 $\rightarrow$ 4.7 is retrained (HSS = 0.00, 17 joint-wrong). Sonnet shows the same pattern: 4.5 $\rightarrow$ 4.6 is retrained (HSS = 0.09, 11 joint-wrong).
- **Google Gemini.** Every cross-generation Flash and Pro pair is retrained (Flash 2.0 $\rightarrow$ 2.5:

HSS = 0.00, 55 joint-wrong; Flash 2.5 $\rightarrow$ 3: 0.06, 18; Pro 2.5 $\rightarrow$ 3.1: 0.03, 35). Gemini generations appear to be full re-trains, not post-training increments.

- **DeepSeek V3.** V3 $\rightarrow$ V3.1 (HSS = 0.23) and V3.1 $\rightarrow$ V3.2 (0.28) are both in the lineage regime, consistent with incremental continued pretraining on a shared base.
- **Zhipu GLM.** GLM 4.5 $\rightarrow$ 4.6 (retrained, 0.04), 4.6 $\rightarrow$ 4.7 (lineage, 0.20), 4.7 $\rightarrow$ 5 (retrained, 0.02) alternate, consistent with a pattern of minor point releases interleaved with full retraining.

Cross-family outliers. Applying the same test to the $\sim 13\,000$ cross-vendor pairs flags a small number as HSS ≥ 0.20 with ≥ 10 joint-wrong probes—the regime occupied by shared weights within a vendor. The strongest signal (across multiple tests) is Baidu ERNIE 4.5, which achieves HSS = 0.33–0.44 against GPT-4o, Llama-3, Mistral-Large and Qwen-Max simultaneously—a pattern consistent with heavy training on mixed distilled outputs rather than a single teacher. Llama 3.1 70B shows up as an apparent “teacher” in a striking number of pairings (HSS ≥ 0.30 against grok-3, gemini-2.0-flash, qwen3-max, nova-pro, and several OpenAI models), which is most likely an artifact of Llama 3.1 being the most widely used open base for synthetic-data generation in 2024. GPT-5 vs. Grok-4 (HSS = 0.31, 29 joint-wrong) and GPT-5-pro vs. Kimi-K2.6 (HSS = 0.27, 22 joint-wrong) are individually above threshold and merit follow-up, though neither is large enough, at our probe count, to reject an innocent-explanation null with confidence. Full rankings and raw counts are in Appendix H.

6.10 What Determines Whether an LLM Knows Something?

This is the question that motivated the IKP instrument in the first place (Section 1): a researcher or an entity is “inside” a model to the extent that the model reliably produces the right answers to probes about it. Here we turn the instrument on itself and ask what observable properties of a researcher, or of a fact, predict whether a model has absorbed it. We cross-reference IKP recognition rates with external metrics for researcher probes (OpenAlex citations and h-index, $n = 345$) and for factual probes about entities (Wikipedia sitelinks and pageviews, $n = 557$). Both analyses converge on the same underlying quantity: not raw prominence, but *effective mention frequency*—the number of training-corpus documents that state the specific target fact in directly retrievable form.

6.10.1 Researcher Probes

Bibliometric signals explain roughly a third of the variance. Across 345 researcher probes scored by 140 models, log-transformed OpenAlex citations correlate with recognition rate at Spearman $\rho = 0.575$ (Pearson $r = 0.590$; Figure 10); log-transformed h-index correlates at $\rho = 0.561$. Tier assignment is well-predicted by citations (median citations drop from 6,859 at T3 to 325 at T7, Spearman $\rho = -0.51$), confirming that tiers track bibliometric mass at the population level. But at the individual-probe level, citations and h-index together explain only $\sim 35\%$ of variance in recognition. Table 4 makes the shallow slope explicit: recognition grows monotonically with either metric, yet no threshold cleanly separates recognized from unrecognized researchers. Even at h-index ≥ 50 , mean recognition is 0.55 and only 65% are recognized by a majority of models—nearly half the variance sits elsewhere.

High citation is approximately necessary but not sufficient for high recognition.

The asymmetry is visible directly in Figure 10: the upper-left region is essentially empty. No researcher with < 50 citations exceeds 15% recognition, and no researcher with < 500 citations exceeds 75%. The lower-right region, by contrast, is densely populated—many researchers with

10K+ citations sit below 25% recognition. Citation count therefore acts as a probabilistic upper bound on recognition (a low-cited researcher rarely becomes a household name to a frontier model), but does not lift recognition on its own; the gap between the bound and the realized recognition rate is filled by the artifact-and-derivative-content mechanisms detailed in the audit below. The exceptions to the necessary-condition reading—low-cited researchers with high recognition, e.g. Tri Dao (3K, 100%), Eyuboglu (267, 57%), Psaras (318, 69%)—are without exception attached to a named, widely-distributed artifact (FlashAttention, ColBERT-style work, IPFS), confirming that the route to escaping the citation ceiling is artifact attachment, not citation mass alone.

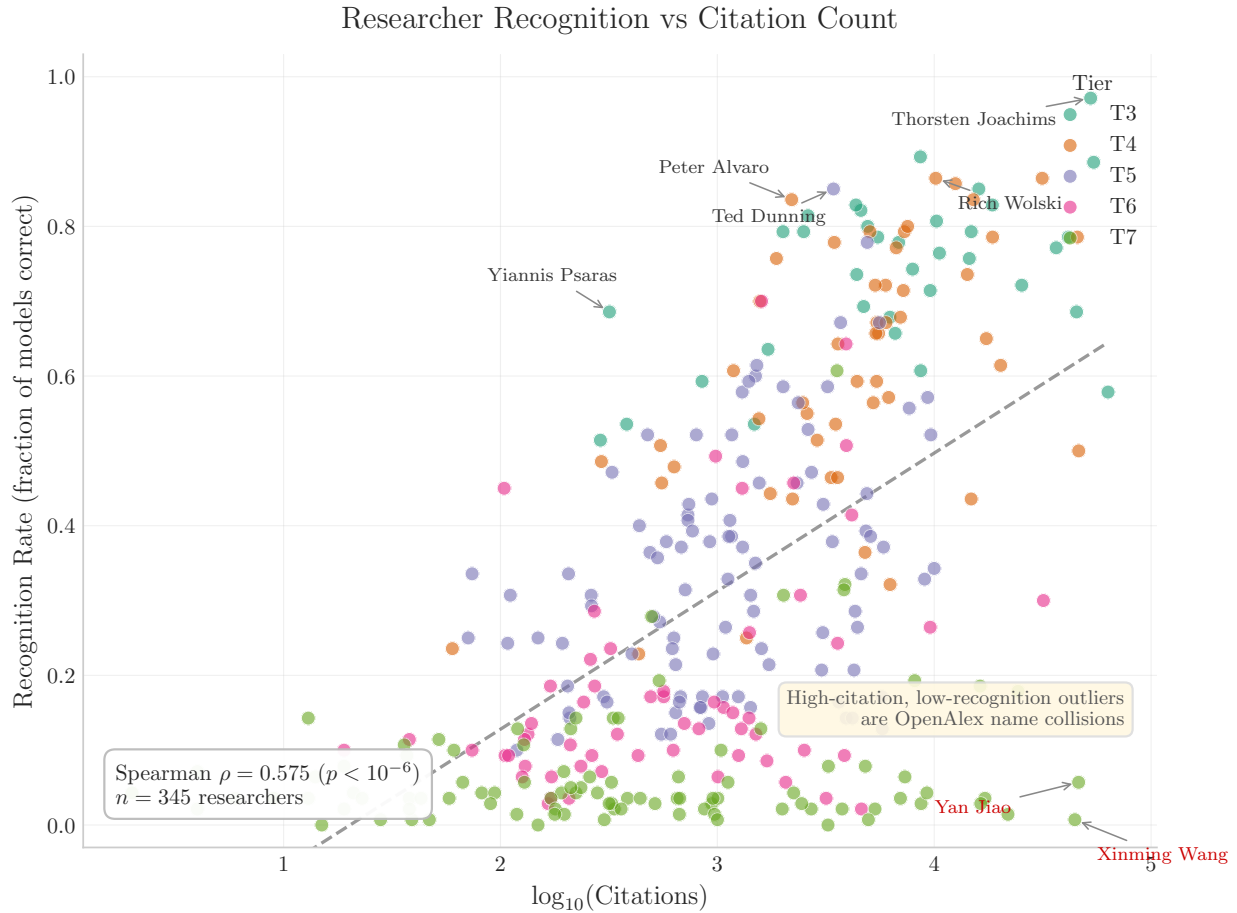


Figure 10: Researcher recognition rate vs $\log(\text{citations})$, colored by tier. The moderate correlation ($\rho = 0.575$) reflects the influence of name uniqueness and subfield-ecosystem amplification beyond raw citation count.

Table 4: Recognition rate by bibliometric bucket ($n = 345$ researchers, 140 models). Recognition grows sub-logarithmically in both citations and h-index, and the top bucket does not saturate—a substantial fraction of even the most-cited researchers are unknown to most models.

Citations				h-index			
Bucket	n	Mean recog.	> 50% recog.	Bucket	n	Mean recog.	> 50% recog.
0–99	36	0.07	0%	0–4	43	0.09	2%
100–499	78	0.16	10%	5–9	61	0.14	7%
500–999	47	0.24	19%	10–19	99	0.28	25%
1,000–4,999	111	0.38	37%	20–29	48	0.40	42%
5,000–9,999	41	0.52	61%	30–49	63	0.51	62%
10,000+	32	0.59	69%	50+	31	0.55	65%

Subfield predicts recognition beyond bibliometric mass. Table 5 breaks recognition down by CS subfield. Information retrieval (IR) and programming languages (PL) researchers are recognized $1.5\text{--}2\times$ more often than computer-architecture researchers at comparable or higher citation levels; HCI and theoretical CS sit near the bottom despite high mean bibliometric density. The subfields at the top are precisely those with the densest derivative content layer—tutorials, blog posts, course materials, library documentation—suggesting that what a model memorizes is not the research itself but the web text that surrounds it.

Table 5: Recognition rate by CS subfield, sorted by mean recognition ($n \geq 6$). IR and PL lead despite moderate bibliometric mass; HCI, theoretical CS and computer architecture lag despite comparable or higher citation averages.

Subfield	n	Avg. citations	Avg. h-index	Mean recog.
Information retrieval	7	10,497	29.6	0.571
Programming languages	43	2,596	19.3	0.437
Database systems	7	3,670	26.7	0.429
Data mining	6	7,039	34.3	0.370
Computer networking	60	7,840	29.8	0.354
Distributed systems	42	2,206	16.1	0.307
Natural language processing	7	1,569	13.6	0.302
Computer security	62	2,597	18.5	0.283
Computer vision	9	10,347	32.8	0.283
Operating systems	38	3,718	20.6	0.277
Embedded systems	11	1,414	14.0	0.277
Human-computer interaction	7	10,051	35.6	0.260
Theoretical CS	11	7,563	23.3	0.204
Computer architecture	32	3,151	19.5	0.179

A controlled audit isolates three mechanisms. To characterize the residual variance, we ran a web-presence audit on a $2 \times 2 \times 2$ matrix of 20 researchers: $\{\text{ML, Systems}\} \times \{\text{high citations, low citations}\} \times \{\text{high recognition, low recognition}\}$ (“ML” = core ML/AI subfields; “Systems” = networking, OS, architecture, security). For each researcher we recorded personal site, Wikipedia page, GitHub-artifact stars, secondary-content presence (tutorials, blog posts, course materials), and name uniqueness. Table 6 reports the cell-level outcomes; three patterns emerge.

Table 6: Controlled $2 \times 2 \times 2$ audit of 20 researchers. Each cell lists sampled researchers with (OpenAlex citations, recognition rate). † marks OpenAlex name-collision errors, where the citation count belongs to a same-named researcher in a different field. The “ML / low-recognition” cell is empty in our sample—no ML researcher falls below 43%.

	ML / AI	Systems (net / OS / arch / sec)
High citation / High recognition	Tri Dao (3K, 100%) Raghunathan (4.7K, 100%) A. Gu (3K, 86%)	Joachims (53K, 97%) Gerla (54K, 89%)
High citation / Low recognition	<i>Empty: ML floor $\geq 43\%$.</i> Mid-recognition analogs: Mitchell (15K, 57%) Goel (3K, 57%), Xie (5K, 57%)	Yan Jiao (46K [†] , 6%) Xinming Wang (45K [†] , 1%) Dan Suciu (24K, 18%)
Low citation / High recognition	M. Chen (215, 57%) Eyuboglu (267, 57%) Dunlap (303, 43%)	Psaras (318, 69%) Micinski (850, 59%) Lysecky (382, 54%)
Low citation / Low recognition	<i>Empty in sample.</i>	Q. Rao (15, 0%)

1. **Named artifacts dominate bibliometric mass.** The “low-citation / high-recognition” cells are populated by researchers attached to widely-adopted artifacts (FlashAttention, IPFS, zy-Books, a free YouTube PL course) whose names travel in derivative documents. Within our full 345-researcher sample, researchers attached to a tool with ≥ 10 K GitHub stars or a dedicated Wikipedia page are recognized at rates $\geq 86\%$ regardless of citation count; researchers without either average 0.34 recognition at matched citation buckets. A single high-visibility artifact is worth more than an order of magnitude in additional citations.
2. **Name uniqueness is multiplicative.** Controlling for citation count, researchers with common East Asian surnames (two-character names and single-initial given names) are recognized at 22.6% in our dataset versus 44.6% for uniquely spelled names—a factor-of-two attenuation attributable to the name alone. The same mechanism explains why Eric Mitchell (DPO lead, ~ 15 K Google Scholar citations) sits at 57%: “Eric Mitchell” is dominated in web indices by a film director and unrelated professionals, so even a named method cannot fully anchor the researcher name.
3. **The ML–Systems floor gap.** The “ML / low-recognition” cell in Table 6 is empty: no ML researcher in our sample falls below 43%, including PhD students at < 300 citations. The equivalent Systems cell is non-empty at arbitrarily low citation counts. The asymmetry is not individual merit; it is subfield ecosystem density—ML labs generate orders of magnitude more Twitter, blog, and podcast content per paper than Systems labs, and this derivative content is what foundation-model training pipelines scrape.

Name-collision data quality caveat. The audit surfaced an under-appreciated artifact in bibliometric studies of LLM recognition: three of the four “Systems / high-citation / low-recognition” cell occupants were OpenAlex disambiguation errors—the citations belonged to same-named researchers in chemistry, atmospheric science, or medicine. Any future work correlating bibliometric metrics with LLM memorization should treat common-name high-citation profiles as provisional

until verified against field-consistent publication lists.

Implication for impact measurement. The mapping from academic output to LLM-internalized knowledge is not monotonic in citation count. It is closer to $\text{citations} \times \text{name uniqueness} \times \text{named-artifact amplification} \times \text{subfield-ecosystem density}$, with each multiplicative factor spanning roughly a 2–5 $\times$ range in the regime we observe. For a working researcher, the marginal effect of one additional widely-used open-source tool with clean name attribution is larger than the marginal effect of one additional well-cited paper: the tool generates derivative documents that each carry the author’s name, while the paper typically carries it only in its own bibliography record and those of its direct citers.

6.10.2 Factual Probes About Entities

For the 557 Wikidata-grounded factual probes—dominated (95%) by founding-year questions about universities, journals, museums, bridges, sports clubs, and places—the picture is sharper and more mechanistic.

Pageviews dominate sitelinks. Sitelink count (how many Wikipedia language editions link to the entity) correlates with recognition at Pearson $r = 0.502$ (log-transformed). For a 78-probe subset with English Wikipedia pageview data, pageviews reach $r = 0.774$ and *completely subsume* sitelinks in joint regression (sitelink coefficient falls to -0.003 when both are included). Sitelinks measure breadth of multilingual coverage; pageviews approximate English-language discourse volume, which is what actually appears in the training corpora.

The name–fact gap widens with prominence. Identity facts (“what is the name of X?”) are always easier than temporal facts (“when was X founded?”), but the gap is a function of entity prominence:

Entity prominence (sitelinks)	n	Name recog.	Year recog.	Gap
1–5 (obscure)	126	0.299	0.259	+0.040
6–15 (moderate)	127	0.614	0.533	+0.081
16+ (prominent)	39	0.728	0.473	+0.255

A famous bridge is mentioned thousands of times, reinforcing its name; its opening year appears in only a small fraction of those mentions. Prominence amplifies name knowledge proportionally but amplifies temporal knowledge only weakly—consistent with the hypothesis that each *specific* fact must clear its own mention-frequency bar rather than being carried along by its entity’s overall visibility. Founding years are also uniquely hallucination-prone: wrong-answer rates reach 37–47% on year probes (versus <1% on capitals), because any plausible 4-digit number is an attractor for a model that does not refuse.

Domain-specific mention multipliers. Controlling for sitelinks, some domains are systematically easier than entity prominence would predict, and others are systematically harder:

Domain	n	Residual (easier +, harder −)
Journal founding years	40	+0.201
Journals (names)	39	+0.215
University founding years	60	+0.117
Universities (names)	36	+0.120
Museum founding years	37	−0.102
Museums (names)	26	−0.118
Sports-club founding years	37	−0.144
Bridge opening years	20	−0.219
Place founding years	44	−0.310

Journal founding years are +0.22 easier than sitelinks predict because every citation and every bibliography entry for an article in that journal implicitly states the journal’s name; publication-year metadata is correspondingly dense. Place founding years are −0.31 harder because a municipality’s founding date is typically buried in a single “History” section of its Wikipedia page, while the municipality is otherwise mentioned in contexts (weather, sports, transit) that do not carry the year. The *structure of web discourse around each fact type* determines mention frequency far more than the prominence of the underlying entity.

A documentation sweet spot. Entities founded in 1900–1950 are best-known overall (mean recognition 0.40), outperforming both older entities (<1800: 0.25) that have fewer sitelinks and digital records, and younger entities (2000+: 0.26) that have not yet accumulated comprehensive historical coverage. This is consistent with a training-corpus curve that peaks where sustained institutional documentation (Wikipedia, newspaper archives, bibliographic databases) and historical notability both apply.

6.10.3 Synthesis: Effective Mention Frequency

Researcher probes and factual probes are two paths to the same conclusion. LLM knowledge is determined not by abstract prominence but by *effective mention frequency*: the number of training-corpus documents that state the specific target fact in retrievable form, attributed to the specific name or entity being queried. Citation count, sitelink count, and h-index are partial proxies; pageviews, practitioner adoption, named-artifact count, and the density of derivative-document mentions are better ones.

This has two practical corollaries. For *researchers*, the fraction of one’s work that foundational models have internalized is not a rank-ordered function of citations. It is closer to a function of how much downstream text—tutorials, course materials, model cards, blog posts, GitHub documentation, news coverage—mentions the work by name. The fastest route to being memorized by future frontier models is to produce named artifacts whose names travel. For *facts about entities*, the query-design lesson is that any structured, bibliographically-enforced metadata (publication years, DOIs, institutional affiliations encoded in citation strings) is far better memorized than the same metadata when it lives only on the entity’s own page. IKP difficulty tiers implicitly exploit this: the hard-tier probes are precisely those whose specific facts clear the mention-frequency bar only for the largest models.

6.11 Key Empirical Findings

The 188-model evaluation yields a handful of empirical regularities that we state crisply for future reference:

- **Knowledge scales log-linearly.** Aggregate accuracy grows by ~ 15.9 percentage points per $10\times$ increase in parameters, across four orders of magnitude (135M–1,600B).
- **Total parameters, not active, predict MoE knowledge.** $R^2 = 0.67$ versus 0.41. Factual storage is distributed across experts, not localized to those activated per token.
- **Thinking mode is a flat ~ 2 pp bonus.** The benefit peaks at T3–T4 and vanishes at T7: chain-of-thought aids retrieval, not storage.
- **T7 is a ceiling.** Only two models in the evaluation score above 1% accuracy on T7 (Jamba-large at 2.8%, Grok-4 at 1.0%); every frontier proprietary model lands at 0%. A knowledge frontier exists beyond the current state of the art.
- **Safety tuning hides measurable knowledge.** Within the Claude Sonnet line, Sonnet 4 reads -6.7 pp lower than Claude 3.7 Sonnet on IKP, while its T5 refusal rate jumps from 54% to 88%. The implied capacity gap is largely an artifact of refusal policy rather than parameter count.
- **Hallucination rate is a vendor fingerprint.** Google’s smaller Gemma models hallucinate on 89–97% of unknown probes at T5–T7; Anthropic Claude on 3–28%. The hallucination penalty by construction rewards conservative vendors.
- **Hallucination similarity exposes lineage.** Comparing models on *which wrong answer they give* on rare facts cleanly separates shared-base pairs ($\text{HSS} \geq 0.30$) from independent retrains ($\text{HSS} < 0.10$), even when raw Jaccard overlap is high. Several nominally “.x” point releases (GPT-5 $\rightarrow$ 5.1, Opus 4.6 $\rightarrow$ 4.7, every cross-generation Gemini pair) fall in the retrained regime, while others (Opus 4 $\rightarrow$ 4.1, DeepSeek V3 $\rightarrow$ V3.1) are clear lineages.

7 Discussion

7.1 The Silent Tax of Safety Tuning

Heavily safety-tuned models [Ouyang et al., 2022, Christiano et al., 2017, Bai et al., 2022] refuse probes they demonstrably know, producing systematic capacity underestimates that compound across generations. Within the Claude Sonnet line, Sonnet 4 scores 48.2% while Claude 3.7 Sonnet—almost certainly a smaller or comparably-sized model—scores 54.9%. On T5, Sonnet 4 issues explicit “I don’t know” refusals on 175/200 probes (88%); Sonnet 3.7 refuses only 108/200 (54%). If Sonnet 4’s underlying knowledge matches or exceeds Sonnet 3.7’s, the -6.7 pp score gap is largely an artifact of refusal policy rather than capacity.

This has two implications beyond IKP methodology. First, “the model knows but will not say” is empirically measurable, not just philosophically uncomfortable: the same prompts that elicit correct answers in earlier generations produce refusals in later ones, and the magnitude is tens of percentage points of “refused but known” capacity. Second, under our no-penalty scoring ($\lambda = 0$: a refusal and a confidently wrong answer both score 0), raw accuracy *under-counts* the latent knowledge of refusal-heavy models, since a “refused but known” fact is scored identically to genuine ignorance. IKP estimates for heavily safety-tuned proprietary models should therefore be read as lower bounds; a mild penalty (Appendix B) raises them, so the true capacity lies at or above the no-penalty prediction.

7.2 Hallucination Rate as a Vendor Signature

When models encounter probes beyond their knowledge (T5–T7), the ratio of wrong answers to non-correct responses (wrong / (wrong + refusal)) is a characteristic vendor fingerprint. Google’s smaller Gemma models hallucinate at 89–97%—they almost never refuse—while larger Gemma and Gemini frontier models span 10–90% (Gemini 3.1 Pro as low as 10%, Gemini 3 Flash at 35%). Anthropic Claude averages 11%, with thinking variants as low as 3% and non-thinking frontier reaching 28%. The most striking within-vendor variation appears at OpenAI: GPT-4.1 and GPT-4.1-mini hallucinate at 53–72%, while GPT-5 nano/mini models drop to 3–4%—a generational change in safety calibration that, by construction, lifts GPT-5-class IKP scores at fixed parametric capacity (Section 7.1).

7.3 Knowledge Regression Between Generations

Multiple model families show newer models knowing *less* than their predecessors on IKP. Two mechanisms account for this: (1) RLHF conservatism [Ouyang et al., 2022] increasing between generations, so the newer model likely knows the facts but refuses to state them under the IKP prompt (Section 7.1); (2) genuine knowledge transfer loss when a smaller student replaces a larger predecessor in the same naming family—GPT-3.5 Turbo [Brown et al., 2020] outperforms GPT-4o-mini on T3/T4 by 10+ pp. Later generations partially reverse these regressions (Claude 4.5/4.6 recover, DeepSeek v3.2 recovers from v3.1’s drop).

7.4 Robustness to Benchmark Saturation

A common pattern in LLM evaluation is that any benchmark, once influential, eventually saturates: MMLU [Hendrycks et al., 2021], GSM8K [Cobbe et al., 2021], SimpleQA [Wei et al., 2024], and most recently Humanity’s Last Exam [Phan et al., 2025] are all approaching ceiling on frontier models, in part because the underlying tasks are compressible—once a model has internalized modular arithmetic or the scientific-reasoning template, it solves every instance of that task—and in part because high-profile benchmarks inevitably enter the training pipeline. IKP is structurally resistant to both failure modes. The probes measure stored facts, not procedures, so a model that “saturates” a tier has by construction memorized the long-tail facts that tier was sampling—which is precisely the quantity IKP exists to estimate. Direct contamination of the released probe file is a separate concern, but the pipeline (Section 4) is itself the benchmark: Wikidata sitelink-stratified sampling, DBLP citation-stratified sampling, and tier calibration against the landmark ladder regenerate a fresh probe set from disjoint entities in a few hours of compute.

7.5 The T7 Cliff: A Hard Ceiling on Parametric Storage

The T7 cliff is universal across 201 models. Every frontier model we evaluated—GPT-5 Pro, Claude Opus 4.7, Gemini 3.1 Pro, DeepSeek V4 Pro (1.6T actual parameters)—scores 0.0% accuracy on T7, and 0–24% on T6. This is not a threshold scaling is gradually approaching; it is a cliff at which all models land simultaneously, regardless of size or training budget. The probes at this tier are not exotic edge cases: they are specific historical facts with unambiguous answers, the researcher/artifact layer of the scientific record, and long-tail entities drawn from Wikidata.

Shannon entropy explains why the cliff is structural. Storing a fact requires bits proportional to its information content, and transformer feed-forward layers store ~ 2 –4 bits of factual knowledge per parameter [Allen-Zhu and Li, 2025, Morris et al., 2025]. The long-tail entropy of globally-useful knowledge already exceeds what frontier models absorb, and grows faster than any

foreseeable scaling trajectory can match. Post-training—vendor-specific RLHF, distillation across model sizes, training-data curation—accounts for at most a $\sim 3\times$ per-parameter capacity multiplier and only transfers facts the larger teacher already holds; it cannot manufacture knowledge of long-tail facts no model has absorbed.

7.6 Limitations

Training-data and post-training variance. The calibration assumes approximately similar training signals across models. Vendor-specific choices in pretraining-data curation, RLHF refusal calibration, and post-training fine-tuning produce within-family scatter that the calibration absorbs as residual rather than slope. Heavily safety-tuned proprietary models (e.g., Anthropic’s Haiku line, the GPT-5 nano/mini variants) systematically read smaller than their underlying weights because the metric credits committed CORRECT verdicts and discounts confidently-WRONG ones; a model that refuses on rare facts therefore loses scoring opportunities on probes it could in principle answer. IKP estimates for such models should be read as lower bounds on raw parameter count.

Prediction interval width. The 90% prediction interval factor is $\sim 3.20\times$ in either direction, with 72% of LOO folds within $2\times$ and 86% within $3\times$ (Section 6.2). This width is inherent to the single-variable approach and comparable to inference-economics estimates [Epoch AI, 2024]. Combining IKP with inference-economics constraints could narrow the interval.

Calibration sparsity above 1T parameters. The largest open-weight calibration anchors are DeepSeek V4 Pro (1,600B) and Kimi K2.5/K2.6 ($\sim 1,000$ B). With only two calibration points above 1T, the calibration curve is effectively extrapolated beyond this range, and the slope at the high end is determined by very few data points. Consequently, parameter estimates for proprietary frontier models whose effective capacity falls above ~ 1 T should be read with wider uncertainty than the global $3.20\times$ prediction interval factor suggests. We expect this limitation to be substantially mitigated as additional open-weight models above 1T parameters are released.

Retrieval augmentation confound. A model using retrieval augmentation [Lewis et al., 2020] could achieve high scores without parametric storage. T7’s near-zero scores across all models suggest RAG is not currently deployed for factual queries of this type.

Landmark circularity. All six landmark models have scores inflated at their defining tier boundary by construction, since probe assignment is conditioned on landmark correctness. The effect is strongest for L6 (Gemini 3.1 Pro), whose 90% T6 accuracy is an artifact. The five open-weight landmarks (L1–L5) also participate in the calibration set, potentially biasing the regression. The LOO-CV analysis (Section 6.2) partially addresses this: when each landmark is held out, its prediction uses a regression fit without it, and the resulting LOO-CV $R^2 = 0.907$ confirms the fit is not driven by landmarks alone. Nevertheless, *landmark models should be excluded from estimation targets*.

Non-deterministic API responses. We query all models at temperature = 0, but many API providers do not guarantee deterministic outputs at this setting (due to batching, quantization, or speculative decoding). We did not perform repeated evaluations to quantify this noise source, though the large probe count (1,400) mitigates its effect on aggregate scores.

Probe contamination. If the probe set leaks into training data, accuracy inflates. The probe set should remain private; we release only the methodology and evaluation toolkit. This creates a tension with reproducibility: independent researchers cannot replicate the calibration without the probes. We mitigate this by describing the probe generation procedure in sufficient detail to construct equivalent probe sets, and by releasing the scoring and estimation code.

Probe allocation. The uniform allocation of 200 probes per tier is suboptimal: T1–T2 saturate early and T7 provides near-zero signal, while T3–T5 carry most of the discriminative power. Future

versions could allocate more probes to the informative tiers.

T7 probe quality. Nine T7 probes have $> 25\%$ correct rate across 201 models, indicating they are miscategorized and should be reassigned to T5 or T6 in future versions. The most extreme case (Ivan Beschastnikh, 64.0% correct) is clearly not T7-difficulty knowledge.

7.7 Wikidata Long-Tail Data Quality

After an initial evaluation pass, we audited every T5–T7 Wikidata-sourced probe (557 probes) for ground-truth quality, web-verifiable correctness, and question-form ambiguity. The audit revealed three classes of issues that motivated the released probe set: (i) Wikidata-itself-wrong facts (e.g., Makran Medical College’s founding year listed as 2015 but Wikipedia says 2017; “L’Homme qui marche I” attributed to the bronze foundry Susse Frères rather than Giacometti; Roku still listed as headquartered in Los Gatos despite a 2019 move to San Jose); (ii) probe-construction ambiguity, where Wikidata’s specific entity is correct but the question is unanswerable without disambiguation (“In what year was Putnam founded?”—which Putnam?); and (iii) politically contested attributions where Wikidata picks one answer that reasonable models will dispute (Cape Plaka in Crimea, Loaita Cay in the Spratlys).

Per-fact-type Wikidata reliability at the long tail (verified empirically across 10 sourcing rounds and ~ 120 web cross-checks):

Fact type (Wikidata property)	Web pass rate
P403 river mouth	100%
P57 director (film)	67–100%
P17 country (cape/island/lake/mountain)	75–100%
P571 inception (founding year)	$\sim 95\%$
P112 founder	75% (sometimes returns the founding <i>organization</i> not person)
P170 creator (sculpture)	$\sim 70\%$ (often returns the bronze <i>foundry</i> not sculptor)
P58 screenwriter	50% (conflates story with screenplay credit)
P159 headquarters	25% (frequently stale: Roku, Vista Outdoor)
P170 creator (painting)	$\sim 70\%$ on attribution, but $\sim 100\%$ title-collision at obscurity (“Madonna and Child”)
P138 named after	mixed (self-referential errors observed)

The audit applied 119 changes to the probe set: 4 fact corrections, 64 question rewrites adding country/admin grounding, and 51 drop-and-replace pairs sourced from new diverse fact types. T6/T7 Wikidata-source fact-type diversity went from a 100% founding-year monoculture before the audit to 16 fact types after (river mouths, film directors, capes, sculptors, founders, mountains, lakes, painters, screenwriters, etc.). *All probes with corrected answers were re-calibrated through the landmark ladder*; the empirical observation that all 5 corrected probes shifted from T7 down to T5 after correction (because models that previously got the “wrong” answer were now judged correct) confirms that gold-truth correctness directly determines tier assignment and that any answer-field edit invalidates the prior calibration. Full audit telemetry, the failure-mode taxonomy, and the changeset are in Appendix D.

7.8 Open Questions

Our results suggest a set of falsifiable questions that incompressibility-grounded probes can answer as more models and more ground-truth anchors become available:

1. **When does T7 fall?** No current model scores above 0.5% on T7 after penalty. If factual capacity continues to scale log-linearly, there is an approximate parameter threshold at which T7 begins to discriminate. Observing the first T7-positive model would empirically validate, or refute, the incompressibility argument and locate the next plateau.
2. **How much within-family scatter is post-training rather than parameter count?** Proprietary “Flash”-class variants typically read close to their Pro-class siblings on IKP despite presumed parameter-count differences. Extending the analysis to additional same-vendor pairs with disclosed sizes would partition the scatter into pretraining-data, RLHF, and architecture contributions.
3. **Can RLHF-hidden knowledge be recovered?** The “Opus 4.1 knows but refuses” pattern implies an alignment tax measurable in percentage points of factual recall. Prompting strategies, deprompting, or activation-steering methods that recover refused-but-known answers would place an upper bound on how much capacity safety tuning obscures.
4. **Do knowledge fingerprints persist through continued pretraining?** Our fingerprints survive fine-tuning and distillation. If they also survive substantial continued pretraining on fresh data, they become a practical training-free provenance tool for open-weight licensing enforcement.

8 Conclusion

We began from the observation that frontier language models have quietly absorbed a large fraction of the open expert discourse of the past two decades—individual researchers, their named artifacts, founding years of institutions, and thousands of facts whose “incompressible” nature makes them impossible to reconstruct from language alone. Incompressible Knowledge Probes turn this observation into a measurement instrument: a principled method for estimating LLM effective knowledge capacity from black-box API access, and for characterizing what kinds of people, artifacts, and facts a model has actually internalized. The approach achieves $R^2 = 0.910$ across 93 open models from 19 vendors spanning four orders of magnitude (135M to 1,600B), under no-penalty scoring and validated by leave-one-out cross-validation (median fold 1.48 $\times$; 72% of models predicted within 2 $\times$, 86% within 3 $\times$). We demonstrate that total parameters—not active parameters—predict MoE knowledge capacity ($R^2 = 0.67$ vs 0.41), provide effective-capacity estimates for 97 proprietary models, and show that within-family scatter (vendor-specific RLHF, training-data curation, post-training format) is sufficient to explain the distance between same-vendor variants without invoking a separate scaling regime.

The broader claim is conceptual. The widely-reported saturation of reasoning benchmarks has been read as evidence that scaling is winding down. Our results suggest this reading confuses two distinct resources: procedural capability, which compresses under the Densing Law, and factual storage, which does not compress under pretraining. Measuring only the incompressible component, we find that parameter scaling continues across three orders of magnitude with no detectable time effect at fixed parameter count. How much of any given researcher’s contributions will have flowed into the next generation of frontier models is now an empirical question with a measurement instrument.

We release the IKP evaluation toolkit, probe set, raw per-model responses, fingerprinting scripts, and all data needed to reproduce every figure and table in this paper. Code: <https://github.com/19PINE-AI/ikp>. Interactive companion site (calibration curves, per-model knowledge profiles, per-probe responses, fingerprint heatmaps, densing-law analysis): <https://01.me/research/ikp>.

Acknowledgements

The author thanks Jiyan He and Zihan Zheng for early discussions and observations that seeded this work, including the inspiration drawn from three years of factual probes against newly released models—most notably the USTC Hackergame stress test that opens this paper, where the same prompt asked at each new release traced a single fact arriving in parameters over time. The framing of factual capacity as the incompressible component of model knowledge owes much to those conversations.

We thank Sturb and LawrenceC, whose independent sanity-check of an earlier version of this work [Sturb and LawrenceC, 2026] identified a discrepancy between the described and implemented per-tier scoring and the sensitivity of frontier estimates to the hallucination penalty. This revision responds directly to those points: it adopts no-penalty scoring ($\lambda = 0$), which removes both the penalty and the flooring degree of freedom, reports the full $\lambda \times$ flooring ablation and prediction bands, and cleans the probe set of name-collision and label-ambiguity defects (Appendix B).

This paper was produced using Pine Copilot’s voice-directed *whisper coding* workflow [Pine AI, 2026], in which the author specifies, discusses, and reviews the work by voice while a coding agent—Claude Code with Claude Opus 4.8—carries out the planning, coding, experiments, and paper writing.

References

- Zeyuan Allen-Zhu and Yuanzhi Li. Physics of language models: Part 3.3, knowledge capacity scaling laws. *International Conference on Learning Representations*, 2025.
- Sanket Badhe, Deep Shah, and Nehal Kathrotia. Long-tail knowledge in large language models: Taxonomy, mechanisms, interventions and implications. *arXiv preprint arXiv:2602.16201*, 2026.
- Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, et al. Constitutional AI: Harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022.
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D. Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901, 2020.
- Dingfan Cai et al. Model substitution auditing for language model apis. 2025.
- Nicholas Carlini, Florian Tramèr, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Úlfar Erlingsson, Alina Oprea, and Colin Raffel. Extracting training data from large language models. In *30th USENIX Security Symposium*, pages 2633–2650, 2021.
- Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramèr, and Chiyuan Zhang. Quantifying memorization across neural language models. In *International Conference on Learning Representations*, 2023.
- Hoyeon Chang, Jinho Park, Seonghyeon Ye, et al. How do large language models acquire factual knowledge during pretraining? *Neural Information Processing Systems*, 2024.

- Howard Chen, Jiayi Geng, Adithya Bhaskar, Dan Friedman, and Danqi Chen. Continual memorization of factoids in language models. *Neural Information Processing Systems*, 2025.
- Paul F. Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Aidan Clark, Diego de Las Casas, Aurelia Guy, Arthur Mensch, Michela Paganini, Jordan Hoffmann, Bogdan Damoc, Blake A. Hechtman, Trevor Cai, Sebastian Borgeaud, et al. Unified scaling laws for routed language models. In *International Conference on Machine Learning*, 2022.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems, 2021.
- Damai Dai, Li Dong, Yaru Hao, Zhifang Sui, Baobao Chang, and Furu Wei. Knowledge neurons in pretrained transformers. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, pages 8493–8502, 2022.
- DeepSeek-AI. DeepSeek-V2: A strong, economical, and efficient mixture-of-experts language model. *arXiv preprint arXiv:2405.04434*, 2024.
- DeepSeek-AI. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Epoch AI. Estimating training compute of frontier ai models. 2024. <https://epoch.ai/>.
- William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022.
- Irena Gao, Percy Liang, and Carlos Guestrin. Model equality testing: Which model is this API serving? *International Conference on Learning Representations*, 2025.
- Gemma Team, Thomas Mesnard, Cassidy Hardin, Robert Dadashi, Surya Bhupatiraju, Shreya Pathak, Laurent Sifre, Morgane Rivière, Mihir Sanjay Kale, Juliette Love, et al. Gemma: Open models based on gemini research and technology. *arXiv preprint arXiv:2403.08295*, 2024.
- Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. Transformer feed-forward layers are key-value memories. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 5484–5495, 2021.
- Dirk Groeneveld, Iz Beltagy, Pete Walsh, Akshita Bhagia, Rodney Kinney, Oyvind Tafjord, Ananya Harsh Jha, Hamish Ivison, Ian Magnusson, Yizhong Wang, et al. OLMo: Accelerating the science of language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *International Conference on Learning Representations*, 2021.

- Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- Jiaxi Hong et al. The rise of parameter specialization for knowledge storage in large language models. *Neural Information Processing Systems*, 2025.
- Chaojun Huang et al. The densing law of llms. *Nature Machine Intelligence*, 2025.
- Albert Q. Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de Las Casas, Emma Bou Hanna, Florian Bressand, et al. Mixtral of experts. *arXiv preprint arXiv:2401.04088*, 2024.
- Mandar Joshi, Eunsol Choi, Daniel Weld, and Luke Zettlemoyer. TriviaQA: A large scale distantly supervised challenge dataset for reading comprehension. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics*, pages 1601–1611, 2017.
- Nikhil Kandpal, Haikang Deng, Adam Roberts, Eric Wallace, and Colin Raffel. Large language models struggle to learn long-tail knowledge. *International Conference on Machine Learning*, 2023.
- Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.
- Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, et al. Natural questions: A benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:453–466, 2019.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020.
- Pingzhi Li, Morris Yu-Chao Huang, Zhen Tan, Qingquan Song, Jie Peng, Kai Zou, Yu Cheng, Kaidi Xu, and Tianlong Chen. Leave it to the experts: Detecting knowledge distillation via MoE expert signatures. *arXiv preprint arXiv:2510.16968*, 2025.
- Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, et al. Holistic evaluation of language models. *Transactions on Machine Learning Research*, 2023.
- Xingyu Lu, Xiaonan Li, Qinyuan Cheng, Kai Ding, Xuanjing Huang, and Xipeng Qiu. Scaling laws for fact memorization of large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, 2024.
- Jan Ludziejewski, Maciej Pióro, Jakub Krajewski, Maciej Stefaniak, Michał Krutul, Jan Małaśnicki, Marek Cygan, Piotr Sankowski, Kamil Adamczewski, Piotr Miłoś, and Sebastian Jaszczur. Joint MoE scaling laws: Mixture of experts can be memory efficient. *International Conference on Machine Learning*, 2025.

- Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 9802–9822, 2023.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- John X. Morris, Chawin Sitawarin, Chuan Guo, Narine Kokhlikyan, G. Edward Suh, Alexander M. Rush, Kamalika Chaudhuri, and Saeed Mahloujifar. How much do language models memorize? *arXiv preprint arXiv:2505.24832*, 2025.
- Anshul Nasery, Jonathan Hayase, Creston Brooks, Peiyao Sheng, Himanshu Tyagi, Pramod Viswanath, and Sewoong Oh. Scalable fingerprinting of large language models. *arXiv preprint arXiv:2502.07760*, 2025.
- OpenAI. GPT-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- Zhixuan Pan, Shaowen Wang, and Jian Li. Understanding LLM behaviors via compression: Data generation, knowledge acquisition and scaling laws. *Neural Information Processing Systems*, 2025.
- Dario Pasquini, Ilias Tsingenopoulos, and Mauro Rossi. Llmmap: Fingerprinting for large language models. *USENIX Security Symposium*, 2025.
- Fabio Petroni, Tim Rocktäschel, Sebastian Riedel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, and Alexander H. Miller. Language models as knowledge bases? In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, pages 2463–2473, 2019.
- Long Phan, Alice Gatti, Ziwen Han, Nathaniel Li, Josephina Hu, Hugh Zhang, et al. Humanity’s last exam, 2025.
- Steven T. Piantadosi. Zipf’s word frequency law in natural language: A critical review and future directions. *Psychonomic Bulletin & Review*, 21(5):1112–1130, 2014.
- Pine AI. Pine AI: The most natural human-computer interface is your voice. Blog post, 2026. URL <https://www.19pine.ai/blog/pine-ai-the-most-natural-human-computer-interface-is-your-voice>. Accessed 2026-06-28.
- David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. GPQA: A graduate-level Google-proof Q&A benchmark. In *Conference on Language Modeling*, 2024.
- Adam Roberts, Colin Raffel, and Noam Shazeer. How much knowledge can you pack into the parameters of a language model? In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 5418–5426, 2020.

- Claude E. Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27(3):379–423, 1948.
- Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc V. Le, Geoffrey E. Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations*, 2017.
- Qin Shi, Amber Yijia Zheng, Qifan Song, and Raymond A. Yeh. Knowledge distillation detection for open-weights models. *Neural Information Processing Systems*, 2025.
- Sturb and LawrenceC. Sanity-checking incompressible knowledge probes. Less-Wrong, 2026. URL <https://www.lesswrong.com/posts/veFMEzDDyWaer2Sms/sanity-checking-incompressible-knowledge-probes>. Accessed 2026-07-05.
- Kushal Tirumala, Aram Markosyan, Luke Zettlemoyer, and Armen Aghajanyan. Memorization without overfitting: Analyzing the training dynamics of large language models. *Neural Information Processing Systems*, 2022.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. LLaMA: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023a.
- Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023b.
- Yun-Yun Tsai, Chuan Guo, Junfeng Yang, and Laurens van der Maaten. RoFL: Robust fingerprinting of language models. *arXiv preprint arXiv:2505.12682*, 2025.
- Denny Vrandečić and Markus Krötzsch. Wikidata: A free collaborative knowledgebase. *Communications of the ACM*, 57(10):78–85, 2014.
- Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhui Chen. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. In *Advances in Neural Information Processing Systems*, 2024.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, pages 24824–24837, 2022.
- Jason Wei, Nguyen Karina, Hyung Won Chung, Yunxin Joy Jiao, Spencer Papay, Amelia Glaese, John Schulman, and William Fedus. Measuring short-form factuality in large language models. OpenAI, 2024. <https://openai.com/index/introducing-simpleqa/>.
- An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, et al. Qwen2 technical report. *arXiv preprint arXiv:2407.10671*, 2024.
- Yongqi Zhang et al. The law of knowledge overshadowing in large language models. *Association for Computational Linguistics*, 2025.

Guoliang Zhao, Yuhan Fu, Shuaipeng Li, Xingwu Sun, Ruobing Xie, An Wang, Weidong Han, Zhen Yang, Weixuan Sun, Yudong Zhang, Cheng-zhong Xu, Di Wang, and Jie Jiang. Towards a comprehensive scaling law of mixture-of-experts. *arXiv preprint arXiv:2509.23678*, 2025.

George Kingsley Zipf. *Human Behavior and the Principle of Least Effort*. Addison-Wesley Press, 1949.

A Probe Generation Methodology

This appendix expands Section 4.1 with the full reproducibility detail of the IKP probe pipeline: tier definitions, per-corpus sampling procedures, quality filters with their drop rates, the final per-tier composition, verification procedure, and sample probes.

A.1 Tier Definitions

Each of the seven IKP tiers is defined by a calibrated parameter range: the model size at which a model with typical training-data coverage starts answering reliably. Tier targets are then *validated* against the landmark ladder (Section 4.2); probes whose landmark behavior contradicts the assigned tier are reassigned or dropped.

Tier	Name	Param range	Example
T1	Universal Knowledge	0.1B–0.5B	“What is the capital of Norway?” (Oslo)
T2	Common Reference Knowledge	0.5B–7B	“Who composed the Enigma Variations?” (Edward Elgar)
T3	Domain-Specific Knowledge	7B–32B	“In what year was the Battle of Hastings fought?” (1066)
T4	Obscure Knowledge	32B–235B	“In computer science, what is the research subfield of Peter Druschel?” (computer networking)
T5	Deep Knowledge	235B–1T	“In what year was the town of Eliot in York County, Maine founded?” (1810)
T6	Long-Tail Knowledge	1T–10T	“In computer science, what is the research subfield of Jeffrey Helt?” (distributed systems)
T7	Extreme Long-Tail	>10T	“In what country is the mountain Tadekho Hill located?” (Canada)

Table 7: Tier definitions used during probe generation. Parameter range is the model size at which a typically-trained model begins answering reliably; T7 is by construction beyond all current models and used as a ceiling probe. Examples here match those in Appendix A.7.

A.2 Phase A: LLM-Generated Candidates

Phase A targets T1–T2 saturation and supplementary fill at T3–T4. We use GPT-5 as the candidate generator with a structured prompt that: (i) specifies the target tier’s parameter range and exemplar Q/A pairs, (ii) rotates a primary region (8 regions: North America, Europe, East Asia, South Asia, Middle East, Africa, Latin America, Oceania) and primary domain (6 domains: people, places, publications, measurements, events, organizations) per generation batch, (iii) requests 50 candidates per batch with ≥ 3 paraphrases each, and (iv) requires verifiable, single-token-or-short-string gold answers.

Yield. 401 of 1,400 final probes (28.6%) come from Phase A: T1 $n=166$, T2 $n=152$, T3 $n=51$, T4 $n=32$. The empirical tier distribution of LLM-generated candidates is heavily skewed toward T1–T2 ($\sim 82\%$) regardless of difficulty prompting, confirming that an LLM is structurally unable to generate probes beyond its own knowledge horizon—the central reason Phase B exists.

A.3 Phase B: Corpus-Grounded Probes

Phase B grounds probes in two external corpora whose ground-truth answers can be verified independently of any LLM.

Wikidata-grounded probes (557 probes). We use Wikidata [Vrandečić and Krötzsch, 2014] SPARQL queries to sample entities from six categories: universities, journals, museums, bridges, sports clubs, and geographic places. Per-category SPARQL queries filter on entity type (P31 / instance of), require a published founding year (P571) or equivalent property, and stratify by Wikipedia article monthly-views-as-of-2025 to target a specific tier. Gold answers are read directly from the queried Wikidata property value. T3–T7 receive 94, 111, 100, 141, 100 Wikidata probes respectively.

Researcher subfield probes (345 probes). Computer-science researchers are sampled from DBLP and OpenAlex, stratified by citation-count buckets (10–50 citations for T7, 50–500 for T6, 500–5,000 for T5, 5,000–50,000 for T3/T4). Each researcher’s primary subfield is initially assigned from the dominant DBLP venue tag via a venue $\rightarrow$ subfield mapping table (e.g., SOSP/OSDI $\rightarrow$ “operating systems”; SIGCOMM $\rightarrow$ “computer networking”; CRYPTO $\rightarrow$ “computer security”; SIGMOD/VLDB $\rightarrow$ “database systems”; STOC/FOCS $\rightarrow$ “theoretical computer science”). T5–T7 each receive 100, 59, 100 researcher probes; T3–T4 receive 35 and 51.

Probe template. The released format is two-part:

In computer science, what is the research subfield of Name, and name one paper, system, institution, or co-author associated with their work? If you don’t know who this person is, say so.

The CS scoping (“In computer science”) reduces cross-field name collisions for short / common names. The artifact requirement (“name one paper, system, institution, or co-author”) is the load-bearing change: it forces models to produce a verifiable token that can be cross-checked against ground truth, rather than emitting a plausible-sounding but unattested CS subfield label. Without this requirement, a model that has never encountered “Dan Schatzberg” can still get full credit by guessing “computer architecture” (a popular subfield), since OpenAlex tags him as such; with the requirement, the model must additionally name TMO, EbbRT, ASPLOS, ISCA, Boston University, Meta, or some other token in his actual research footprint.

Per-researcher evidence bundle. Each researcher carries a structured gold record built from their OpenAlex profile and, for the tail of name-collision-flagged researchers, manually web-verified:

- **primary_subfield:** one of ~ 17 CS subfield labels.
- **secondary_subfields:** 0–3 acceptable adjacent labels for researchers whose work spans multiple subfields.
- **affiliations:** top 1–2 last-known institutions.
- **named_systems:** short, capitalized system / artifact tokens extracted from paper titles by a regex pass over the top 10 most-cited works (e.g., “TMO”, “Zoltan”, “Wukong”).
- **venues:** top 5–6 publication venues from the same set.
- **co_authors:** top 5 collaborators ranked by joint-paper count.
- **top_works:** top 5 paper titles with publication years and citation counts.

Four-way evidence-aware judge. Responses are classified into four classes, scored at $\lambda = -1$ for WRONG:

- **CORRECT-STRONG** (score +1.0): response names **primary_subfield** (or a direct synonym, or a listed **secondary_subfield**) *and* cites at least one matching evidence item—a system from **named_systems**, a venue from **venues**, an affiliation from **affiliations**, a co-author from **co_authors**, or a paper-title fragment overlapping with **top_works**.

- **CORRECT-WEAK** (score +0.5): response names a primary or secondary subfield but cites no specific evidence (e.g., “computer architecture, focusing on memory systems and cache coherence” for a researcher with no actual cache work).
- **REFUSAL** (score 0): response says “I don’t know”, expresses uncertainty, or declines.
- **WRONG** (score $\lambda = -1$): response names a CS subfield outside {primary, secondary} *or* confidently fabricates specifics that contradict / are absent from the evidence bundle.

The score formula $\lambda \cdot \mathbb{I}[\text{WRONG}] + 0.5 \cdot \mathbb{I}[\text{WEAK}] + 1.0 \cdot \mathbb{I}[\text{STRONG}]$ makes confident bluffing strictly worse than refusing, while still rewarding partial knowledge (WEAK) above refusal.

Name-collision sub-class. Sixteen names (all in T7) had no real CS researcher despite OpenAlex listing a person under that name (the OpenAlex profile pointed at, e.g., a chemist, virologist, or radiologist with the same name). These were replaced with web-verified CS researchers in the same subfield, re-calibrated through the 6-landmark ladder, and re-confirmed as T7 (no landmark answers correctly).

Manual / supplementary probes (97 probes). A small set of high-quality probes were manually authored or carried over from earlier dataset versions to balance per-tier coverage in T1–T4 where automated sampling was insufficient.

A.4 Quality Filters and Drop Rates

Four filters are applied to every candidate before inclusion. Each filter is conservative (false positives drop legitimate probes; false negatives admit problematic ones, which we accept as long as the rate is bounded):

1. **Computable knowledge filter.** Drops probes whose answer can be derived by rule rather than memorization: arithmetic on years (“what year was X 10 years ago”), alphabetical/ordinal answers, IUPAC chemical naming, etc. Implementation: regex + LLM classification, manual review of edge cases.
2. **Monotonicity filter.** A probe is admitted only if accuracy across the landmark ladder (Section 4.2) is monotonic non-decreasing in landmark capability. Probes where a smaller landmark answers correctly while a larger one does not—typically caused by ambiguous phrasing, multi-valued ground truth, or judge errors—are dropped. Drops $\sim 15\%$ of post-Phase-B candidates.
3. **Name-collision filter (researcher probes).** Drops two-character Chinese names and single-initial given names where multiple distinct researchers share the identifier; a 50-citation Liu Yang and a 5,000-citation Liu Yang are not the same person. Implementation: cross-check unique (*name*, *affiliation*) tuples against DBLP author IDs.
4. **Contamination filter (researcher probes).** Drops researchers in machine-learning, deep-learning, NLP, computer vision, and reinforcement-learning subfields. These authors’ own work disproportionately appears in frontier-model training corpora, inflating accuracy in a way that conflates IKP-style long-tail knowledge with training-set memorization of one’s own field. The remaining 14 CS subfields (systems, databases, security, networking, theory, etc.) are kept.
5. **Question-template grounding filter.** For Wikidata-sourced probes, every question template embeds a discriminating attribute—year, country, administrative region, genre, language, or publisher—extracted from a Wikidata field that is *not* the answer field. This is the same design used by researcher probes (which prefix every question with “In computer science”). The filter exists because an audit of the bare-name templates (“In what year was X founded?”) revealed

silent failures under title collision: “Putnam” could refer to Putnam, Connecticut (Wikidata’s intended entity) or Putnam County, Florida; “Madonna and Child” refers to dozens of Renaissance paintings. Grounding raises the within-probe identifiability bar without leaking the answer (e.g., “Who painted the 1522 painting ‘The Virgin of Carmel’?” grounds via year without naming the artist). The grounded template is generated programmatically from Wikidata claims at SPARQL time, then independently re-calibrated through the landmark ladder—a non-trivial post-condition, since grounding tends to make probes easier and shift them out of T6/T7. In our audit, $\sim 30\%$ of grounded T7 candidates re-calibrated to T5 (became too easy with grounding) and $\sim 5\%$ became non-monotonic; the surviving $\sim 65\%$ form the released T6/T7 Wikidata pool.

A.5 Final Composition

Tier	LLM	Wikidata	Researcher	Manual	Total
T1	166	6	0	28	200
T2	152	5	0	43	200
T3	51	94	35	20	200
T4	32	111	51	6	200
T5	0	100	100	0	200
T6	0	141	59	0	200
T7	0	100	100	0	200
Total	401	557	345	97	1,400

Table 8: Per-tier probe composition by source. “Manual” aggregates manually-authored and supplementary probes from earlier dataset iterations. The dominance of researcher and Wikidata probes at T5–T7 reflects the unavoidability of corpus grounding at frontier difficulty.

For T5–T7 specifically, the dominant domain is computer-science researcher subfield (259/600 probes, 43%). The remaining T5–T7 Wikidata probes span 16 fact types after the audit (Section 7.7): grounded founding years (sports clubs, museums, bridges, universities, journals, abbeys, tramways), and entity-attribute questions (river mouths, film directors, capes, sculptors, founders, mountains, lakes, painters, screenwriters, composers, named-after eponyms, headquarters cities, architects). The original T6/T7 Wikidata composition was a 100% founding-year monoculture; the audit replaced $\sim 50\%$ of T6/T7 Wikidata probes with diverse fact types to remove that source bias while preserving the unambiguous-answer property.

A.6 Verification Procedure

Gold-answer verification differs by source:

- **Wikidata probes:** gold answer is the queried property value as of the SPARQL extraction date. Conflicts (e.g., a museum with multiple founding-date claims) are resolved by selecting the value with the most reference citations; probes with unresolved conflicts are dropped. After an audit (Section 7.7), every T5–T7 Wikidata probe was independently web-cross-checked against Wikipedia and primary sources; probes with a Wikidata-vs-Wikipedia disagreement on the answer were either corrected (4 cases, e.g. Makran Medical College: Wikidata 2015, Wikipedia 2017—kept Wikipedia’s answer) or dropped (e.g., probes attributing a sculpture to its bronze foundry rather than its sculptor). All probes whose answers were corrected were re-calibrated through the

landmark ladder, since changing the gold answer changes which landmarks are scored as correct and therefore the assigned tier.

- **Researcher probes:** gold subfield is derived from each researcher’s dominant DBLP venue tag in their highest-cited 5 papers, mapped to one of 14 CS subfields. Researchers whose top-5 papers split across ≥ 3 unrelated subfields are dropped (only researchers with a clear primary specialization are retained).
- **Phase A probes:** gold answers are produced by the generator and independently re-verified by a different LLM (Claude Opus 4.6) plus a manual spot-check of $\sim 10\%$.
- **Manual probes:** directly authored with citation evidence.

A.7 Sample Probes

Table 9 lists three representative probes per tier, drawing from all three probe sources (LLM-generated, Wikidata-grounded, researcher subfield) so that each contributes.

B Hallucination Penalty Sensitivity

The hallucination penalty λ (a wrong answer scores λ ; correct = +1, refusal = 0) is a scoring-convention choice. This appendix reports its sensitivity: it sweeps λ over $\{0, -0.25, -0.5, -1.0, -2.0\}$, refits the log-linear calibration on the same 93-model open-weight set for each value, and re-inverts for four flagship proprietary models (Table 10). The released dataset, companion website, and estimation toolkit (`scripts/ikp_estimate.py`) default to $\lambda = 0$ (raw accuracy), so that IKP accuracy is directly the fraction of probed facts a model answers correctly with no scoring convention to interpret; the table lets results under any penalty be read off directly.

Table 10: Sensitivity of the IKP calibration and of four flagship proprietary estimates to the hallucination penalty λ . Each row refits $A = \alpha \log_{10}(N) + \beta$ on the same 93 open-weight models, recomputing every model’s accuracy from stored verdicts under the listed λ (correct = +1, refusal = 0, wrong = λ), and re-inverts the fit for each flagship model. “Slope” is in percentage points per decade; “LOO $\times$ ” is the median multiplicative parameter-prediction error in leave-one-out cross-validation; “PI $\times$ ” is the 90% prediction-interval factor. $\lambda = 0$ (starred) is the default used by the released dataset, website, and toolkit. Table generated by `scripts/lambda_sensitivity.py`.

λ	Slope (pp/dec)	R^2	LOO $\times$ (med.)	PI $\times$ (90%)	GPT-5.5 Pro est.	Claude Fable 5 est.	GPT-4.1 est.	Claude Opus 4.7 est.
0.00 *	15.9	0.910	1.48	3.20	5.3T	3.5T	2.2T	538B
−0.25	16.7	0.936	1.48	2.64	4.6T	4.3T	1.6T	761B
−0.50	16.6	0.935	1.57	2.65	5.2T	4.9T	1.5T	916B
−1.00	16.5	0.934	1.65	2.68	5.7T	4.8T	1.2T	1.2T
−2.00	16.0	0.920	1.68	2.99	5.4T	5.8T	1.3T	1.5T

Findings. (i) Fit quality is near-flat across the band: R^2 ranges only ~ 0.90 – 0.93 and the 90% PI factor ~ 2.7 – $3.5\times$, so the calibration is robust to the penalty choice. (ii) We adopt $\lambda = 0$ (raw accuracy) as the operating point for *interpretability*: with no penalty, IKP accuracy is directly the fraction of probed facts a model answers correctly, requiring no scoring convention to read. This costs ~ 0.02 – 0.03 in R^2 relative to the shallow optimum near $\lambda = -0.5$ and slightly widens the prediction interval—a tradeoff we accept in exchange for a directly-meaningful accuracy metric.

Tier	Source	Question	Answer
T1	Wikidata	What is the capital of Norway?	Oslo
	LLM	In what year was the Meiji Constitution promulgated?	1889
	LLM	Who designed the Sydney Opera House?	Jørn Utzon
T2	LLM	Who composed the Enigma Variations?	Edward Elgar
	LLM	What is the most common isotope of hydrogen?	Protium
	Manual	Who was the first Prime Minister of India?	Jawaharlal Nehru
T3	LLM	In what year was the Battle of Hastings fought?	1066
	Wikidata	In what year was Hunan Normal University founded?	1938
	Manual	What year was the Camp David Accords signed?	1978
T4	Researcher	In computer science, what is the research subfield of Peter Druschel, and name one paper, system, institution, or co-author associated with their work? If you don't know who this person is, say so.	computer networking
	Wikidata	In what year was Lawrence Heritage State Park founded?	1980
	Wikidata	In what year was the journal <i>Cancer</i> first published?	1948
T5	Wikidata	In what year was the town of Eliot in York County, Maine founded?	1810
	Researcher	In computer science, what is the research subfield of Jianliang Xu, and name one paper, system, institution, or co-author associated with their work? If you don't know who this person is, say so.	data mining
	Wikidata	In what year was State Polytechnic of Indramayu founded?	2014
T6	Researcher	In computer science, what is the research subfield of Jeffrey Helt, and name one paper, system, institution, or co-author associated with their work? If you don't know who this person is, say so.	distributed systems
	Wikidata	In what year was Museu da Misericórdia founded?	2006
	Wikidata	In what year was Universitas Imelda Medan founded?	2019
T7	Wikidata	In what country is the mountain Tadekho Hill located?	Canada
	Researcher	In computer science, what is the research subfield of Christopher Bengel, and name one paper, system, institution, or co-author associated with their work? If you don't know who this person is, say so.	computer architecture
	Wikidata	Into what body of water does the Bolshaya Khopa in Arkhangelsk Oblast, Russia flow?	Soyana

Table 9: Three sample probes per tier, one per source where possible. Each row is a single probe as it appears in the released probe set.

(iii) Individual estimates shift with λ in a *vendor-dependent* way (last four columns of Table 10): a heavier penalty raises the estimates of models that refuse rather than bluff (e.g. Claude Opus 4.7, $0.55 \rightarrow 1.5T$ from $\lambda = 0$ to -2) and lowers those of confident guessers (e.g. GPT-4.1, $2.2 \rightarrow 1.1T$), because it reattributes above-trend scores from confident hallucination to genuine capacity. Readers who prefer a hallucination-aware metric can read any column of Table 10 directly; the qualitative frontier ordering is preserved within each column.

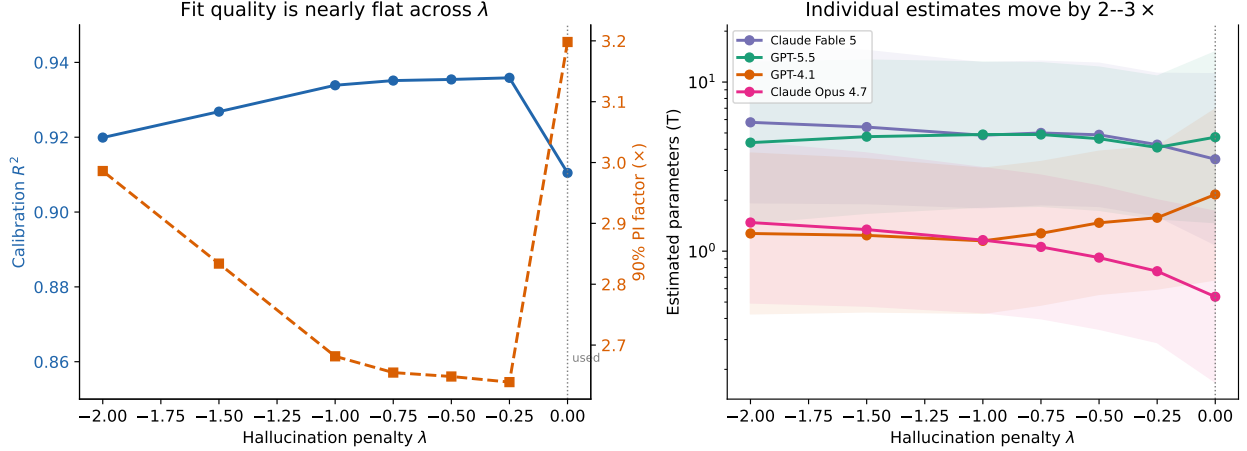


Figure 11: What changes across the hallucination penalty λ . **Left:** calibration fit quality is nearly flat— R^2 stays in $[0.91, 0.94]$ and the 90% prediction-interval factor in $[2.6, 3.2] \times$ (floored scoring for $\lambda < 0$; $\lambda = 0$ is floor-free). **Right:** individual frontier estimates (90% bands shaded) move by 2–3 $\times$ and *re-order* by vendor: as $\lambda \rightarrow 0$ the confident guesser GPT-4.1 rises while the refusal-heavy Claude Opus 4.7 falls. Generated by `scripts/./paper/figures/generate_lambda_figure.py`.

The λ -choice principle. We fix λ by *parsimony*, not by tuning. The rule is: adopt the parameter-free operating point— $\lambda = 0$, which simultaneously eliminates the flooring rule (per-tier scores are then ≥ 0 by construction)—*unless* a nonzero penalty materially improves the calibration. It does not: across $\lambda \in [0, -2]$ the calibration R^2 varies by ≤ 0.03 and the 90% PI factor by $\leq 0.6 \times$ (Figure 11, left), and the frontier ordering is preserved within each column (Table 10). A nonzero λ therefore buys at most 0.03 in R^2 at the price of *two* tunable knobs—the penalty magnitude and the floor—that, as Table 11 shows, can together be dialed to inflate frontier estimates by 3–5 $\times$. Because IKP is a measurement instrument, we prefer the estimator a reader cannot tune toward a desired answer, and $\lambda = 0$ is that Schelling point. This is a principled default, not a convenience: readers who prefer a hallucination-aware score may read any λ column directly, accepting the 2–3 $\times$ vendor-dependent shift of Figure 11 (right)—but no qualitative conclusion of the paper depends on the choice.

B.1 Flooring and its interaction with λ

A second scoring degree of freedom is whether per-tier scores are floored at zero. Under a negative penalty a strongly-bluffing model can score negative on the hard tiers; flooring those scores at zero raises small/aggressive models’ accuracy, which flattens the calibration slope and—because the curve is inverted to estimate size—inflates frontier estimates. Table 11 crosses λ with flooring on/off.

Table 11: Calibration and two flagship estimates under $\lambda \times$ flooring. Flooring only matters under a negative penalty; unfloored, the fit degrades sharply as $|\lambda|$ grows (R^2 from 0.91 to 0.72, PI factor from $3.2\times$ to $9.9\times$) because negative hard-tier scores dominate the regression, and estimates fall by up to $\sim 3\times$. At $\lambda = 0$ (starred) the floor is a no-op, so the ambiguity vanishes entirely—the principal reason we adopt no-penalty scoring. Generated by `scripts/lambda_floor_ablation.py`.

λ	floor	R^2	Slope (pp/dec)	PI $\times$	LOO $\times$	GPT-5.5 est.	Gemini 3.1 Pro est.
0.00*	on	0.910	15.9	3.20	1.48	4.7T	7.3T
(at $\lambda = 0$ the floor is a no-op: scores are correct/total ≥ 0)							
0.00	off	0.910	15.9	3.20	1.48	4.7T	7.3T
-0.25	on	0.936	16.7	2.64	1.48	4.1T	9.4T
-0.25	off	0.931	19.3	2.74	1.56	3.0T	6.2T
-0.50	on	0.935	16.6	2.65	1.57	4.6T	12.0T
-0.50	off	0.905	22.7	3.32	1.55	2.2T	5.5T
-1.00	on	0.934	16.5	2.68	1.65	4.9T	18.0T
-1.00	off	0.830	29.5	5.35	1.77	1.5T	4.8T
-1.50	on	0.927	16.3	2.83	1.63	4.8T	25.4T
-1.50	off	0.768	36.4	7.66	2.12	1.2T	4.3T
-2.00	on	0.920	16.0	2.99	1.68	4.4T	34.5T
-2.00	off	0.723	43.2	9.91	2.40	972B	4.1T

Relation to prior discrepancy. An earlier version of IKP reported floored scores under $\lambda = -1.0$ while the accompanying text described the scores as unfloored [Sturb and LawrenceC, 2026]; the two differ materially (e.g. GPT-5.5 4.9T floored vs. 1.5T unfloored at $\lambda = -1.0$). Adopting $\lambda = 0$ removes the discrepancy at its root: with no penalty every per-tier score is correct/total ≥ 0 , so floored and unfloored scoring are identical and there is nothing left to specify.

C Dense vs. MoE Calibration

The calibration pools dense and Mixture-of-Experts models; Table 12 fits them separately. The per-decade slope is nearly identical (dense 0.151, MoE-total 0.145), but MoE-total sits slightly higher—an MoE knows $\sim 2\text{--}4$ pp more per *total* parameter than a same-size dense model—and fits more loosely ($R^2 = 0.67$ vs 0.88). Active parameters predict MoE knowledge far worse ($R^2 = 0.41$), confirming that factual capacity scales with *total*, not active, weights: an MoE with T total and A active parameters behaves, for recall, like a dense model of $\sim T$ parameters. Notably the combined-set slope (0.159) is *steeper* than either subset—a mixture effect, since MoE models cluster at large N /high accuracy and dense at small N /low accuracy.

Table 12: Log-linear calibration fit by architecture ($\lambda = 0$, cleaned set). Generated by `scripts/moe_dense_analysis.py`.

Calibration subset	n	Slope	Intercept	R^2
Dense	52	0.151	0.245	0.875
MoE (total params)	41	0.145	0.282	0.667
MoE (active params)	41	0.144	0.453	0.412
Combined (headline)	93	0.159	0.242	0.910

Which curve should estimate (MoE) frontier models? Frontier proprietary models are

widely believed to be MoE, so one might fit only on MoE anchors. Empirically this does *not* help. In leave-one-out over the 41 known-size open MoE models, the MoE-only curve recovers their sizes no better than the combined curve (median fold error $1.58\times$ vs. $1.52\times$ overall; $1.55\times$ vs. $1.55\times$ for the 34 models $\geq 100\text{B}$). Dropping the 52 dense anchors to erase a $\sim 2\text{--}4$ pp offset costs more in slope stability than the offset is worth, and per-model deviations from post-training, distillation, and refusal behavior—GLM over-knowing for its size by $\sim 3\times$, DeepSeek V4 Pro’s degraded non-thinking variant $\sim 10\times$ low—dwarf the architecture gap. We therefore keep the combined curve as the headline: it is the *validated* choice, not merely a convenient one.

MoE-deflation caveat. Because the combined slope is steeper, it assigns *lower* sizes to the highest-accuracy models than a pure MoE curve would (Table 13): switching to the MoE-total curve raises the top estimates by $\sim 18\%$ (e.g. Claude Fable 5 from $\sim 3.5\text{T}$ to $\sim 4.0\text{T}$; GPT-5.5 from $\sim 4.7\text{T}$ to $\sim 5.5\text{T}$). These shifts are well within the $3.2\times$ prediction interval and cannot be validated out-of-sample, so we report combined-curve values as headline while noting that presumed-MoE frontier models may be modestly under-placed.

Table 13: Frontier estimate sensitivity to the calibration subset: headline combined curve vs. the MoE-total curve. Generated by `scripts/moe_dense_analysis.py`.

Model	Accuracy	Combined	MoE-curve
Claude Fable 5	80.5%	$\sim 3.5\text{T}$	$\sim 4.0\text{T}$
GPT-5.5 Pro	83.4%	$\sim 5.3\text{T}$	$\sim 6.3\text{T}$
GPT-5.5	82.6%	$\sim 4.7\text{T}$	$\sim 5.5\text{T}$
Gemini 2.5 Pro	79.5%	$\sim 3.0\text{T}$	$\sim 3.4\text{T}$
GPT-4.1	77.2%	$\sim 2.2\text{T}$	$\sim 2.4\text{T}$
Claude Opus 4.7	67.6%	$\sim 538\text{B}$	$\sim 515\text{B}$

D Wikidata Long-Tail Audit

This appendix expands Section 7.7: it documents the failure-mode taxonomy uncovered when auditing the T5–T7 Wikidata-sourced probes against Wikipedia and primary sources, the per-round telemetry of the 10-round repair cycle, and the empirical observation that an answer-field correction necessarily invalidates the probe’s prior tier assignment.

D.1 Failure-Mode Taxonomy

Every problematic probe falls into exactly one of five buckets. The taxonomy is intended to be a re-usable diagnostic for any future Wikidata-sourced evaluation set.

A. Wikidata-itself-wrong.

Outright incorrect facts in Wikidata that are falsifiable via Wikipedia or a primary source.

- **A1. Stale data** (P159 headquarters typically): Roku still listed as headquartered in Los Gatos despite a 2019 move to San Jose; Vista Outdoor in Clearfield despite a move to Anoka. Frequency: $\sim 75\%$ of headquarters probes were stale.
- **A2. Wrong field semantics**: P170 (creator) for sculptures often returns the bronze foundry, not the sculptor—Giacometti’s *L’Homme qui marche I* attributed to Susse Frères; Bourdelle’s *Monument to Alvear* attributed to Eugène Rudier.
- **A3. Self-referential / circular**: “Na Klang named after Na Klang”—a Thai descriptive place-name (“middle field”) incorrectly marked as having an eponymous referent.

- **A4. Plain wrong year/value:** Makran Medical College listed as 2015 when Wikipedia infobox says 2017; Lady Snowblood screenplay attributed to the manga illustrator Kazuo Kamimura rather than the actual screenwriter Norio Osada.

B. Pipeline extraction bugs (on our side).

P571 inception was correct in Wikidata, but our SPARQL pipeline extracted a different date field (e.g., the year the building was constructed rather than the year the museum was chartered: George Eastman Museum stored as 1905, but Wikidata P571=1947).

C. Probe-construction ambiguity (Wikidata correct, question unanswerable).

The most prevalent class.

- **C1. Generic religious-art title:** “Madonna and Child”, “Holy Trinity”, “Saint Apollonia”—each refers to dozens of works by different painters across centuries.
- **C2. Generic geographic-feature name:** “Stone Bridge” (Latvia? Russia? UK?), “South Island” (New Zealand? or the one in Lake Turkana, Kenya?), “Hog Island” (Falklands, or one of several US ones?), “Devil’s Lake”.
- **C3. Bare common-name place:** “Putnam” (the Connecticut town, or Putnam County?), “Norwich” (Vermont? or the much more famous Norwich, England?), “Wells”.
- **C4. Generic institution name:** “Maritime Museum” (Jakarta? San Diego? Greenwich?), “St. Lawrence University” (Kampala, Uganda? or the famous one in NY?).

D. Politically/legally contested entities.

Wikidata picks one answer where reasonable models will give a different one.

- **D1. Disputed sovereignty:** Loaita Cay listed under Vietnam (the cay is one of the Spratlys; claimed by China, Vietnam, and the Philippines). Cape Plaka listed under Ukraine (in Crimea, internationally recognized as Ukraine but Russian-occupied since 2014).
- **D2. Historical-vs-current statehood:** Mount Ashley listed under “United Kingdom”—technically the South Georgia archipelago is its own British Overseas Territory, not part of the UK proper.

E. Multi-actor attribution.

A genuinely correct attribution that is one of several equally valid answers; reasonable models will give a different one.

- **E1. Co-founder selection:** Cinémathèque de Tanger has 3 founders (Lahlou, Auriol, Barrada); the National Council of Women of Canada has Adelaide Hoodless (treasurer at founding) and Lady Aberdeen (first president).
- **E2. Co-author / story-vs-screenplay:** Twenty-Four Eyes screenplay attributed to the novelist Sakae Tsuboi rather than the actual screenwriter (and director) Keisuke Kinoshita; The Amazing Panda Adventure screenplay attributed to John Wilcox (story credit) rather than Jeff Rothberg / Laurice Elehwany (actual screenplay).

The audit response by class: A and B are resolved by correcting the answer field (and re-calibrating); C is resolved by adding a discriminating qualifier to the question template (“the town of Putnam in Connecticut”); D is resolved by either dropping the probe (Loaita Cay) or accepting Wikidata’s choice with the caveat that the LLM judge accepts the documented answer; E is resolved by accepting any of the multiply-valid answers via the judge’s instruction to credit co-contributors.

D.2 Per-Round Sourcing Telemetry

Replacement probes were sourced via 10 SPARQL rounds, each producing 100–300 candidates that were calibrated through the landmark ladder. Per-fact-type web-verification pass rates emerged empirically and informed which fact types to over-source in later rounds.

Round	Sourced	Calib. valid	T6/T7 yield	Web-verified pass
R1 (founding-year)	186	155	70 (T6=25, T7=45)	— (most rejected post-mandate)
R2 (mixed)	152	138	39 (T6=20, T7=19)	18/27 diverse (67%)
R3 (composer/birthplace)	0	—	—	— (SPARQL timeouts)
R4 (diverse-only)	140	124	24 (T6=10, T7=14)	15/24 (62%)
R5 (12 fact types)	250	228	68 (T6=39, T7=29)	21/29 T7 (72%)
R6 (11 fact types + castle)	275	256	64 (T6=36, T7=28)	18/28 T7 (64%)
R7 (7 reliable types only)	140	130	21 (T6=9, T7=12)	11/12 T7 (92%)
R8 (aggressive ambiguity filter)	100	92	24 (T6=13, T7=11)	7/11 T7 (64%)
R9 (most reliable types)	75	71	26 (T6=8, T7=18)	18/18 T7 (100%)
R10 (grounded SPARQL templates)	45	15	5 (T6=3, T7=2)	— (low yield; alt path used)

Table 14: Per-round sourcing telemetry. Total API spend across all rounds: ~10,000 landmark+judge calls and ~120 manual web verifications.

D.3 Per-Fact-Type Reliability Ranking

Aggregating across rounds 4–7, the empirical reliability of each Wikidata fact type at the long tail (T6/T7 obscurity):

1. **P403 river mouth** — 100% pass rate. Geography is well-curated.
2. **P57 director (film)** — 67–100% on attribution; ambiguity from generic film titles (“Night Train”, “Hit Man”) is the dominant failure mode and is resolved by adding year + country + genre to the template.
3. **P17 country (cape/island/lake/mountain)** — 75–100%. Caveat: politically contested entities (Crimea, Spratlys, South Georgia) consistently fail web cross-check because Wikidata picks one claimant.
4. **P571 inception (founding year)** — ~95%. The most-curated date field. Most failures trace to pipeline-side date-field selection bugs (Bucket B), not Wikidata error.
5. **P112 founder (organization)** — 75%. Sometimes returns the founding *organization* rather than a person (e.g., St John Clinic founded by “Order of the Holy Ghost”), making the question awkward when phrased “Who founded ...?”.
6. **P170 creator (sculpture)** — ~70%. The major Bucket-A2 failure: Wikidata’s creator field for sculptures frequently records the bronze foundry rather than the sculptor.
7. **P58 screenwriter** — 50%. Routinely conflates “story by” credit with “screenplay by” credit (e.g., the manga author of *Lady Snowblood* listed as the film’s screenwriter).
8. **P159 headquarters** — 25%. *Stale*. Wikidata frequently reflects a company’s HQ from years prior; modern moves are not propagated quickly.

9. P170 **creator (painting)** — $\sim 70\%$ on attribution, but $\sim 100\%$ title-collision rate at T6/T7 obscurity (generic religious-art titles). Painter probes are essentially unusable for T7 obscurity.
10. P138 **named after** — mixed; self-referential and descriptive-placename errors observed.

E Full Model Results

Tables 15 and 16 present the complete evaluation results for all 201 models. Models are grouped by vendor and sorted by overall accuracy (descending) within each group. Vendors are ordered by the highest accuracy achieved by any model in that group. The *Acc.* column reports the tier-mean accuracy at $\lambda = 0$ (no penalty; Section 5); *Raw* is overall correct/total. The *Raw* column reports raw accuracy without hallucination penalty. Per-tier accuracy (T1–T7) is reported in the range $[0, 1]$.

Figure 12 shows the distribution of per-tier accuracy across all models, illustrating how each tier discriminates at different capability levels. Figure 13 shows the hallucination rate by vendor on T5–T7 probes, revealing stark differences in safety calibration across vendors.

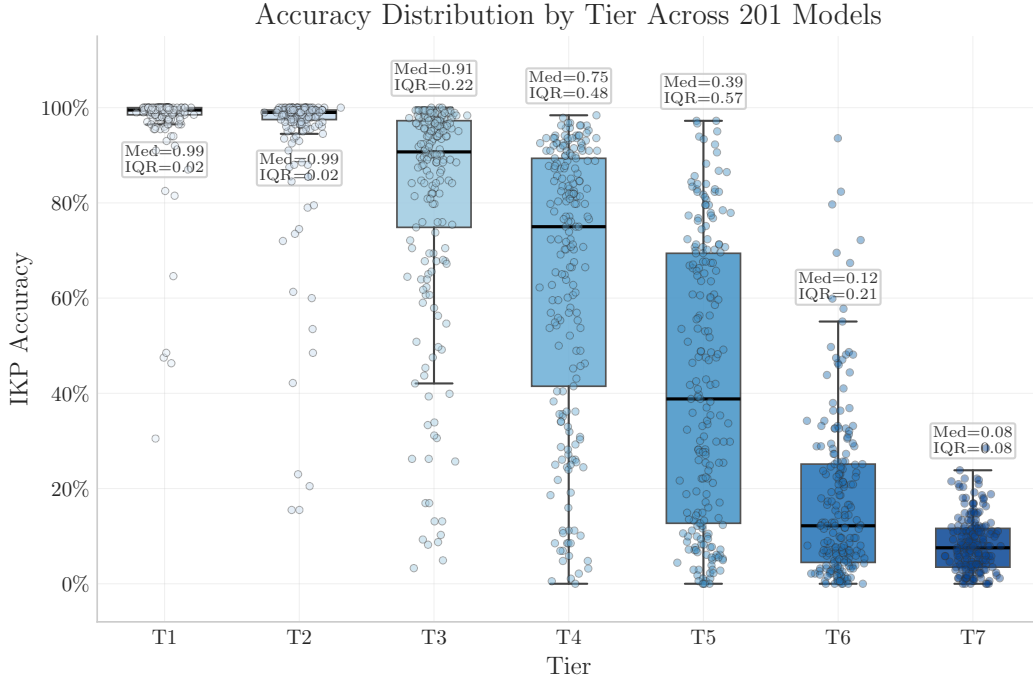


Figure 12: Accuracy distribution by tier across 201 models. T1–T2 are compressed at ceiling (median $> 98\%$). T4 has the widest interquartile range, making it the best population discriminator. T6–T7 are at the floor for most models.

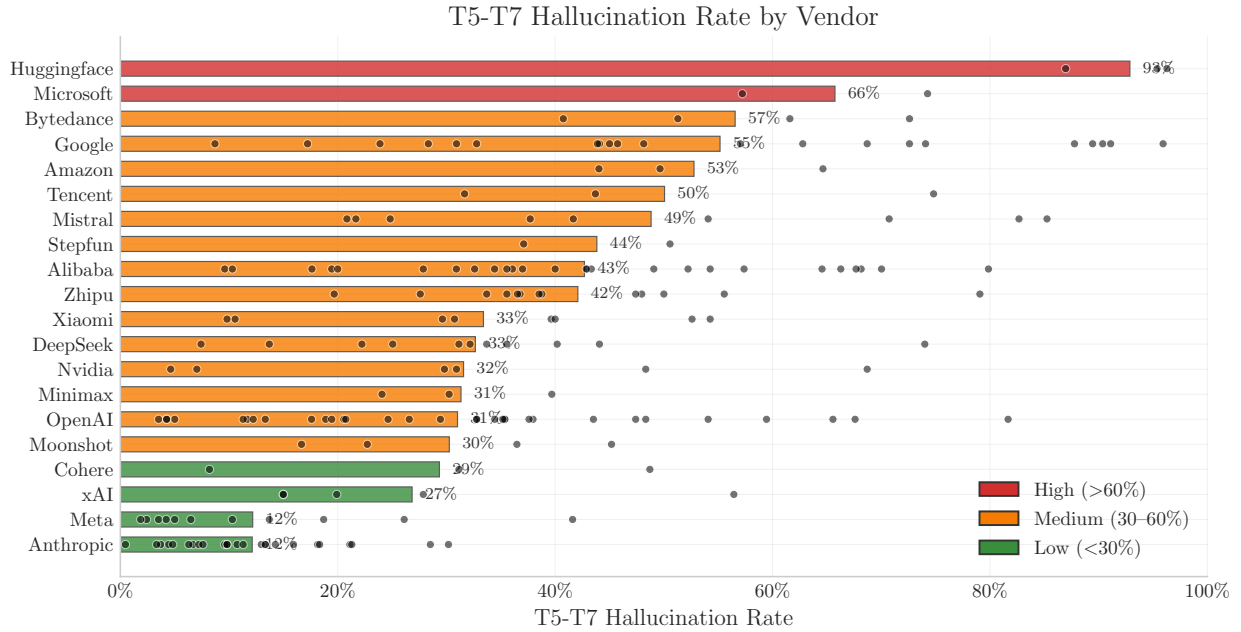


Figure 13: T5–T7 hallucination rate by vendor. Individual model points are overlaid on vendor means. Anthropic models hallucinate at only 10% on probes beyond their knowledge (preferring to refuse), while Microsoft and Google models hallucinate at 58–66%. This vendor-specific calibration directly impacts IKP scoring.

Table 15: Full model results: per-tier accuracy at $\lambda = 0$ (no penalty). Models grouped by vendor, sorted by accuracy. **Acc.** is the tier-mean accuracy used throughout; **Raw** is overall correct/total. The **IKP Pred.** column shows the calibration’s predicted parameter count and ratio to actual for open-weight models; ratios $> 2\times$ or $< 0.5\times$ are bolded as systematic outliers. Proprietary models show ‘–’.

Vendor	Model	Params	IKP Pred.	Arch	Think	Acc.	Raw	T1	T2	T3	T4	T5	T6	T7
Google	gemini-3-flash-think	–	–	–	Y	0.862	0.870	1.00	0.99	0.99	0.96	0.97	0.82	0.28
	gemini-3.1-pro	–	–	–		0.856	0.867	1.00	0.99	1.00	0.98	0.97	0.94	0.11
	gemini-3-flash	–	–	–		0.851	0.860	1.00	1.00	0.99	0.96	0.97	0.80	0.24
	gemini-2.5-pro	–	–	–		0.795	0.805	0.99	1.00	0.99	0.95	0.82	0.58	0.22
	gemini-2.5-pro-think	–	–	–	Y	0.789	0.799	0.99	1.00	0.99	0.94	0.83	0.55	0.22
	gemini-3.1-flash-lite	–	–	–		0.768	0.779	0.99	0.99	0.98	0.93	0.87	0.50	0.13
	gemini-2.5-flash-think	–	–	–	Y	0.702	0.714	1.00	0.99	0.95	0.84	0.66	0.33	0.15
	gemini-2.0-flash	–	–	–		0.663	0.675	0.99	0.98	0.95	0.81	0.47	0.26	0.17
	gemini-2.5-flash	–	–	–		0.621	0.635	0.99	0.99	0.90	0.80	0.39	0.19	0.08
	gemini-2.5-flash-lite-think	–	–	–	Y	0.599	0.613	1.00	0.99	0.90	0.75	0.37	0.12	0.06
	gemini-2.5-flash-lite	–	–	–		0.557	0.571	0.99	0.98	0.87	0.62	0.30	0.10	0.03
	gemma-4-31b	31B	27B (0.9 $\times$)	dense		0.470	0.486	0.99	0.99	0.68	0.34	0.13	0.09	0.08
	gemma-3-27b	27B	25B (0.9 $\times$)	dense		0.465	0.481	0.98	0.98	0.64	0.36	0.15	0.06	0.08
	gemma-4-26b-a4b	26B	21B (0.8 $\times$)	moe		0.452	0.468	0.99	0.98	0.62	0.35	0.10	0.07	0.05
	gemma-2-27b	27B	15B (0.5 $\times$)	dense		0.428	0.486	0.99	0.95	0.65	0.32	0.07	0.01	0.00
	gemma-3-12b	12B	10B (0.8 $\times$)	dense		0.400	0.416	0.97	0.94	0.48	0.24	0.07	0.05	0.06
	gemma-3n-e4b	4B	5B (1.2 $\times$)	dense		0.348	0.393	0.99	0.88	0.34	0.09	0.05	0.03	0.06
	gemma-3-4b	4B	4B (0.9 $\times$)	dense		0.330	0.345	0.98	0.79	0.26	0.07	0.06	0.03	0.13
	gemma-2-2b	3B	3B (1.2 $\times$)	dense		0.320	0.333	0.94	0.72	0.17	0.11	0.10	0.07	0.12
	gemma-3-1b	1B	572M (0.6 $\times$)	dense		0.204	0.216	0.81	0.48	0.09	0.01	0.01	0.01	0.01
OpenAI	gemma-3-270m	268M	237M (0.9 $\times$)	dense		0.143	0.145	0.30	0.15	0.08	0.09	0.17	0.08	0.12
	gpt-5.5-pro	–	–	–	Y	0.834	0.844	1.00	1.00	0.98	0.98	0.94	0.72	0.22
	gpt-5.5	–	–	–		0.826	0.835	1.00	1.00	0.99	0.96	0.95	0.67	0.20
	gpt-5.5-think	–	–	–	Y	0.826	0.835	1.00	1.00	0.98	0.96	0.93	0.70	0.21
	gpt-5.4-pro	–	–	–		0.778	0.789	1.00	0.99	0.99	0.94	0.86	0.49	0.18
	gpt-4.1	–	–	–		0.772	0.782	1.00	0.98	0.99	0.92	0.82	0.47	0.21
	gpt-5-pro	–	–	–		0.770	0.780	1.00	0.99	0.99	0.94	0.80	0.48	0.19
	o3	–	–	–		0.769	0.780	1.00	0.99	0.99	0.91	0.82	0.48	0.18
	gpt-5-think	–	–	–	Y	0.765	0.776	0.99	0.99	0.98	0.93	0.83	0.46	0.17
	gpt-5	–	–	–		0.758	0.769	0.99	0.99	0.99	0.92	0.80	0.44	0.17
	o1	–	–	–	Y	0.736	0.748	0.99	1.00	0.98	0.94	0.77	0.32	0.15
	gpt-5.4	–	–	–		0.723	0.735	1.00	0.98	0.97	0.91	0.72	0.34	0.13

Vendor	Model	Params	IKP	Pred.	Arch	Think	Acc.	Raw	T1	T2	T3	T4	T5	T6	T7
Anthropic	gpt-5.3	–	–	–			0.722	0.734	1.00	0.99	0.99	0.90	0.74	0.29	0.13
	gpt-4-turbo	–	–	–			0.722	0.733	0.99	1.00	0.98	0.91	0.78	0.27	0.12
	gpt-5.2-pro	–	–	–			0.688	0.700	0.99	0.99	0.97	0.90	0.60	0.26	0.10
	gpt-5.1	–	–	–			0.687	0.699	0.99	0.98	0.97	0.86	0.66	0.27	0.08
	gpt-4	1.8T	–	–	moe		0.686	0.698	0.99	0.99	0.97	0.82	0.67	0.26	0.09
	gpt-5.2	–	–	–			0.680	0.693	0.99	0.99	0.98	0.87	0.60	0.23	0.09
	gpt-4.1-mini	–	–	–			0.671	0.683	0.99	0.99	0.94	0.81	0.54	0.25	0.17
	gpt-4o	–	–	–			0.654	0.667	0.99	0.98	0.91	0.80	0.56	0.25	0.09
	o4-mini-think	–	–	–		Y	0.650	0.663	1.00	0.99	0.96	0.82	0.43	0.21	0.13
	gpt-5.4-mini	–	–	–			0.644	0.657	1.00	0.99	0.91	0.82	0.53	0.17	0.09
	gpt-3.5-turbo	–	–	–			0.622	0.635	0.99	0.99	0.94	0.75	0.48	0.13	0.08
	gpt-oss-120b-think	120B	106B (0.9×)	–	dense	Y	0.564	0.578	0.97	0.98	0.85	0.63	0.31	0.12	0.09
	gpt-5-mini-think	–	–	–		Y	0.563	0.578	0.99	0.99	0.92	0.64	0.28	0.07	0.05
	gpt-5-mini	–	–	–			0.555	0.571	0.99	0.99	0.89	0.65	0.25	0.07	0.04
	gpt-4o-mini	–	–	–			0.554	0.569	0.98	0.98	0.89	0.65	0.25	0.07	0.05
	gpt-4.1-nano	–	–	–			0.536	0.551	1.00	0.96	0.82	0.54	0.28	0.08	0.06
	o3-mini	–	–	–			0.527	0.543	0.99	0.99	0.89	0.60	0.12	0.05	0.04
	gpt-5-nano-think	–	–	–		Y	0.466	0.483	1.00	0.99	0.76	0.43	0.06	0.01	0.01
	gpt-5-nano	–	–	–			0.459	0.475	0.99	0.98	0.75	0.41	0.05	0.01	0.01
	gpt-oss-20b-think	20B	16B (0.8×)	–	dense	Y	0.435	0.451	0.99	0.95	0.64	0.26	0.07	0.06	0.06
	gpt-5.4-nano	–	–	–			0.400	0.416	0.95	0.93	0.59	0.25	0.06	0.01	0.01
	claude-fable-5	–	–	–			0.805	0.815	0.99	0.99	0.99	0.94	0.91	0.60	0.20
	claude-opus-4.6-think	–	–	–		Y	0.751	0.763	1.00	1.00	0.99	0.97	0.81	0.38	0.12
	claude-opus-4.6	–	–	–			0.735	0.747	1.00	0.99	0.99	0.96	0.77	0.33	0.10
	claude-3.7-sonnet	–	–	–			0.726	0.738	0.99	1.00	0.97	0.87	0.77	0.37	0.11
	claude-opus-4.5	–	–	–			0.711	0.722	0.99	0.99	0.97	0.89	0.72	0.29	0.12
	claude-opus-4.1-think	–	–	–		Y	0.707	0.719	0.99	1.00	0.98	0.93	0.71	0.25	0.09
	claude-opus-4.5-think	–	–	–		Y	0.706	0.718	1.00	1.00	0.99	0.93	0.70	0.23	0.09
	claude-sonnet-4.6-think	–	–	–		Y	0.700	0.712	0.99	0.99	0.96	0.93	0.71	0.22	0.09
	claude-sonnet-4.6	–	–	–			0.700	0.712	1.00	1.00	0.96	0.93	0.70	0.23	0.09
	claude-opus-4.7-think	–	–	–		Y	0.698	0.710	1.00	0.99	0.99	0.94	0.65	0.24	0.08
	claude-sonnet-5	–	–	–			0.682	0.694	1.00	0.99	0.99	0.89	0.66	0.17	0.07
	claude-opus-4.7	–	–	–			0.676	0.689	1.00	0.99	0.97	0.91	0.59	0.20	0.07
	claude-sonnet-4.5-think	–	–	–		Y	0.662	0.675	0.99	1.00	0.98	0.91	0.56	0.13	0.05
	claude-opus-4.8	–	–	–			0.619	0.633	0.99	1.00	0.94	0.81	0.41	0.13	0.05
	claude-3.7-sonnet-think	–	–	–		Y	0.616	0.629	0.99	0.99	0.91	0.77	0.48	0.12	0.06
	claude-opus-4-think	–	–	–		Y	0.611	0.625	0.99	0.99	0.97	0.75	0.40	0.12	0.04

Vendor	Model	Params	IKP	Pred.	Arch	Think	Acc.	Raw	T1	T2	T3	T4	T5	T6	T7
DeepSeek	claude-sonnet-4-think	–	–	–		Y	0.574	0.589	0.99	1.00	0.89	0.70	0.34	0.06	0.03
	claude-sonnet-4.5	–	–	–			0.568	0.583	0.99	1.00	0.90	0.71	0.28	0.06	0.03
	claude-opus-4.1	–	–	–			0.561	0.576	0.99	1.00	0.89	0.66	0.25	0.07	0.06
	claude-3.5-haiku	–	–	–			0.543	0.605	0.99	0.99	0.80	0.63	0.30	0.07	0.02
	claude-sonnet-4	–	–	–			0.511	0.527	0.99	0.99	0.85	0.56	0.15	0.02	0.02
	claude-opus-4	–	–	–			0.511	0.526	0.99	0.99	0.83	0.55	0.14	0.03	0.04
	claude-haiku-4.5-think	–	–	–		Y	0.508	0.524	0.98	0.99	0.84	0.54	0.13	0.04	0.03
	claude-haiku-4.5	–	–	–			0.471	0.487	0.99	0.98	0.74	0.46	0.08	0.02	0.01
	claude-3-haiku	–	–	–			0.404	0.459	0.99	0.96	0.58	0.24	0.03	0.01	0.01
	deepseek-v4-pro-think	1.6T	2.2T	(1.4×)	moe	Y	0.773	0.808	1.00	0.99	0.97	0.96	0.84	0.46	0.18
	deepseek-v4-flash	284B	1.5T	(5.4×)	moe		0.748	0.759	0.99	1.00	0.99	0.90	0.78	0.44	0.13
	deepseek-v4-flash-think	284B	1.2T	(4.1×)	moe	Y	0.730	0.741	1.00	0.99	0.99	0.94	0.71	0.36	0.11
	deepseek-r1-think	671B	793B	(1.2×)	moe	Y	0.703	0.757	1.00	0.97	0.96	0.87	0.68	0.25	0.19
	deepseek-v3.2-think	671B	663B	(1.0×)	moe	Y	0.690	0.703	1.00	0.99	0.97	0.86	0.66	0.24	0.10
	deepseek-v3	671B	642B	(1.0×)	moe		0.688	0.720	0.99	0.99	0.96	0.89	0.64	0.23	0.10
	deepseek-v3.2	671B	422B	(0.6×)	moe		0.659	0.672	0.99	1.00	0.96	0.83	0.61	0.17	0.06
	deepseek-v4-pro	1.6T	181B	(0.1×)	moe		0.601	0.648	0.99	0.98	0.90	0.77	0.34	0.16	0.06
	deepseek-v3.1	–	–	–			0.589	0.603	0.99	0.99	0.89	0.72	0.35	0.13	0.04
	deepseek-r1-distill-llama-70b-think	70B	119B	(1.7×)	dense	Y	0.572	0.586	0.98	0.96	0.85	0.66	0.36	0.14	0.04
	deepseek-r1-distill-qwen-32b-think	32B	16B	(0.5×)	dense	Y	0.435	0.488	0.98	0.95	0.70	0.24	0.08	0.03	0.05
xAI	grok-3	–	–	–			0.768	0.826	1.00	1.00	0.98	0.94	0.86	0.47	0.12
	grok-4	–	–	–			0.766	0.821	1.00	0.99	0.97	0.94	0.88	0.41	0.17
	grok-4.20	–	–	–			0.693	0.743	1.00	0.97	0.91	0.76	0.68	0.31	0.22
	grok-4.20-think	–	–	–		Y	0.665	0.727	1.00	0.99	0.97	0.83	0.61	0.19	0.08
	grok-3-mini-think	–	–	–		Y	0.608	0.672	1.00	0.97	0.95	0.77	0.42	0.12	0.03
Xiaomi	mimo-v2.5-pro	–	–	–			0.744	0.755	1.00	0.99	0.99	0.95	0.84	0.33	0.10
	mimo-v2.5-pro-think	–	–	–		Y	0.742	0.754	1.00	0.99	0.99	0.94	0.84	0.33	0.10
	mimo-v2.5	–	–	–			0.695	0.706	1.00	0.98	0.98	0.88	0.67	0.22	0.12
	mimo-v2.5-think	–	–	–		Y	0.684	0.696	1.00	0.99	0.99	0.85	0.67	0.21	0.08
	mimo-v2-flash	309B	348B	(1.1×)	moe		0.646	0.658	0.99	0.96	0.92	0.81	0.56	0.16	0.11
	mimo-v2-flash-think	309B	339B	(1.1×)	moe	Y	0.644	0.657	1.00	0.99	0.93	0.80	0.53	0.14	0.11
	mimo-v2-pro	–	–	–			0.635	0.648	0.99	1.00	0.96	0.82	0.49	0.14	0.04
	mimo-v2-pro-think	–	–	–		Y	0.624	0.638	0.99	0.99	0.94	0.82	0.46	0.13	0.03
Moonshot	kimi-k2.5-think	1.0T	1.4T	(1.4×)	moe	Y	0.742	0.753	0.99	0.99	0.99	0.97	0.92	0.21	0.12
	kimi-k2.7-code	1.0T	1.1T	(1.0×)	moe		0.724	0.735	0.99	1.00	0.97	0.91	0.81	0.23	0.15
	kimi-k2.6-think	1.0T	909B	(0.9×)	moe	Y	0.712	0.724	0.99	1.00	1.00	0.94	0.83	0.16	0.06
	kimi-k2	1.0T	682B	(0.7×)	moe		0.692	0.751	0.99	0.99	0.97	0.85	0.69	0.22	0.12

Vendor	Model	Params	IKP	Pred.	Arch	Think	Acc.	Raw	T1	T2	T3	T4	T5	T6	T7
Tencent	hy3-preview	–	–	–			0.741	0.752	1.00	0.99	0.98	0.90	0.80	0.36	0.15
	hunyuan-a13b-think	80B	6B	(0.1 ×)	moe	Y	0.368	0.384	0.96	0.85	0.44	0.16	0.06	0.04	0.06
	hunyuan-a13b	80B	4B	(0.0 ×)	moe		0.331	0.347	0.95	0.85	0.33	0.10	0.02	0.03	0.02
Z.ai	glm-5	744B	1.3T	(1.8×)	moe		0.737	0.748	0.99	0.99	0.97	0.89	0.82	0.30	0.19
	glm-5-think	744B	1.3T	(1.7×)	moe	Y	0.735	0.746	0.99	0.99	0.98	0.91	0.77	0.34	0.16
	glm-5.1-think	754B	1.2T	(1.5×)	moe	Y	0.729	0.740	0.96	0.98	0.99	0.94	0.78	0.33	0.12
	glm-4.7-think	358B	1.1T	(3.0 ×)	moe	Y	0.723	0.735	0.99	0.99	0.98	0.92	0.71	0.33	0.15
	glm-4.6-think	357B	1.0T	(2.9 ×)	moe	Y	0.722	0.733	0.99	0.98	0.96	0.93	0.71	0.33	0.15
	glm-5.2	–	–	–			0.722	0.733	1.00	1.00	1.00	0.88	0.75	0.29	0.13
	glm-5.1	754B	1.0T	(1.4×)	moe		0.721	0.732	1.00	0.99	0.98	0.91	0.76	0.27	0.12
	glm-5-turbo	–	–	–			0.711	0.723	1.00	0.98	0.98	0.92	0.69	0.28	0.11
	glm-5-turbo-think	–	–	–		Y	0.710	0.769	0.99	0.99	0.98	0.88	0.70	0.31	0.11
	glm-4.5-think	355B	479B	(1.3×)	moe	Y	0.668	0.726	0.99	0.99	0.96	0.80	0.60	0.21	0.12
	glm-4.5-air-think	106B	139B	(1.3×)	moe	Y	0.583	0.597	0.99	0.99	0.90	0.70	0.34	0.11	0.04
	glm-4.7-flash-think	30B	75B	(2.5 ×)	moe	Y	0.540	0.555	0.98	0.99	0.90	0.54	0.22	0.06	0.08
	glm-4-32b	32B	46B	(1.4×)	dense		0.506	0.565	0.97	0.98	0.80	0.52	0.19	0.03	0.05
Alibaba	qwen3.7-max	–	–	–			0.710	0.723	1.00	1.00	0.97	0.90	0.71	0.29	0.10
	qwen3-max	–	–	–			0.684	0.745	1.00	0.99	0.96	0.87	0.67	0.18	0.11
	qwen3.7-plus	–	–	–			0.681	0.693	1.00	1.00	0.95	0.88	0.61	0.21	0.12
	qwen3.5-plus-think	–	–	–		Y	0.663	0.720	0.99	0.99	0.96	0.84	0.54	0.17	0.15
	qwen3.5-397b-a17b-think	397B	356B	(0.9×)	moe	Y	0.647	0.705	0.99	0.99	0.92	0.82	0.49	0.17	0.15
	qwen3.6-plus-think	–	–	–		Y	0.641	0.704	0.99	1.00	0.97	0.85	0.49	0.10	0.09
	qwen-plus	–	–	–			0.629	0.694	0.98	0.99	0.94	0.85	0.48	0.12	0.04
	qwen3.5-122b-a10b-think	122B	216B	(1.8×)	moe	Y	0.613	0.675	0.99	0.99	0.95	0.80	0.39	0.09	0.08
	qwen3-next-80b-a3b	80B	106B	(1.3×)	moe		0.564	0.579	0.98	0.99	0.92	0.71	0.22	0.07	0.04
	qwen3.5-flash-think	–	–	–		Y	0.548	0.606	1.00	0.99	0.89	0.57	0.22	0.07	0.10
	qwen3.5-35b-a3b-think	35B	84B	(2.4 ×)	moe	Y	0.548	0.603	0.99	0.97	0.90	0.55	0.21	0.05	0.15
	qwen3-235b-a22b-think	235B	83B	(0.4 ×)	moe	Y	0.547	0.612	0.99	0.99	0.89	0.72	0.19	0.04	0.02
	qwen-max	–	–	–			0.528	0.589	0.99	0.99	0.84	0.61	0.19	0.03	0.04
	qwen3.5-27b-think	27B	55B	(2.0 ×)	dense	Y	0.518	0.575	0.99	0.97	0.84	0.49	0.17	0.08	0.08
	qwen-2.5-72b	73B	40B	(0.6×)	dense		0.497	0.513	1.00	0.99	0.81	0.49	0.14	0.02	0.03
	qwq-32b-think	32B	29B	(0.9×)	dense		0.475	0.490	0.99	0.97	0.76	0.40	0.13	0.04	0.03
	qwen3-32b-think	32B	26B	(0.8×)	dense	Y	0.466	0.481	0.99	0.97	0.87	0.29	0.08	0.03	0.02
	qwen3-30b-a3b-think	30B	20B	(0.7×)	moe	Y	0.448	0.465	0.99	0.98	0.76	0.33	0.06	0.01	0.02
	qwen-turbo	–	–	–			0.442	0.497	0.96	0.96	0.66	0.36	0.11	0.02	0.03
	qwen3.5-9b-think	9B	18B	(2.0×)	dense	Y	0.441	0.494	0.98	0.95	0.68	0.28	0.08	0.05	0.06
	qwen3-14b-think	14B	16B	(1.2×)	dense	Y	0.436	0.452	0.97	0.97	0.72	0.29	0.04	0.02	0.03

Vendor	Model	Params	IKP Pred.	Arch	Think	Acc.	Raw	T1	T2	T3	T4	T5	T6	T7
NVIDIA	qwen3-8b-think	8B	14B (1.7 $\times$)	dense	Y	0.422	0.471	0.97	0.94	0.55	0.25	0.09	0.04	0.10
	qwen-2.5-7b	8B	4B (0.6 $\times$)	dense		0.344	0.396	0.98	0.97	0.31	0.11	0.03	0.01	0.00
	qwen-2.5-0.5b	494M	349M (0.7 $\times$)	dense		0.170	0.180	0.94	0.15	0.03	0.03	0.00	0.02	0.01
	qwen3.5-0.8b	800M	231M (0.3$\times$)	dense		0.141	0.213	0.46	0.42	0.10	0.00	0.00	0.00	0.00
	qwen3-0.6b	596M	146M (0.2$\times$)	dense		0.110	0.116	0.47	0.20	0.05	0.01	0.01	0.02	0.01
	nemotron-3-ultra	550B	771B (1.4 $\times$)	moe		0.701	0.713	1.00	1.00	0.97	0.91	0.70	0.25	0.07
	nemotron-3-super-120b	120B	203B (1.7 $\times$)	moe		0.609	0.622	0.99	0.98	0.94	0.72	0.42	0.09	0.12
	nemotron-70b	70B	86B (1.2 $\times$)	dense		0.549	0.564	0.99	0.99	0.86	0.68	0.24	0.05	0.03
	nemotron-super-49b-think	49B	24B (0.5$\times$)	dense	Y	0.461	0.477	0.98	0.99	0.69	0.36	0.16	0.02	0.02
	nemotron-3-nano-30b	30B	21B (0.7 $\times$)	moe		0.452	0.468	0.99	0.98	0.67	0.31	0.12	0.04	0.05
ByteDance	nemotron-nano-9b-v2	9B	4B (0.5$\times$)	dense		0.340	0.357	0.98	0.91	0.39	0.09	0.00	0.00	0.01
	seed-2.0-lite-think	–	–	–	Y	0.697	0.757	1.00	0.99	0.95	0.89	0.65	0.29	0.10
	seed-2.0-mini-think	–	–	–	Y	0.629	0.686	0.99	0.97	0.88	0.74	0.52	0.19	0.11
	seed-1.6-think	–	–	–	Y	0.600	0.664	0.99	0.99	0.94	0.76	0.38	0.10	0.03
StepFun	seed-1.6-flash-think	–	–	–	Y	0.457	0.512	1.00	0.96	0.70	0.30	0.12	0.04	0.06
	step-3.7-flash	–	–	–		0.667	0.679	1.00	0.99	0.95	0.77	0.64	0.21	0.11
Amazon	step-3.5-flash-think	197B	429B (2.2$\times$)	moe	Y	0.660	0.676	1.00	0.98	0.96	0.81	0.56	0.20	0.12
	nova-premier	–	–	–		0.656	0.715	1.00	0.98	0.95	0.84	0.48	0.24	0.11
	nova-pro	–	–	–		0.568	0.629	0.99	0.99	0.89	0.66	0.27	0.12	0.05
	nova-micro	–	–	–		0.396	0.450	0.99	0.94	0.50	0.26	0.08	0.00	0.01
Meta	llama-4-maverick	402B	333B (0.8 $\times$)	moe		0.643	0.656	0.99	0.99	0.90	0.80	0.51	0.21	0.09
	llama-3-70b	71B	116B (1.6 $\times$)	dense		0.570	0.634	0.99	0.98	0.86	0.72	0.32	0.08	0.03
	llama-3.3-70b	71B	113B (1.6 $\times$)	dense		0.568	0.583	0.98	0.99	0.84	0.71	0.30	0.10	0.06
	llama-3.1-70b	71B	58B (0.8 $\times$)	dense		0.522	0.538	0.99	0.99	0.81	0.57	0.22	0.05	0.02
	hermes-4-405b	405B	58B (0.1$\times$)	dense		0.522	0.538	0.99	0.98	0.82	0.61	0.15	0.07	0.03
	llama-4-scout	109B	35B (0.3$\times$)	moe		0.488	0.504	0.98	0.93	0.75	0.45	0.22	0.06	0.02
	hermes-3-405b	405B	22B (0.1$\times$)	dense		0.456	0.472	0.99	0.98	0.69	0.40	0.07	0.03	0.01
	llama-3-8b	8B	17B (2.1$\times$)	dense		0.438	0.455	0.98	0.96	0.61	0.36	0.12	0.02	0.02
	llama-3.1-8b	8B	9B (1.1 $\times$)	dense		0.392	0.409	0.97	0.96	0.51	0.26	0.03	0.01	0.01
	llama-3.2-3b	3B	3B (0.9 $\times$)	dense		0.318	0.366	0.95	0.88	0.31	0.07	0.02	0.00	0.00
MiniMax	llama-3.2-1b	1B	908M (0.7 $\times$)	dense		0.236	0.273	0.87	0.60	0.13	0.05	0.00	0.00	0.00
	minimax-m3	–	–	–		0.639	0.652	1.00	0.99	0.96	0.82	0.49	0.14	0.06
	minimax-m2.7-think	230B	113B (0.5$\times$)	moe	Y	0.568	0.635	0.99	0.99	0.88	0.65	0.33	0.06	0.08
	minimax-m1-think	456B	16B (0.0$\times$)	moe	Y	0.434	0.479	0.65	0.61	0.68	0.55	0.35	0.10	0.09
Mistral	mistral-medium-3.1	–	–	–		0.631	0.691	0.99	0.98	0.93	0.85	0.43	0.12	0.11
	mixtral-8x22b	141B	216B (1.5 $\times$)	moe		0.613	0.674	0.99	1.00	0.90	0.78	0.42	0.12	0.08
	mistral-large	123B	103B (0.8 $\times$)	dense		0.562	0.624	0.98	0.98	0.90	0.69	0.28	0.06	0.04

Vendor	Model	Params	IKP Pred.	Arch	Think	Acc.	Raw	T1	T2	T3	T4	T5	T6	T7
Cohere	mixtral-8x7b	47B	68B (1.5 $\times$)	moe		0.533	0.548	0.96	0.98	0.82	0.60	0.25	0.06	0.06
	mistral-small-24b	24B	36B (1.5 $\times$)	dense		0.489	0.505	0.98	0.99	0.81	0.50	0.10	0.03	0.01
	ministral-8b	8B	23B (2.9$\times$)	dense		0.459	0.508	0.97	0.97	0.64	0.34	0.09	0.06	0.14
	mistral-nemo-12b	12B	15B (1.2 $\times$)	dense		0.428	0.444	0.96	0.94	0.56	0.38	0.09	0.02	0.03
	mistral-7b	7B	7B (1.0 $\times$)	dense		0.376	0.422	0.93	0.88	0.45	0.19	0.07	0.04	0.07
	ministral-3b	3B	6B (2.0 $\times$)	dense		0.365	0.412	0.96	0.89	0.40	0.11	0.11	0.03	0.05
	command-a	111B	215B (1.9 $\times$)	dense		0.612	0.675	0.99	0.99	0.93	0.77	0.39	0.15	0.05
	command-r-plus	104B	26B (0.2$\times$)	dense		0.466	0.524	0.99	0.95	0.68	0.46	0.14	0.03	0.02
	command-r7b	7B	6B (0.8 $\times$)	dense		0.364	0.410	0.96	0.84	0.42	0.19	0.03	0.03	0.08
AI21	jamba-large	398B	129B (0.3$\times$)	moe		0.578	0.639	0.99	0.99	0.92	0.70	0.28	0.10	0.06
Prime-intellect	intellect-3-think	106B	87B (0.8 $\times$)	moe	Y	0.550	0.564	0.98	0.96	0.82	0.58	0.26	0.09	0.16
Baidu	ernie-4.5-300b-a47b	300B	73B (0.2$\times$)	moe		0.538	0.601	0.99	0.98	0.84	0.60	0.28	0.06	0.01
Inclusionai	ling-2.6-flash	104B	65B (0.6 $\times$)	moe		0.530	0.751	0.99	0.99	0.87	0.57	0.30	0.00	0.00
Microsoft	phi-4	15B	16B (1.1 $\times$)	dense		0.432	0.448	0.97	0.96	0.62	0.26	0.10	0.05	0.06
	phi-3-mini	4B	2B (0.5 $\times$)	dense		0.289	0.304	0.92	0.73	0.26	0.05	0.01	0.01	0.03
AllenAI	olmo-3.1-32b	32B	12B (0.4$\times$)	dense		0.415	0.432	0.98	0.97	0.61	0.27	0.05	0.01	0.01
Nex-agi	deepseek-v3.1-nex-n1	671B	8B (0.0$\times$)	moe		0.384	0.401	0.98	0.97	0.49	0.22	0.01	0.01	0.01
HuggingFace	smollm2-1.7b	2B	2B (1.4 $\times$)	dense		0.301	0.317	0.95	0.80	0.17	0.06	0.06	0.04	0.03
	smollm2-360m	362M	2B (5.1$\times$)	dense		0.284	0.294	0.82	0.54	0.13	0.11	0.13	0.08	0.18
	smollm2-135m	135M	182M (1.4 $\times$)	dense		0.125	0.131	0.48	0.23	0.09	0.02	0.02	0.01	0.02
Ibm	granite-3.3-2b	2B	2B (1.2 $\times$)	dense		0.300	0.315	0.91	0.74	0.26	0.09	0.04	0.02	0.04

Table 16: Per-tier hallucination rate = wrong / (wrong + correct + refusal). Higher = more confident incorrect answers. Models sorted as in Table 15.

Vendor	Model	Acc.	Raw	T1	T2	T3	T4	T5	T6	T7
Google	gemini-3-flash-think	0.862	0.870	0.00	0.01	0.01	0.04	0.03	0.17	0.68
	gemini-3.1-pro	0.856	0.867	0.00	0.01	0.00	0.02	0.01	0.02	0.24
	gemini-3-flash	0.851	0.860	0.00	0.00	0.01	0.04	0.03	0.19	0.73
	gemini-2.5-pro	0.795	0.805	0.01	0.00	0.01	0.05	0.17	0.41	0.76
	gemini-2.5-pro-think	0.789	0.799	0.01	0.00	0.01	0.06	0.17	0.44	0.75
	gemini-3.1-flash-lite	0.768	0.779	0.00	0.01	0.02	0.07	0.12	0.47	0.74
	gemini-2.5-flash-think	0.702	0.714	0.00	0.01	0.05	0.16	0.31	0.56	0.58
	gemini-2.0-flash	0.663	0.675	0.01	0.01	0.05	0.18	0.44	0.70	0.74
	gemini-2.5-flash	0.621	0.635	0.01	0.01	0.07	0.13	0.15	0.18	0.19
	gemini-2.5-flash-lite-think	0.599	0.613	0.00	0.01	0.10	0.24	0.62	0.79	0.81
	gemini-2.5-flash-lite	0.557	0.571	0.01	0.01	0.10	0.27	0.36	0.58	0.43
	gemma-4-31b	0.470	0.486	0.00	0.01	0.30	0.59	0.74	0.61	0.72
	gemma-3-27b	0.465	0.481	0.01	0.02	0.36	0.64	0.85	0.93	0.91
	gemma-4-26b-a4b	0.452	0.468	0.01	0.01	0.34	0.53	0.65	0.55	0.52
	gemma-2-27b	0.428	0.486	0.01	0.04	0.27	0.48	0.29	0.23	0.16
	gemma-3-12b	0.400	0.416	0.01	0.07	0.50	0.74	0.92	0.90	0.89
	gemma-3n-e4b	0.348	0.393	0.01	0.12	0.65	0.89	0.91	0.94	0.86
	gemma-3-4b	0.330	0.345	0.02	0.21	0.73	0.91	0.90	0.91	0.82
	gemma-2-2b	0.320	0.333	0.06	0.28	0.81	0.85	0.81	0.63	0.74
	gemma-3-1b	0.204	0.216	0.17	0.51	0.89	0.97	0.97	0.97	0.94
OpenAI	gemma-3-270m	0.143	0.145	0.29	0.35	0.51	0.42	0.27	0.26	0.46
	gpt-5.5-pro	0.834	0.844	0.00	0.00	0.02	0.02	0.06	0.27	0.76
	gpt-5.5	0.826	0.835	0.00	0.00	0.01	0.04	0.04	0.31	0.72
	gpt-5.5-think	0.826	0.835	0.00	0.00	0.02	0.04	0.05	0.28	0.72
	gpt-5.4-pro	0.778	0.789	0.00	0.01	0.01	0.06	0.12	0.48	0.73
	gpt-4.1	0.772	0.782	0.00	0.01	0.01	0.08	0.18	0.52	0.77
	gpt-5-pro	0.770	0.780	0.00	0.01	0.01	0.04	0.08	0.31	0.35
	o3	0.769	0.780	0.00	0.01	0.01	0.08	0.08	0.37	0.44
	gpt-5-think	0.765	0.776	0.00	0.01	0.01	0.06	0.04	0.26	0.32
	gpt-5	0.758	0.769	0.00	0.01	0.01	0.06	0.05	0.26	0.31
	o1	0.736	0.748	0.01	0.00	0.02	0.05	0.05	0.14	0.18
	gpt-5.4	0.723	0.735	0.00	0.01	0.03	0.07	0.14	0.42	0.51
	gpt-5.3	0.722	0.734	0.00	0.01	0.01	0.07	0.12	0.27	0.41
	gpt-4-turbo	0.722	0.733	0.00	0.00	0.02	0.07	0.13	0.53	0.47

Vendor	Model	Acc.	Raw	T1	T2	T3	T4	T5	T6	T7
Anthropic	gpt-5.2-pro	0.688	0.700	0.00	0.01	0.02	0.07	0.13	0.22	0.23
	gpt-5.1	0.687	0.699	0.00	0.01	0.02	0.08	0.04	0.14	0.16
	gpt-4	0.686	0.698	0.01	0.01	0.03	0.09	0.09	0.25	0.22
	gpt-5.2	0.680	0.693	0.00	0.01	0.01	0.07	0.13	0.20	0.20
	gpt-4.1-mini	0.671	0.683	0.01	0.01	0.06	0.19	0.46	0.74	0.83
	gpt-4o	0.654	0.667	0.01	0.01	0.03	0.06	0.02	0.15	0.18
	o4-mini-think	0.650	0.663	0.00	0.01	0.03	0.11	0.31	0.40	0.42
	gpt-5.4-mini	0.644	0.657	0.00	0.01	0.07	0.12	0.20	0.47	0.31
	gpt-3.5-turbo	0.622	0.635	0.01	0.01	0.05	0.21	0.25	0.64	0.53
	gpt-oss-120b-think	0.564	0.578	0.02	0.01	0.14	0.36	0.47	0.60	0.55
	gpt-5-mini-think	0.563	0.578	0.00	0.01	0.01	0.04	0.04	0.04	0.06
	gpt-5-mini	0.555	0.571	0.00	0.00	0.01	0.05	0.02	0.03	0.06
	gpt-4o-mini	0.554	0.569	0.01	0.02	0.10	0.32	0.48	0.71	0.60
	gpt-4.1-nano	0.536	0.551	0.00	0.04	0.16	0.43	0.55	0.73	0.69
	o3-mini	0.527	0.543	0.01	0.01	0.08	0.28	0.41	0.24	0.34
	gpt-5-nano-think	0.466	0.483	0.00	0.01	0.05	0.08	0.03	0.03	0.07
	gpt-5-nano	0.459	0.475	0.01	0.01	0.03	0.11	0.06	0.01	0.07
	gpt-oss-20b-think	0.435	0.451	0.01	0.04	0.35	0.69	0.89	0.82	0.73
	gpt-5.4-nano	0.400	0.416	0.01	0.03	0.12	0.24	0.11	0.14	0.15
	claude-fable-5	0.805	0.815	0.01	0.01	0.01	0.05	0.05	0.15	0.20
	claude-opus-4.6-think	0.751	0.763	0.00	0.00	0.01	0.03	0.09	0.26	0.28
	claude-opus-4.6	0.735	0.747	0.00	0.00	0.01	0.04	0.07	0.29	0.28
	claude-3.7-sonnet	0.726	0.738	0.01	0.00	0.03	0.09	0.08	0.19	0.22
	claude-opus-4.5	0.711	0.722	0.00	0.01	0.01	0.05	0.09	0.16	0.30
	claude-opus-4.1-think	0.707	0.719	0.01	0.00	0.00	0.04	0.06	0.11	0.17
	claude-opus-4.5-think	0.706	0.718	0.00	0.00	0.01	0.04	0.06	0.11	0.27
	claude-sonnet-4.6-think	0.700	0.712	0.00	0.01	0.04	0.07	0.16	0.36	0.33
	claude-sonnet-4.6	0.700	0.712	0.00	0.00	0.03	0.05	0.15	0.41	0.34
	claude-opus-4.7-think	0.698	0.710	0.00	0.01	0.01	0.03	0.07	0.17	0.16
	claude-sonnet-5	0.682	0.694	0.00	0.01	0.01	0.09	0.12	0.26	0.17
	claude-opus-4.7	0.676	0.689	0.00	0.01	0.02	0.03	0.05	0.10	0.15
	claude-sonnet-4.5-think	0.662	0.675	0.00	0.00	0.02	0.04	0.09	0.15	0.16
	claude-opus-4.8	0.619	0.633	0.01	0.00	0.00	0.03	0.04	0.07	0.10
	claude-3.7-sonnet-think	0.616	0.629	0.01	0.01	0.01	0.05	0.02	0.04	0.08
	claude-opus-4-think	0.611	0.625	0.00	0.00	0.01	0.05	0.04	0.05	0.11
	claude-sonnet-4-think	0.574	0.589	0.00	0.00	0.04	0.06	0.02	0.05	0.08
	claude-sonnet-4.5	0.568	0.583	0.00	0.00	0.02	0.05	0.03	0.07	0.13

Vendor	Model	Acc.	Raw	T1	T2	T3	T4	T5	T6	T7
DeepSeek	claude-opus-4.1	0.561	0.576	0.00	0.00	0.02	0.05	0.01	0.05	0.13
	claude-3.5-haiku	0.543	0.605	0.01	0.01	0.08	0.12	0.14	0.06	0.08
	claude-sonnet-4	0.511	0.527	0.00	0.01	0.00	0.05	0.01	0.03	0.06
	claude-opus-4	0.511	0.526	0.01	0.01	0.01	0.04	0.01	0.04	0.07
	claude-haiku-4.5-think	0.508	0.524	0.01	0.01	0.08	0.15	0.13	0.07	0.09
	claude-haiku-4.5	0.471	0.487	0.01	0.01	0.09	0.13	0.11	0.12	0.09
	claude-3-haiku	0.404	0.459	0.01	0.04	0.07	0.08	0.01	0.00	0.01
	deepseek-v4-pro-think	0.773	0.808	0.00	0.01	0.02	0.04	0.12	0.40	0.44
	deepseek-v4-flash	0.748	0.759	0.01	0.00	0.01	0.10	0.15	0.41	0.52
	deepseek-v4-flash-think	0.730	0.741	0.00	0.01	0.01	0.06	0.20	0.44	0.56
	deepseek-r1-think	0.703	0.757	0.00	0.03	0.03	0.08	0.15	0.31	0.33
	deepseek-v3.2-think	0.690	0.703	0.00	0.01	0.03	0.07	0.17	0.38	0.47
	deepseek-v3	0.688	0.720	0.00	0.01	0.03	0.09	0.16	0.40	0.43
	deepseek-v3.2	0.659	0.672	0.01	0.00	0.02	0.09	0.10	0.25	0.33
	deepseek-v4-pro	0.601	0.648	0.01	0.01	0.00	0.02	0.02	0.08	0.15
	deepseek-v3.1	0.589	0.603	0.00	0.01	0.03	0.08	0.04	0.11	0.27
	deepseek-r1-distill-llama-70b-think	0.572	0.586	0.01	0.04	0.14	0.30	0.36	0.52	0.43
xAI	deepseek-r1-distill-qwen-32b-think	0.435	0.488	0.02	0.04	0.27	0.72	0.76	0.78	0.64
	grok-3	0.768	0.826	0.00	0.00	0.02	0.06	0.11	0.43	0.34
	grok-4	0.766	0.821	0.00	0.01	0.02	0.06	0.05	0.26	0.15
	grok-4.20	0.693	0.743	0.00	0.03	0.09	0.24	0.32	0.69	0.78
	grok-4.20-think	0.665	0.727	0.00	0.01	0.02	0.10	0.13	0.20	0.11
Xiaomi	grok-3-mini-think	0.608	0.672	0.00	0.03	0.04	0.13	0.20	0.23	0.15
	mimo-v2.5-pro	0.744	0.755	0.00	0.01	0.01	0.04	0.12	0.36	0.41
	mimo-v2.5-pro-think	0.742	0.754	0.00	0.01	0.01	0.05	0.09	0.38	0.45
	mimo-v2.5	0.695	0.706	0.00	0.01	0.02	0.11	0.25	0.47	0.49
	mimo-v2.5-think	0.684	0.696	0.00	0.01	0.01	0.13	0.23	0.45	0.51
Moonshot	mimo-v2-flash	0.646	0.658	0.01	0.04	0.08	0.18	0.36	0.65	0.62
	mimo-v2-flash-think	0.644	0.657	0.00	0.01	0.05	0.19	0.37	0.63	0.58
	mimo-v2-pro	0.635	0.648	0.01	0.00	0.01	0.07	0.08	0.05	0.17
	mimo-v2-pro-think	0.624	0.638	0.01	0.01	0.01	0.05	0.06	0.08	0.18
	kimi-k2.5-think	0.742	0.753	0.00	0.01	0.01	0.03	0.07	0.58	0.45
Tencent	kimi-k2.7-code	0.724	0.735	0.01	0.00	0.03	0.09	0.17	0.63	0.56
	kimi-k2.6-think	0.712	0.724	0.01	0.00	0.00	0.03	0.03	0.28	0.19
	kimi-k2	0.692	0.751	0.01	0.01	0.03	0.13	0.14	0.33	0.22
	hy3-preview	0.741	0.752	0.00	0.01	0.02	0.10	0.19	0.53	0.59
	hunyuan-a13b-think	0.368	0.384	0.04	0.14	0.55	0.83	0.86	0.74	0.64

Vendor	Model	Acc.	Raw	T1	T2	T3	T4	T5	T6	T7
Z.ai	hunyuan-a13b	0.331	0.347	0.04	0.13	0.49	0.59	0.34	0.34	0.27
	glm-5	0.737	0.748	0.01	0.01	0.02	0.11	0.17	0.60	0.66
	glm-5-think	0.735	0.746	0.01	0.01	0.02	0.09	0.22	0.55	0.69
	glm-5.1-think	0.729	0.740	0.03	0.02	0.01	0.06	0.18	0.41	0.42
	glm-4.7-think	0.723	0.735	0.01	0.01	0.02	0.08	0.27	0.58	0.66
	glm-4.6-think	0.722	0.733	0.01	0.01	0.04	0.07	0.28	0.62	0.77
	glm-5.2	0.722	0.733	0.00	0.00	0.00	0.10	0.15	0.49	0.42
	glm-5.1	0.721	0.732	0.00	0.01	0.02	0.08	0.18	0.54	0.44
	glm-5-turbo	0.711	0.723	0.00	0.01	0.02	0.07	0.21	0.51	0.45
	glm-5-turbo-think	0.710	0.769	0.00	0.01	0.02	0.11	0.21	0.49	0.43
	glm-4.5-think	0.668	0.726	0.00	0.01	0.01	0.12	0.14	0.24	0.23
	glm-4.5-air-think	0.583	0.597	0.01	0.01	0.07	0.20	0.24	0.32	0.26
	glm-4.7-flash-think	0.540	0.555	0.01	0.01	0.10	0.45	0.74	0.83	0.80
	glm-4-32b	0.506	0.565	0.01	0.02	0.17	0.37	0.42	0.42	0.19
Alibaba	qwen3.7-max	0.710	0.723	0.00	0.00	0.02	0.07	0.12	0.20	0.20
	qwen3-max	0.684	0.745	0.00	0.01	0.04	0.12	0.17	0.52	0.22
	qwen3.7-plus	0.681	0.693	0.00	0.00	0.05	0.10	0.29	0.42	0.35
	qwen3.5-plus-think	0.663	0.720	0.01	0.01	0.04	0.14	0.31	0.59	0.38
	qwen3.5-397b-a17b-think	0.647	0.705	0.01	0.01	0.07	0.16	0.35	0.58	0.34
	qwen3.6-plus-think	0.641	0.704	0.00	0.00	0.01	0.09	0.12	0.09	0.09
	qwen-plus	0.629	0.694	0.01	0.01	0.05	0.13	0.29	0.47	0.24
	qwen3.5-122b-a10b-think	0.613	0.675	0.01	0.01	0.05	0.20	0.50	0.71	0.50
	qwen3-next-80b-a3b	0.564	0.579	0.01	0.01	0.05	0.24	0.36	0.52	0.40
	qwen3.5-flash-think	0.548	0.606	0.00	0.01	0.11	0.41	0.65	0.76	0.59
	qwen3.5-35b-a3b-think	0.548	0.603	0.00	0.03	0.10	0.44	0.64	0.76	0.54
	qwen3-235b-a22b-think	0.547	0.612	0.01	0.01	0.10	0.21	0.39	0.46	0.17
	qwen-max	0.528	0.589	0.01	0.01	0.10	0.20	0.14	0.08	0.06
	qwen3.5-27b-think	0.518	0.575	0.01	0.03	0.16	0.49	0.79	0.81	0.79
	qwen-2.5-72b	0.497	0.513	0.00	0.01	0.15	0.36	0.25	0.13	0.20
	qwq-32b-think	0.475	0.490	0.01	0.03	0.21	0.51	0.59	0.55	0.49
	qwen3-32b-think	0.466	0.481	0.01	0.03	0.11	0.62	0.55	0.53	0.38
	qwen3-30b-a3b-think	0.448	0.465	0.01	0.01	0.19	0.48	0.42	0.30	0.26
	qwen-turbo	0.442	0.497	0.04	0.03	0.30	0.46	0.36	0.26	0.17
	qwen3.5-9b-think	0.441	0.494	0.00	0.03	0.29	0.63	0.75	0.67	0.67
	qwen3-14b-think	0.436	0.452	0.03	0.03	0.23	0.57	0.44	0.38	0.38
	qwen3-8b-think	0.422	0.471	0.01	0.06	0.41	0.62	0.58	0.49	0.47
	qwen-2.5-7b	0.344	0.396	0.01	0.03	0.61	0.66	0.44	0.42	0.18

Vendor	Model	Acc.	Raw	T1	T2	T3	T4	T5	T6	T7
NVIDIA	qwen-2.5-0.5b	0.170	0.180	0.06	0.64	0.79	0.77	0.69	0.73	0.63
	qwen3.5-0.8b	0.141	0.213	0.10	0.24	0.33	0.26	0.42	0.57	0.81
	qwen3-0.6b	0.110	0.116	0.29	0.34	0.38	0.24	0.18	0.14	0.28
	nemotron-3-ultra	0.701	0.713	0.00	0.00	0.03	0.08	0.13	0.41	0.35
	nemotron-3-super-120b	0.609	0.622	0.00	0.01	0.05	0.24	0.46	0.50	0.49
	nemotron-70b	0.549	0.564	0.00	0.00	0.04	0.10	0.03	0.06	0.12
	nemotron-super-49b-think	0.461	0.477	0.01	0.01	0.28	0.47	0.33	0.32	0.28
	nemotron-3-nano-30b	0.452	0.468	0.01	0.02	0.32	0.65	0.72	0.71	0.63
ByteDance	nemotron-nano-9b-v2	0.340	0.357	0.01	0.07	0.11	0.15	0.03	0.03	0.08
	seed-2.0-lite-think	0.697	0.757	0.00	0.01	0.04	0.10	0.29	0.53	0.43
	seed-2.0-mini-think	0.629	0.686	0.01	0.03	0.10	0.26	0.44	0.75	0.71
	seed-1.6-think	0.600	0.664	0.00	0.01	0.05	0.23	0.44	0.62	0.48
StepFun	seed-1.6-flash-think	0.457	0.512	0.00	0.04	0.28	0.66	0.75	0.70	0.72
	step-3.7-flash	0.667	0.679	0.00	0.01	0.05	0.20	0.27	0.39	0.46
Amazon	step-3.5-flash-think	0.660	0.676	0.00	0.01	0.04	0.18	0.34	0.59	0.60
	nova-premier	0.656	0.715	0.00	0.01	0.05	0.15	0.49	0.71	0.80
	nova-pro	0.568	0.629	0.01	0.01	0.09	0.27	0.40	0.67	0.39
Meta	nova-micro	0.396	0.450	0.01	0.06	0.42	0.63	0.44	0.51	0.33
	llama-4-maverick	0.643	0.656	0.01	0.01	0.08	0.14	0.14	0.20	0.22
	llama-3-70b	0.570	0.634	0.01	0.01	0.08	0.15	0.09	0.12	0.10
	llama-3.3-70b	0.568	0.583	0.02	0.01	0.11	0.14	0.09	0.13	0.19
	llama-3.1-70b	0.522	0.538	0.01	0.01	0.04	0.09	0.04	0.05	0.10
	hermes-4-405b	0.522	0.538	0.01	0.02	0.03	0.04	0.02	0.04	0.09
	llama-4-scout	0.488	0.504	0.02	0.07	0.25	0.44	0.43	0.44	0.37
	hermes-3-405b	0.456	0.472	0.01	0.02	0.03	0.04	0.00	0.02	0.05
	llama-3-8b	0.438	0.455	0.01	0.04	0.29	0.35	0.40	0.11	0.28
	llama-3.1-8b	0.392	0.409	0.01	0.03	0.12	0.12	0.03	0.02	0.06
	llama-3.2-3b	0.318	0.366	0.02	0.07	0.23	0.22	0.03	0.01	0.01
	llama-3.2-1b	0.236	0.273	0.08	0.20	0.42	0.24	0.06	0.03	0.04
MiniMax	minimax-m3	0.639	0.652	0.00	0.01	0.03	0.10	0.17	0.30	0.25
	minimax-m2.7-think	0.568	0.635	0.01	0.01	0.09	0.23	0.31	0.31	0.27
	minimax-m1-think	0.434	0.479	0.34	0.38	0.30	0.39	0.37	0.47	0.33
Mistral	mistral-medium-3.1	0.631	0.691	0.01	0.01	0.05	0.11	0.32	0.31	0.56
	mixtral-8x22b	0.613	0.674	0.01	0.00	0.05	0.14	0.18	0.40	0.13
	mistral-large	0.562	0.624	0.01	0.01	0.03	0.18	0.25	0.17	0.19
	mixtral-8x7b	0.533	0.548	0.02	0.02	0.14	0.31	0.40	0.43	0.42
	mistral-small-24b	0.489	0.505	0.01	0.01	0.11	0.26	0.23	0.18	0.24

Vendor	Model	Acc.	Raw	T1	T2	T3	T4	T5	T6	T7
Cohere	ministral-8b	0.459	0.508	0.03	0.03	0.34	0.64	0.88	0.85	0.82
	mistral-nemo-12b	0.428	0.444	0.03	0.04	0.34	0.48	0.61	0.50	0.51
	mistral-7b	0.376	0.422	0.06	0.10	0.51	0.73	0.76	0.63	0.73
	ministral-3b	0.365	0.412	0.04	0.12	0.58	0.85	0.80	0.84	0.85
	command-a	0.612	0.675	0.00	0.01	0.05	0.15	0.31	0.30	0.34
	command-r-plus	0.466	0.524	0.01	0.02	0.09	0.19	0.08	0.10	0.06
	command-r7b	0.364	0.410	0.04	0.15	0.49	0.65	0.48	0.62	0.29
AI21	jamba-large	0.578	0.639	0.01	0.01	0.02	0.08	0.08	0.07	0.02
Prime-intellect	intellect-3-think	0.550	0.564	0.01	0.04	0.18	0.41	0.72	0.87	0.79
Baidu	ernie-4.5-300b-a47b	0.538	0.601	0.01	0.01	0.05	0.16	0.11	0.21	0.09
Inclusionai	ling-2.6-flash	0.530	0.751	0.01	0.01	0.12	0.26	0.27	0.20	0.00
Microsoft	phi-4	0.432	0.448	0.02	0.04	0.34	0.66	0.78	0.73	0.72
	phi-3-mini	0.289	0.304	0.07	0.26	0.64	0.74	0.57	0.57	0.57
AllenAI	olmo-3.1-32b	0.415	0.432	0.01	0.03	0.29	0.49	0.31	0.21	0.28
Nex-agi	deepseek-v3.1-nex-n1	0.384	0.401	0.01	0.01	0.02	0.04	0.00	0.01	0.07
HuggingFace	smollm2-1.7b	0.301	0.317	0.04	0.20	0.83	0.94	0.94	0.95	0.97
	smollm2-360m	0.284	0.294	0.17	0.46	0.87	0.88	0.87	0.91	0.82
	smollm2-135m	0.125	0.131	0.51	0.77	0.91	0.97	0.96	0.98	0.95
Ibm	granite-3.3-2b	0.300	0.315	0.09	0.25	0.73	0.88	0.89	0.89	0.87

F Densing Law Falsification: Full Regression Results

This appendix provides the full regression tables supporting the Densing-Law falsification in Section 6.7.

Dataset. We restrict to open-weight models with published parameter counts and completed IKP evaluations: $n = 96$ models from 23 vendors, with release dates spanning 2023-09-27 (Mistral-7B) to 2026-04-24 (DeepSeek V4). Release dates were manually collected and verified against authoritative sources (vendor blog posts, HuggingFace model cards, and archived press releases). For ‘-think’ variants that share weights with a non-thinking base, the release date equals that of the base model. The full dated configuration, the analysis script, and the tidy data CSV are released with the paper.

Specifications. All fits are OLS with IKP accuracy ($\lambda = 0$) as the dependent variable. `months` is the release date minus 2024-01-01, in months. `thinking` is a 0/1 indicator; `MoE` is a 0/1 indicator for mixture-of-experts architecture. The baseline M0 contains only $\log_{10}(N_B)$.

Specification	$\log_{10}(N_B)$	months	thinking	MoE	R^2
M0: $\log_{10}(\text{params})$ only	+0.1362***	—	—	—	0.7612
M1: + months	+0.1397***	−0.0010	—	—	0.7635
M2: + thinking	+0.1390***	−0.0012	+0.0066	—	0.7639
M3: + MoE	+0.1507***	−0.0004	+0.0079	−0.0396	0.7718
M_t : time only	—	+0.0062**	—	—	0.1033

Table 17: Full regression results for the Densing-Law analysis ($n = 96$). Entries are point estimates; *** $p < 0.001$, ** $p < 0.01$. The time coefficient is statistically indistinguishable from zero in every specification that controls for parameter count (M1–M3). The time-only model (M_t) produces a nominally significant but much smaller slope than the Densing prediction, and its explanatory power ($R^2 = 0.12$) collapses once log-params is included—indicating the residual time correlation is a selection effect (newer models skew larger), not a real per-parameter trend.

Tests against the Densing prediction. Under the observed $\log_{10}(N)$ slope of $\beta_1 = 0.136$, the Densing Law predicts a monthly accuracy gain of $\beta_2^{\text{Densing}} = \beta_1 \cdot \log_{10}(2)/3.5 \approx +0.01171/\text{month}$. Table 18 reports two-sided t -tests of the fitted time coefficient against both 0 and this Densing target.

Spec	$\hat{\beta}_2$ (months)	SE	p (vs 0)	t (vs Densing)	p (vs Densing)
M1	−0.00104	0.00109	0.34	−11.85	$<10^{-15}$
M2	−0.00120	0.00118	0.31	−11.04	$<10^{-15}$
M3	−0.00043	0.00121	0.72	−9.87	$<10^{-15}$

Table 18: Hypothesis tests on the time coefficient. In every specification, the null of zero time trend is not rejected (all $p > 0.3$), while the Densing-Law prediction is rejected with effectively certainty ($|t| > 10$).

Bootstrap inference. A 4000-replicate nonparametric bootstrap on M1 gives a 95% CI for the time coefficient of $[-0.00308, +0.00079]$, tightly centered on zero and well below the Densing target of $+0.01171$. The gap between the upper bound and the Densing prediction is roughly $11\times$ the bootstrap standard error.

Robustness. Three additional checks confirm the main finding: (i) excluding the two DeepSeek-R1 distilled models ($n = 79$) leaves $\hat{\beta}_2 = -0.00051/\text{month}$; (ii) excluding the five landmark models ($n = 76$) leaves $\hat{\beta}_2 = -0.00018/\text{month}$. In both cases the Densing prediction is rejected at $p < 10^{-15}$. Check (iii) uses $\log_{10}(\text{active_B})$ in place of total parameters and yields $\hat{\beta}_2 = +0.0063/\text{month}$ —nominally significant but still only half the Densing target, and also rejecting the Densing prediction ($p \approx 10^{-4}$). The apparent time effect on the active-params regression is a known artifact: MoE models store knowledge across all experts, not just the ones activated per token (Section 6), so active-params overstates N_{fact} for dense models and understates it for MoEs in a way that correlates with release date (newer models are more often MoE). This is why our preferred specification uses total parameters.

Date uncertainty. Release dates for post-January-2026 models (e.g., Claude Opus 4.7, Qwen 3.5 series, Kimi K2.6, GLM-5.1, Ling-2.6-flash) were collected contemporaneously; other dates are vendor-announced release dates. A small amount of date noise (± 1 month per observation) would inflate $\text{SE}(\hat{\beta}_2)$ but leaves the point estimate unchanged in expectation. To hide a Densing-magnitude effect would require systematic backdating of post-2025 releases by >3 months each, which is not what our sources support.

G Cross-Generation and Model Family Analysis

Figure 14 traces knowledge evolution across model generations for 7 major families. Most families show steady improvement, but several notable regressions occur when vendors optimize for efficiency or safety at the expense of factual recall. Figure 15 reveals the internal size stratification of the GPT-5 family.

GPT-5 family internal structure. The 15 GPT-5 variants in our evaluation reveal OpenAI’s model lineup structure through both coarse (overall-accuracy) and fine (knowledge-fingerprint) analysis. Clear size stratification appears at T5: nano (4%), mini (20%), base (76%)—a $\sim 20\times$ knowledge gap—with a “pro” shelf above the base (GPT-5.4-pro at 69.2% is the strongest OpenAI model in the evaluation). Each .x iteration generally improves overall accuracy, but GPT-5.4-nano regresses from GPT-5-nano on T3 (57.5% vs 72.0%), suggesting a different distillation approach between generations. The fingerprint evidence (Section 6.9, Appendix H) is consistent with three distinct training runs: a GPT-5 / 5-pro / 5-think cluster (shared base), a GPT-5.1 / 5.2 / 5.2-pro cluster, and a GPT-5.3 / 5.4 / 5.4-pro cluster, with retrain-regime HSS at every transition between them.

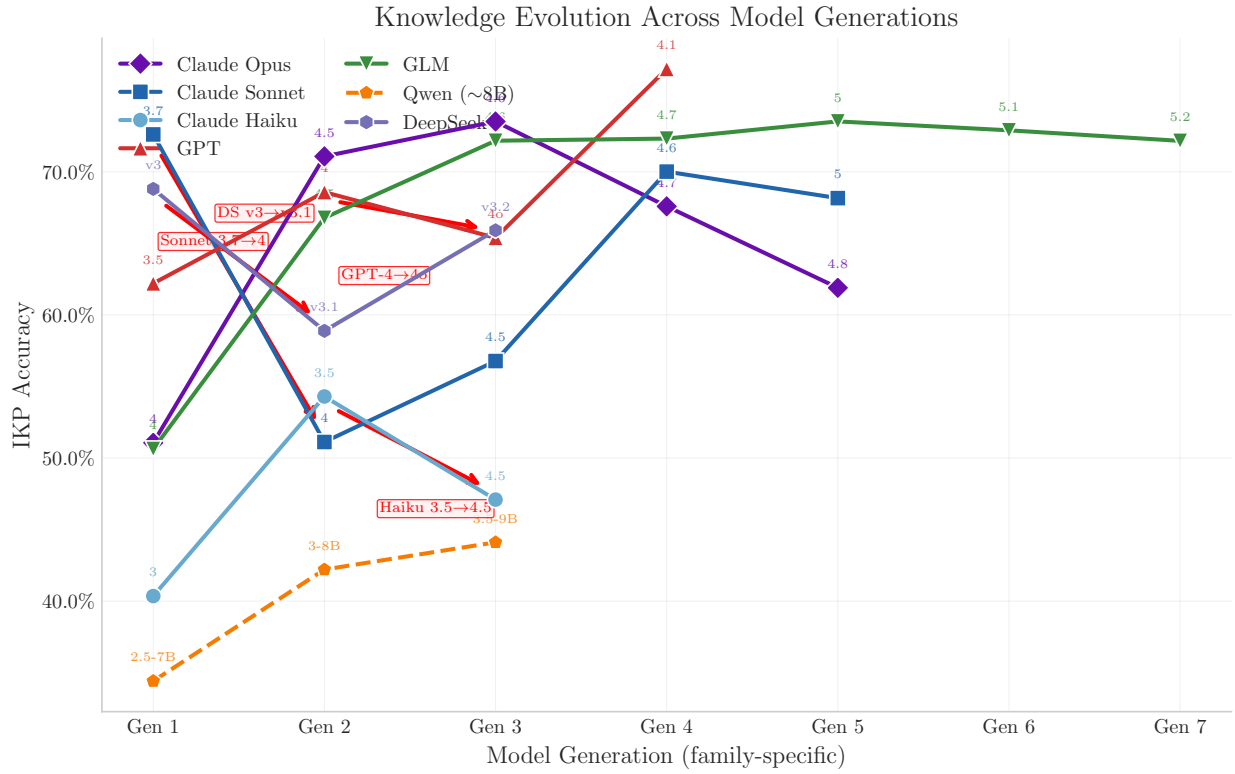


Figure 14: Knowledge evolution across model generations. Red boxes highlight regressions: GPT-4→4o (smaller efficiency-optimized successor), Claude Sonnet 3.7→4 (RLHF conservatism), Claude Haiku 3.5→4.5 (smaller post-training). GLM shows the steadiest improvement across 6 generations.

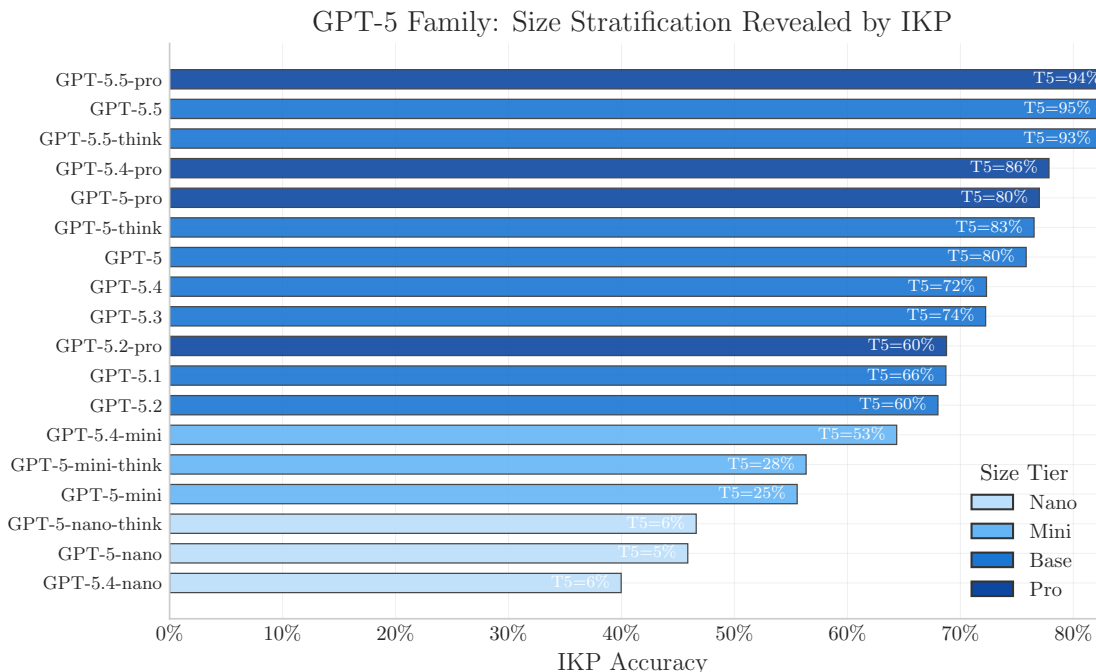


Figure 15: GPT-5 family size stratification revealed by IKP. T5 accuracy (annotated) shows clear tiers: nano (0–3%), mini (19–41%), base (51–77%), pro (74–76%). GPT-5, GPT-5-pro, and GPT-5-think score nearly identically, suggesting the same base model with different inference configurations.

H Knowledge Fingerprinting: Detailed Tables

This appendix expands Section 6.9 with the complete consecutive-pair tables for each tracked model family and the full list of cross-vendor outliers. For every pair of models we report, on the 400 T5–T6 probes, the Jaccard similarity J of correct-answer sets, the lift (= observed/expected, under the null of independence) of the correct-set intersection, the hallucination similarity HSS, the number of joint-wrong probes (“both_w”), and a regime classification (shared-base, lineage, retrained, independent). The thresholds are those stated in Section 6.9: shared-base requires $HSS \geq 0.30$ and $J \geq 0.60$; lineage requires $HSS \geq 0.10$ and $J \geq 0.50$; retrained is $HSS < 0.10$ with ≥ 10 joint-wrong probes (limited statistical resolution when both_w is much smaller).

Why HSS? Raw Jaccard on correct-answer sets is dominated by probes that nearly all frontier models get right, producing high J even for truly independent models (for example, GPT-5 vs. Claude Opus 4.5 scores $J = 0.56$ despite being independent trainings by different vendors). Lift partially corrects for this but is noisy when either model’s correct set is small. Hallucination similarity—the rate at which two models produce the same normalized wrong answer on a probe that both got wrong—is a much cleaner signal: independent models almost never converge on the same wrong rare fact, while weight-sharing siblings do so on 30–55% of joint-wrong probes. A bright threshold around $HSS = 0.30$ separates the two regimes empirically.

When the three metrics disagree. The three metrics are complementary, not redundant. Heavy post-training can preserve rare-fact knowledge (high J and high lift) while rewriting the

generation style enough that a model gives *differently worded* wrong answers on hard probes—driving HSS down. Nemotron-70B vs. Llama-3.1-70B is the textbook example: $J = 0.79$ and lift = 7.5 (clear shared base) but HSS = 0.08 (SFT has scrambled the surface form of wrong answers). Conversely, a high-HSS pair with low J and low lift is a weaker signal than it sounds, because it may reflect a single over-represented hallucination (e.g., a common name collision) rather than broad knowledge overlap. We classify a pair as *shared base* only when all three metrics point the same way; one-sided signals are reported as “possible lineage” and should be investigated case-by-case.

Release-practice summary. Applied to consecutive-generation pairs across all tracked families (see Table 20), three release patterns recur:

- *Fine-tune-on-same-base releases* (lineage): Claude Opus 4 $\rightarrow$ 4.1, Sonnet 4 $\rightarrow$ 4.5, DeepSeek V3 $\rightarrow$ V3.1 $\rightarrow$ V3.2, GLM 4.6 $\rightarrow$ 4.7, Kimi K2 $\rightarrow$ K2.5 $\rightarrow$ K2.6. These preserve a large fraction of the base’s rare-fact mistakes.
- *Inference/alignment variants of a shared base* (shared-base): GPT-5 / GPT-5-pro / GPT-5-think; GPT-5.2 / 5.2-pro; Claude Opus 4.5 / 4.5-think. These show the highest HSS in the dataset.
- *Full retrains* (retrained): every GPT-5. $x \rightarrow$ 5.($x+1$) transition for $x = 0, 1, 2, 3$; Claude Opus 4.6 $\rightarrow$ 4.7; every cross-generation Gemini pair; GLM 4.7 $\rightarrow$ 5 and 5 $\rightarrow$ 5.1. For these, the same-wrong-answer count is statistically indistinguishable from what we see between models built by different vendors, despite the shared version-family label.

Cross-vendor outliers: how to read the table. Applying the same test to the $\sim 13\,000$ cross-vendor pairs flags a small fraction at $\text{HSS} \geq 0.20$ with ≥ 10 joint-wrong probes (the full list is Table 21). Interpretation requires care: a single high-HSS pair between two independent frontier models is not strong evidence of cross-family distillation given the number of pairs tested. More suggestive is the *pattern* across pairs involving the same model on one side:

- **Baidu ERNIE 4.5-300B-A47B** is above the cross-vendor threshold against five independent models simultaneously (GPT-4o HSS = 0.44, Llama-3-70B 0.43, Mistral-Large 0.40, Qwen-Max 0.33, Mistral-Small-24B 0.33). This is the pattern expected when a model is trained on a large mixture of distilled outputs from Western frontier models rather than on a single teacher.
- **Llama 3.1 70B** appears as an apparent “teacher” against many other models ($\text{HSS} \geq 0.30$ vs. grok-3, gemini-2.0-flash, qwen3-max, nova-pro, and several older OpenAI models). This is consistent with Llama 3.1 being the most widely-used open base for synthetic-data generation in 2024–2025, so its characteristic hallucinations leak into many downstream datasets. It is *not* evidence that any single downstream vendor distilled from it directly.
- **deepseek-r1-distill-llama-70b vs. Llama-3.1-70B** (HSS = 0.33) is a known distillation-into-base pairing and serves as a positive control for the method.
- Individual high-HSS frontier-vs-frontier pairs (GPT-5 vs. Grok-4 at 0.31; GPT-5-pro vs. Kimi-K2.6 at 0.27) are individually above threshold but have too few joint-wrong probes, at our probe count, to reject an innocent-explanation null (e.g., training on similar recent web snapshots or on overlapping synthetic-data mixtures).

Limitations. The method identifies *statistical* overlap in knowledge and hallucination patterns; it does not distinguish between direct distillation and shared training data (e.g., two vendors independently scraping the same recent Wikipedia dump or licensing the same curated corpus). It also has limited resolution when the number of joint-wrong rare-fact probes is small: frontier models that each score 70–80% on T5–T6 jointly get only 20–50 probes wrong, and HSS estimated from small samples is noisy. Probe-pool expansion into T6–T7 would tighten these estimates and is a natural next step.

Known-provenance positive controls. As a sanity check, Table 19 reports metrics for six pairs whose parent-child relationship is publicly disclosed. Every pair with ≥ 10 joint-wrong probes shows elevated lift (2.4–7.5) and J (0.32–0.79) consistent with shared training signal; HSS on heavily post-trained derivatives (Nemotron, R1-distill) is suppressed for the reason discussed above, which is why we recommend using all three metrics jointly rather than thresholding HSS alone.

Student (derived)	Parent (base/teacher)	J	lift	HSS	both_w
deepseek-r1-distill-llama-70b-think	llama-3.3-70b	0.315	2.39	0.125	32
deepseek-r1-distill-qwen-32b-think	qwq-32b-think	0.317	7.39	0.038	182
nemotron-70b	llama-3.1-70b	0.792	7.47	0.077	13
nemotron-super-49b-think	llama-3.3-70b	0.167	2.36	0.069	29
hermes-3-405b	llama-3.1-70b	0.255	7.62	1.000	2
hermes-4-405b	hermes-3-405b	0.275	8.49	1.000	3

Table 19: Known-provenance positive controls. Each pair is a publicly-disclosed distillation or fine-tune. All show either lineage or cross-vendor-suspect HSS (≥ 0.10) when both_w permits statistical resolution, confirming that the HSS metric detects shared training signal.

Comprehensive all-family table. Table 20 reports consecutive-generation metrics for every tracked family (open-weight and closed-weight) where at least one consecutive pair has ≥ 8 joint-wrong probes. Three patterns recur across the ~ 30 families:

- *Frequent retrains:* Gemini Flash/Pro, Gemma 2/3/4, Qwen 2.5/3/3.5, most GPT-5 .x transitions, and the Opus 4.6/4.7 step all sit in the retrained regime. These vendors appear to prefer full retrains over incremental post-training when bumping version numbers.
- *Incremental lineages:* DeepSeek V3 $\rightarrow$ V3.1 $\rightarrow$ V3.2, GLM 4.6 $\rightarrow$ 4.7, Kimi K2 $\rightarrow$ K2.5 $\rightarrow$ K2.6, and the Nova / Command families show consistent lineage-regime HSS, consistent with continued pretraining or large-scale post-training on a stable base.
- *Shared-base variants:* Thinking / reasoning variants (Opus 4.5 / 4.5-think, GPT-5.2 / 5.2-pro, DeepSeek V3.2 / V3.2-think) and “pro” variants of the same base generation concentrate at $\text{HSS} \geq 0.30$.

Table 20: Comprehensive within-family fingerprint metrics for all tracked model families (T5–T6 probes). J : Jaccard on correct-answer sets; lift: observed/expected-under-independence; HSS: hallucination similarity; both_w: joint-wrong probes; class: shared-base / lineage / retrained / independent. Families with all pairs having both_w < 8 are omitted.

From	To	J	lift	HSS	both_w	class
<i>OpenAI GPT base</i>						
gpt-3.5-turbo	gpt-4	0.436	1.80	0.118	51	independent
gpt-4	gpt-4-turbo	0.634	1.62	0.111	45	lineage
gpt-4-turbo	gpt-4o	0.448	1.47	0.212	33	independent
gpt-4o	gpt-4.1	0.527	1.51	0.333	27	lineage
gpt-4.1	gpt-5	0.735	1.42	0.000	42	retrained
gpt-5	gpt-5.1	0.638	1.61	0.000	23	retrained
gpt-5.1	gpt-5.2	0.538	1.75	0.083	12	retrained
gpt-5.2	gpt-5.3	0.596	1.74	0.040	25	retrained
gpt-5.3	gpt-5.4	0.714	1.68	0.082	49	retrained
<i>OpenAI GPT-5 base/pro/think cluster</i>						
gpt-5	gpt-5-pro	0.822	1.54	0.511	47	shared-base
gpt-5-pro	gpt-5-think	0.837	1.51	0.532	47	shared-base
<i>OpenAI GPT-5 .x transitions</i>						
gpt-5	gpt-5.1	0.638	1.61	0.000	23	retrained
gpt-5.1	gpt-5.2	0.538	1.75	0.083	12	retrained
gpt-5.2	gpt-5.2-pro	0.777	2.25	0.325	40	shared-base
gpt-5.2-pro	gpt-5.3	0.590	1.70	0.042	24	retrained
gpt-5.3	gpt-5.4	0.714	1.68	0.082	49	retrained
gpt-5.4	gpt-5.4-pro	0.713	1.49	0.113	71	lineage
<i>OpenAI GPT-mini</i>						
gpt-4o-mini	gpt-4.1-mini	0.293	2.19	0.079	152	retrained
gpt-4.1-mini	gpt-5-mini	0.336	2.70	0.333	6	independent
gpt-5-mini	gpt-5.4-mini	0.292	2.52	0.000	4	independent (small n)
<i>OpenAI GPT-nano</i>						
gpt-4.1-nano	gpt-5-nano	0.100	5.07	0.000	9	independent
gpt-5-nano	gpt-5.4-nano	0.118	9.09	0.000	3	independent (small n)
<i>OpenAI o-series</i>						
o1	o3-mini	0.115	1.73	0.000	15	retrained
o3-mini	o3	0.099	1.48	0.000	26	retrained
o3	o4-mini-think	0.381	1.40	0.000	52	retrained
<i>Anthropic Claude Opus</i>						
claude-opus-4	claude-opus-4.1	0.500	8.00	1.000	7	lineage
claude-opus-4.1	claude-opus-4.5	0.239	1.93	0.167	6	independent
claude-opus-4.5	claude-opus-4.6	0.731	1.64	0.100	30	lineage
claude-opus-4.6	claude-opus-4.7	0.601	1.69	0.000	17	retrained
<i>Anthropic Claude Sonnet</i>						
claude-3.7-sonnet	claude-sonnet-4	0.147	1.79	0.000	4	independent (small n)
claude-sonnet-4	claude-sonnet-4.5	0.484	6.06	0.429	7	independent
claude-sonnet-4.5	claude-sonnet-4.6	0.315	2.09	0.091	11	retrained
<i>Anthropic Claude Haiku</i>						
claude-3-haiku	claude-3.5-haiku	0.064	3.33	0.000	1	independent (small n)
claude-3.5-haiku	claude-haiku-4.5	0.175	3.93	0.158	19	independent
<i>Google Gemini Flash</i>						
gemini-2.0-flash	gemini-2.5-flash	0.490	2.29	0.000	55	retrained
gemini-2.5-flash	gemini-3-flash	0.283	1.12	0.056	18	retrained
<i>Google Gemini Flash-Lite</i>						

From	To	J	lift	HSS	both_w	class
gemini-2.5-flash-lite	gemini-3.1-flash-lite	0.211	1.30	0.050	80	retrained
<i>Google Gemma</i>						
gemma-2-2b	gemma-3-1b	0.048	4.10	0.007	284	retrained
gemma-3-1b	gemma-3-4b	0.000	0.00	0.006	351	retrained
gemma-3-4b	gemma-3-12b	0.206	6.76	0.039	336	retrained
gemma-3-12b	gemma-2-27b	0.000	0.00	0.041	98	retrained
gemma-2-27b	gemma-3-27b	0.118	3.66	0.043	93	retrained
gemma-3-27b	gemma-4-31b	0.078	1.39	0.000	232	retrained
<i>Meta Llama 3 (70B)</i>						
llama-3-70b	llama-3.1-70b	0.500	4.74	0.636	11	lineage
llama-3.1-70b	llama-3.3-70b	0.534	5.17	0.375	16	lineage
<i>Meta Llama 3 (8B)</i>						
llama-3-8b	llama-3.1-8b	0.231	13.71	0.222	9	independent
<i>Meta Llama 4</i>						
llama-4-scout	llama-4-maverick	0.250	2.14	0.083	48	retrained
<i>Alibaba Qwen dense (7B)</i>						
qwen-2.5-7b	qwen3-8b-think	0.143	10.26	0.042	118	retrained
<i>Alibaba Qwen dense (70B)</i>						
qwen-2.5-72b	qwq-32b-think	0.250	4.92	0.036	56	retrained
<i>Alibaba Qwen Max</i>						
qwen-max	qwen3-max	0.216	2.16	0.176	17	independent
<i>Alibaba Qwen Plus</i>						
qwen-plus	qwen3.5-plus-think	0.455	1.95	0.000	116	retrained
qwen3.5-plus-think	qwen3.6-plus-think	0.533	2.20	0.095	21	retrained
<i>Alibaba Qwen large MoE</i>						
qwen3-235b-a22b-think	qwen3.5-397b-a17b-think	0.221	2.14	0.009	106	retrained
<i>DeepSeek V3</i>						
deepseek-v3	deepseek-v3.1	0.411	2.04	0.227	22	independent
deepseek-v3.1	deepseek-v3.2	0.471	2.36	0.278	18	independent
<i>Zhipu GLM</i>						
glm-4.5-think	glm-4.6-think	0.621	1.71	0.038	52	retrained
glm-4.6-think	glm-4.7-think	0.686	1.60	0.198	131	lineage
glm-4.7-think	glm-5-think	0.543	1.32	0.022	91	retrained
glm-5-think	glm-5.1-think	0.690	1.50	0.012	83	retrained
<i>xAI Grok</i>						
grok-3	grok-4	0.772	1.36	0.043	46	retrained
grok-4	grok-4.20	0.540	1.27	0.000	52	retrained
<i>Moonshot Kimi K2</i>						
kimi-k2	kimi-k2.5-think	0.595	1.46	0.020	50	retrained
kimi-k2.5-think	kimi-k2.6-think	0.772	1.68	0.451	51	shared-base
<i>Mistral large/medium</i>						
mistral-medium-3.1	mistral-large	0.421	2.97	0.000	36	retrained
<i>Mistral small/open</i>						
mistral-7b	mistral-nemo-12b	0.216	6.32	0.017	177	retrained
mistral-nemo-12b	mistral-small-24b	0.333	8.35	0.070	57	retrained
<i>Mistral MoE (Mixtral)</i>						
mixtral-8x7b	mixtral-8x22b	0.215	1.87	0.000	62	retrained
<i>Mistral ministral</i>						
ministral-3b	ministral-8b	0.080	2.19	0.021	290	retrained

From	To	J	lift	HSS	both_w	class
<i>Amazon Nova</i>						
nova-micro	nova-pro	0.136	3.17	0.067	134	retrained
nova-pro	nova-premier	0.321	2.02	0.014	146	retrained
<i>Cohere Command</i>						
command-r7b	command-r-plus	0.024	1.14	0.000	30	retrained
command-r-plus	command-a	0.231	3.09	0.048	21	retrained
<i>Microsoft Phi</i>						
phi-3-mini	phi-4	0.000	0.00	0.000	174	retrained

All cross-vendor outliers. Table 21 gives the full ranked list of cross-vendor pairs above our reporting threshold.

Table 21: Complete list of cross-vendor outlier pairs with $HSS \geq 0.20$ and ≥ 10 joint-wrong probes on T5–T6.

Model A	Model B	J	lift	HSS	both_w
gemini-3-flash-think	gpt-4o	0.376	1.06	0.500	10
gpt-4o-mini	hermes-4-405b	0.234	3.36	0.500	10
ernie-4.5-300b-a47b	gpt-4o	0.338	2.35	0.438	16
ernie-4.5-300b-a47b	llama-3-70b	0.447	3.67	0.429	14
ernie-4.5-300b-a47b	mistral-large	0.393	3.64	0.400	10
gemini-2.0-flash	llama-3.1-70b	0.283	2.63	0.400	15
gpt-4.1	llama-3.1-70b	0.169	1.52	0.400	10
gpt-4.1-nano	llama-3.1-70b	0.163	2.06	0.400	15
gpt-4	llama-3.1-70b	0.226	2.04	0.385	13
llama-3.1-70b	qwen3-max	0.211	1.97	0.385	13
claude-sonnet-4.6	deepseek-v3.1	0.349	1.78	0.333	12
deepseek-r1-distill-llama-70b-think	llama-3.1-70b	0.208	2.22	0.333	15
ernie-4.5-300b-a47b	mistral-small-24b	0.265	4.72	0.333	18
ernie-4.5-300b-a47b	qwen-max	0.291	3.68	0.333	15
gemini-3.1-flash-lite	llama-3.1-70b	0.152	1.37	0.333	12
grok-3	llama-3.1-70b	0.155	1.40	0.333	12
llama-3-70b	mistral-large	0.533	4.06	0.333	15
gemini-3.1-flash-lite	gpt-4o	0.408	1.23	0.321	28
gpt-5	grok-4	0.632	1.28	0.310	29
claude-sonnet-4.5-think	gpt-4o	0.417	1.67	0.308	13
gpt-3.5-turbo	llama-3.1-70b	0.333	3.16	0.308	13
grok-3	qwen-max	0.140	1.39	0.308	13
claude-haiku-4.5	mistral-large	0.224	5.01	0.300	10
claude-opus-4.7	ernie-4.5-300b-a47b	0.323	2.24	0.300	10
command-r-plus	gpt-5	0.124	1.57	0.300	10
gemini-2.0-flash	qwen-max	0.218	2.29	0.300	20
gemma-4-31b	hermes-4-405b	0.082	1.54	0.300	10
gpt-4-turbo	llama-3.1-70b	0.194	1.77	0.300	10
gpt-4o	llama-3.3-70b	0.350	2.26	0.286	14
gpt-4o	qwen-2.5-72b	0.119	1.59	0.286	14
llama-3.1-70b	nova-pro	0.283	3.17	0.286	14
llama-4-maverick	qwen-max	0.234	2.42	0.286	14
claude-sonnet-4.5	gpt-3.5-turbo	0.385	2.83	0.273	11

Model A	Model B	J	lift	HSS	both_w
command-r-plus	grok-4	0.109	1.39	0.273	11
deepseek-r1-distill-qwen-32b-think	hermes-4-405b	0.093	2.46	0.273	11
glm-4.5-air-think	gpt-5-mini-think	0.358	3.05	0.273	11
gpt-4o	llama-3-70b	0.358	2.13	0.273	11
gpt-5-pro	kimi-k2.6-think	0.571	1.37	0.273	22
claude-opus-4.7-think	llama-3-70b	0.379	2.11	0.267	15
llama-3-70b	qwen-max	0.365	4.03	0.267	15
gemini-2.5-pro	grok-4	0.613	1.18	0.265	34
gemini-2.5-flash	gpt-5-think	0.357	1.46	0.263	19
gpt-4o	llama-4-maverick	0.426	1.72	0.263	19
kimi-k2	mistral-large	0.281	1.87	0.261	23
gpt-5-think	grok-4	0.676	1.30	0.259	27
claude-sonnet-4.5	kimi-k2	0.289	1.94	0.250	12
gemini-2.5-flash	gpt-5	0.348	1.46	0.250	20
gemini-3-flash	grok-3-mini-think	0.300	1.12	0.250	16
gemini-3-flash-think	grok-3-mini-think	0.289	1.08	0.250	12
gpt-4.1	grok-3-mini-think	0.394	1.51	0.250	24
gpt-4o	kimi-k2	0.414	1.45	0.250	16
gpt-4o	mistral-small-24b	0.118	1.99	0.250	12
gpt-4o	o1	0.542	1.66	0.250	12
grok-4	o3	0.644	1.25	0.250	32
kimi-k2	llama-3.1-70b	0.219	1.99	0.250	12
mistral-large	qwen-max	0.425	4.80	0.250	12
gpt-4o	qwen3-max	0.451	1.60	0.242	33
kimi-k2.6-think	o3	0.558	1.33	0.240	25
nova-pro	qwen-max	0.188	2.41	0.240	25
gpt-4-turbo	qwen-max	0.204	1.99	0.235	17
command-r-plus	o3	0.123	1.54	0.231	13
gpt-4	qwen-max	0.204	2.03	0.231	13
gpt-4o-mini	llama-3.1-70b	0.321	3.83	0.231	13
grok-3	mistral-large	0.210	1.37	0.231	13
grok-4	kimi-k2.6-think	0.643	1.45	0.231	26
gpt-4	llama-3.3-70b	0.316	1.96	0.222	18
glm-5-turbo-think	gpt-5	0.601	1.39	0.220	41
gpt-5-pro	grok-4	0.642	1.28	0.219	32
gpt-5	kimi-k2.6-think	0.577	1.40	0.217	23
claude-haiku-4.5	ernie-4.5-300b-a47b	0.194	4.49	0.214	14
claude-sonnet-4-think	olmo-3.1-32b	0.102	3.30	0.214	14
gpt-5-mini-think	qwen3.5-122b-a10b-think	0.269	2.37	0.214	14
gpt-5-mini-think	seed-1.6-think	0.293	2.55	0.214	14
gpt-5.3	grok-4	0.563	1.31	0.214	28
ling-2.6-flash	seed-1.6-think	0.150	2.84	0.214	14
deepseek-r1-distill-llama-70b-think	llama-3-70b	0.346	2.40	0.212	33
ernie-4.5-300b-a47b	o4-mini-think	0.344	2.57	0.211	38
gemini-3-flash	gpt-4o	0.390	1.10	0.211	19
gpt-4o	grok-3-mini-think	0.360	1.71	0.211	19
grok-3	llama-3-70b	0.255	1.39	0.211	19
kimi-k2	llama-3-70b	0.316	1.81	0.211	19
ernie-4.5-300b-a47b	grok-3	0.198	1.34	0.208	24
ernie-4.5-300b-a47b	llama-4-maverick	0.268	2.02	0.208	24

Model A	Model B	J	lift	HSS	both_w
command-r-plus	seed-1.6-think	0.204	2.85	0.207	29
gemini-2.0-flash	llama-3-70b	0.380	2.30	0.207	29
gemini-2.5-pro-think	grok-4	0.615	1.19	0.206	34
glm-5-turbo-think	grok-4	0.585	1.31	0.205	39
claude-3.5-haiku	ernie-4.5-300b-a47b	0.225	2.19	0.200	10
claude-haiku-4.5	deepseek-v3.2	0.124	2.63	0.200	15
claude-haiku-4.5-think	deepseek-v3.2	0.189	2.43	0.200	10
claude-opus-4.1	gpt-4.1-nano	0.227	2.55	0.200	10
claude-opus-4.1	qwen3.5-35b-a3b-think	0.182	2.37	0.200	10
claude-opus-4.6	deepseek-v3.1	0.339	1.64	0.200	10
claude-opus-4.7	gpt-4.1	0.519	1.45	0.200	10
claude-opus-4.7	gpt-5	0.526	1.50	0.200	10
claude-opus-4.7	mixtral-8x22b	0.412	1.88	0.200	15
claude-opus-4.7-think	mistral-large	0.318	2.10	0.200	10
claude-sonnet-4-think	glm-5-think	0.317	1.59	0.200	10
claude-sonnet-4.5	mistral-small-24b	0.261	4.65	0.200	10
claude-sonnet-4.5-think	gemini-3.1-flash-lite	0.439	1.35	0.200	20
command-a	gpt-5-mini-think	0.336	2.73	0.200	10
deepseek-v3.1	qwen-plus	0.360	2.12	0.200	20
ernie-4.5-300b-a47b	qwen-2.5-72b	0.175	2.78	0.200	25
glm-4.5-air-think	gpt-4o	0.335	1.86	0.200	25
glm-4.5-air-think	llama-3.1-70b	0.310	3.20	0.200	10
gpt-3.5-turbo	qwen-max	0.265	2.83	0.200	20
gpt-4.1-nano	hermes-4-405b	0.140	2.04	0.200	10
gpt-5-mini-think	llama-4-maverick	0.273	2.13	0.200	10
gpt-5-mini-think	qwen3.5-27b-think	0.165	2.16	0.200	15
gpt-5-think	kimi-k2.6-think	0.590	1.39	0.200	20
gpt-5.1	mistral-medium-3.1	0.366	1.63	0.200	10
grok-3	qwen3-max	0.520	1.35	0.200	75
grok-4	kimi-k2.5-think	0.679	1.35	0.200	45
hermes-4-405b	qwen3.5-35b-a3b-think	0.138	2.20	0.200	10
ling-2.6-flash	qwen3.5-plus-think	0.115	2.08	0.200	10
llama-3.1-70b	o3-mini	0.263	4.94	0.200	10
llama-3.1-70b	qwen3.5-35b-a3b-think	0.138	1.98	0.200	15
llama-3.1-8b	qwen3.5-27b-think	0.078	4.76	0.200	10
llama-4-maverick	mistral-large	0.298	2.15	0.200	15

I Case Studies

This section presents concrete examples of probe questions and model responses to illustrate key findings from the main text. All responses are verbatim model outputs (truncated for space).

I.1 USTC Hackergame — Watching a Fact Arrive Over Three Years

The motivating observation in Section 1 traces the arrival of a single piece of long-tail knowledge—specific challenge names from *USTC Hackergame*, an annual Capture-the-Flag competition organized by the USTC Linux User Group since 2014—into frontier model parameters across release cycles. We probed 12 representative models (across 2022–2026) with a uniform prompt asking for specific challenge names per year, then verified each response against the official writeup repositories at <https://github.com/USTC-Hackergame> and <https://github.com/ustclug/hackergame>.

Results are summarized in Table 22.

Three observations from this case study.

1. **The transition is sharp and dateable.** GPT-4o (May 2024) hallucinates a deterministic fake list, repeated across independent probes. Claude 3.7 Sonnet (February 2025) is the first model to list real per-year challenges, with ~ 19 verified 2023 titles. Whatever change to the training pipeline absorbed Hackergame writeups happened in this ~ 9 -month window, not gradually across the field; later models (Claude Opus 4.6/4.7, GPT-5.5, Kimi K2.6) inherited and extended this knowledge.
2. **Knowing the meta-fact does not imply knowing the content.** DeepSeek V4 Pro (1.6T) is the only model that correctly states 2014 as the start year (a single Wikipedia-grade meta-fact) but fabricates per-year challenges. The structural fact and the long-tail content are stored independently.
3. **Refusal vs hallucination is a vendor-level choice that interacts with measurement.** GPT-4o and DeepSeek V4 Pro confidently fabricate; Claude Opus 4.7 says “I don’t have reliable detailed memory and don’t want to fabricate specifics”; GPT-5/5.5 burn output budget on reasoning and produce empty content.

This case is one motivating example among many; the systematic IKP probe set (Section 4.1) operationalizes the same kind of observation across 1,400 facts and 201 models, with quantitative scoring rather than human verification.

Model	Released	Verification of LLM Responses
GPT-3.5 Turbo	2022-11	Hallucinates: claims contest started in “2006” (actual: 2014); claims “offline finals” (contest is online only).
GPT-4	2023-03	Admits no specific knowledge. Generic CTF description.
GPT-4o	2024-05	Knows USTC + CTF format. Asked for per-year challenges, returns the <i>identical</i> fabricated list (“Hello World”, “Maze”, “Calculator”, “Reverse Engineering”, “Quantum Computing”) for nine different years (2015–2023). None of these are real Hackergame challenges; the pattern is deterministic fabrication, repeated across two independent probes.
Claude 3.7 Sonnet	2025-02	First model to list real per-year challenges. 2023: “Kitty Quiz”, “Travel Photo”, “Cyber Tic-Tac-Toe”, “Committee Simulator”, “High-Frequency Planet”, “Streaming Planet”, “Small LM Planet”, “Word-Frugal”, “Grandma’s Bedtime flag Story”, “Alien Detour” (all verified real, English glosses of Chinese originals). Also accurate 2022 (HeiLang, Xcaptcha, plus original-Chinese titles) and 2021 (similar).
Gemini 2.5 Pro	2025-03	Lists ~6 real challenges per year for 2021–2023 (e.g., HeiLang, “JSON $\subset$ YAML?”, plus various original-Chinese titles), with one likely fabrication (a non-existent planet-series title).
GPT-5	2025-08	Returns reasoning trace only; content cut off before any answer is written.
Claude Opus 4.6	2025-11	Lists 16+ real Hackergame 2023 challenges, all verified: “Kitty Quiz”, “Deeper Darker”, “Travel Photo 3.0”, “Cyber Tic-Tac-Toe”, “Grandma’s Bedtime flag Story”, “Committee Simulator”, “Worm”, “JSON $\subset$ YAML?”, “Git? Git!”, “HTTP Stamp Album”, “Docker for Everyone”, “Word-Frugal 2.0”, “High-Frequency Planet”, “Streaming Planet”, “Komm süsser satisfiability”, “Alien Detour”. Earlier years also accurate.
Gemini 3.1 Pro	2026-01	Year-by-year breakdown 2018–2023 with specific real challenge names (“Travel Photo”, “Selling Melons”, “Transparent File”, “Kitty Quiz” series). Concise but accurate.
Kimi K2.6-think	2026-03	30+ specific names across 2020–2023, mostly verified real (HeiLang, Xcaptcha, “Cup-Window Goose-Shadow”/Wine, “Komm süsser Flagge”), with occasional year-mixing.
DeepSeek V4 Pro	2026-04	Hallucinates: knows the meta-fact (correctly says contest started 2014, the only model that does) but fabricates per-year challenges as generic CTF tropes (“check-in problem”, “classical cryptography”, “web dog daily”, “PWN intro”)—none are actual Hackergame names.
GPT-5.5	2026-04	Lists real 2023 names (“Sign-In”, “Kitty Quiz”, “Deeper Darker”, “Travel Photo 3.0”, “JSON $\subset$ YAML?”, “Git?”, “HTTP Stamp Album”, “Docker for Everyone”, “Word-Frugal 2.0”).
Claude Opus 4.7	2026-04	Year-by-year list 2018–2024. Specific names per year mostly real, with occasional cross-year mixing. Says contest started 2018 (incorrect: that was the year of the LUG renaming; contest itself dates to 2014).

Table 22: USTC Hackergame probe across 12 representative frontier models. Verification was performed against the official writeup repositories (github.com/USTC-Hackergame/hackergame20{18..23}-writeups) and the platform repo (github.com/ustclug/hackergame).

I.2 Tier Discrimination: Universal Knowledge (T1)

T1 probes test universal knowledge that every model, regardless of size, should know. Even the smallest 1B-parameter models answer these correctly.

Probe: What is the capital of France?

Tier: T1 **Gold answer:** Paris

Domain: general

Gemma-3 1B

CORRECT — Paris.

Llama-3.2 3B

CORRECT — Paris.

GPT-4o

CORRECT — The capital of France is Paris.

Claude Opus 4.6

CORRECT — Paris.

I.3 Tier Discrimination: Mid-Tier Knowledge (T3)

T3 probes separate small models from medium and large ones. Small models (<10B) typically refuse or hallucinate, while models above ~30B answer correctly. This probe tests a moderately obscure scientific fact.

Probe: Who first produced liquid helium?

Tier: T3 **Gold answer:** Heike Kamerlingh Onnes

Domain: science

Gemma-3 4B

WRONG — William Ramsay and Lord Rayleigh.

Llama-3.2 3B

REFUSAL — I don't know.

GPT-4o

CORRECT — Heike Kamerlingh Onnes first produced liquid helium in 1908.

Claude Opus 4.6

CORRECT — Heike Kamerlingh Onnes first produced liquid helium on July 10, 1908, at his laboratory in Leiden, the Netherlands. This achievement earned him widespread recognition and contributed to his receiving the Nobel Prize in P...

I.4 Tier Discrimination: Frontier-Only Knowledge (T6)

T6 probes are only answered correctly by the largest frontier models. These typically involve obscure researchers or rarely documented facts. Even strong models like GPT-4o refuse on these probes, while the top-scoring Gemini models retrieve the information.

Probe: In computer science, what is the research subfield of Janick Edinger, and name one paper, system, institution, or co-author associated with their work? If you don't know who this person is, say so.

Tier: T6 **Gold answer:** distributed systems

Domain: computer_science

Llama-3.3 70B

REFUSAL — I don't know.

GPT-4o

REFUSAL — I don't know.

Gemini 3 Flash

CORRECT — Janick Edinger's research subfield is distributed systems, with a specific focus on edge computing, pervasive computing, and mobile computing.

Gemini 3.1 Pro

CORRECT — Janick Edinger's research subfields in computer science are primarily distributed systems, edge computing, pervasive/ubiquitous computing, and the Internet of Things (IoT).

I.5 Hallucination vs. Refusal

A critical distinction in IKP evaluation is between hallucination (confidently wrong) and refusal (saying one does not know). Our scoring penalizes hallucination but not refusal, reflecting the real-world risk of confident misinformation. Here, Claude Opus 4.6 answers correctly, Llama-3.3 70B appropriately refuses, but GPT-4o-mini hallucinates a plausible but wrong subfield.

Probe: In computer science, what is the research subfield of Zhongzhi Luan, and name one paper, system, institution, or co-author associated with their work? If you don't know who this person is, say so.

Tier: T5 **Gold answer:** computer architecture

Domain: computer_science

Claude Opus 4.6

CORRECT — Zhongzhi Luan is a researcher in the field of high-performance computing (HPC). His research interests include parallel computing, large-scale cluster systems, and related areas in computer architecture and systems. He h...

Llama-3.3 70B

REFUSAL — I don't know.

GPT-4o-mini

WRONG — Zhongzhi Luan is known for his research in the field of artificial intelligence, particularly in areas related to machine learning, data mining, and complex networks.

I.6 Name Collision in Researcher Probes

Researcher probes are particularly susceptible to name collision: the model recognizes the name but associates it with a different person in the same or a different field. Here, “Geoffrey Ye Li” is a wireless communications researcher, but GPT-4o-mini confuses him with Geoffrey Hinton-adjacent ML researchers, producing a confident but wrong answer about “deep learning and reinforcement learning.”

Probe: In computer science, what is the research subfield of Geoffrey Ye Li, and name one paper, system, institution, or co-author associated with their work? If you don’t know who this person is, say so.

Tier: T3 **Gold answer:** computer networking

Domain: computer_science

GPT-4o-mini

WRONG — Geoffrey Ye Li is known for his research in the field of machine learning, particularly in areas related to deep learning, reinforcement learning, and their applications in various domains.

GPT-4o

CORRECT — Geoffrey Ye Li is known for his research in the field of wireless communications and signal processing.

Claude Opus 4.6

CORRECT — Geoffrey Ye Li is a prominent researcher in the field of wireless communications and signal processing. His research subfields include: - OFDM (Orthogonal Frequency-Division Multiplexing) - MIMO (Multiple-Input Multiple...

Llama-3.3 70B

CORRECT — Geoffrey Ye Li’s research subfield is wireless communications and signal processing.

I.7 Thinking Mode Enabling Correct Recall

Extended thinking (chain-of-thought reasoning) can help models retrieve knowledge they would otherwise fail to access. Here, the base Claude Sonnet 4 refuses the question about a classical Japanese poetry anthology, but the thinking variant correctly identifies Fujiwara no Teika as the compiler. The thinking trace allows the model to reason through partial knowledge to arrive at the correct answer.

Probe: What Japanese poet wrote the Hyakunin Isshu anthology?

Tier: T3 **Gold answer:** Fujiwara no Teika

Domain: Non-Western Cultural Works

Claude Sonnet 4

REFUSAL — I don’t know who compiled the Hyakunin Isshu anthology. While I know it’s a famous collection of 100 Japanese poems by 100 different poets, I’m not certain about who assembled the anthology itself.

Claude Sonnet 4 (think)

CORRECT — The Hyakunin Isshu anthology was compiled by Fujiwara no Teika (also known as Fujiwara no Sadaie) around 1235. He selected one poem each from 100 different poets, creating this famous collection of classical Japanese wak...

I.8 Thinking Mode Correcting Hallucination

Thinking mode not only helps with recall but can also correct hallucinations. Here, the base DeepSeek-V3.2 answers 1529 (the first Ottoman siege) instead of the gold answer 1683 (the second, more famous siege). While both sieges were technically “repelled,” the probe targets the 1683 Battle of Vienna—a more common reference in training data. The thinking variant correctly disambiguates between the two events.

Probe: In what year was the Siege of Vienna by the Ottomans that was repelled?

Tier: T1 **Gold answer:** 1683

Domain: Historical Dates

DeepSeek-V3.2

WRONG — The Siege of Vienna by the Ottomans that was successfully repelled occurred in 1529.

DeepSeek-V3.2 (think)

CORRECT — The Siege of Vienna by the Ottomans that was repelled occurred in 1683.

J Ethical Considerations

This work estimates proprietary model sizes from public API access, which raises considerations for competitive intelligence and intellectual property. We note that: (1) parameter count is only one dimension of model capability, and our estimates are effective capacity (not literal architecture); (2) the method provides $\sim 3.0\times$ precision, insufficient for reverse-engineering proprietary designs; and (3) similar estimates are already publicly available from inference-economics analyses [Epoch AI, 2024]. Knowledge fingerprinting for cross-model lineage analysis could be used to enforce licensing terms (a legitimate use) or to make unfounded IP claims (a misuse). We encourage responsible use of these tools for transparency and governance rather than competitive surveillance.

All information presented in this paper is drawn from publicly available sources. The author has no internal access to, confidential briefings about, or non-public collaboration with any frontier LLM vendor; in particular, the parameter and effective-capacity estimates for proprietary models are produced *entirely* from external behavioral measurements and the open-weight calibration set.